

Physical world and Measurements

Question1

The least count of a screw guage is 0.01 mm . If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, the new least count will be _____ $\times 10^{-3}$ mm

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Answer: 35

Solution:

The least count (L.C.) of a screw gauge is given by:

$$\text{L.C.} = \frac{\text{Pitch}}{\text{Number of Divisions on Circular Scale}}$$

Let the original pitch be p and the original number of divisions be n . Then, the original least count is:

$$\frac{p}{n} = 0.01 \text{ mm}$$

After modifications:

The pitch is increased by 75%, so the new pitch is:

$$p_{\text{new}} = p + 0.75p = 1.75p$$

The number of divisions is reduced by 50%, so the new number of divisions is:

$$n_{\text{new}} = 0.5n$$

The new least count is then:

$$\text{L.C.}_{\text{new}} = \frac{p_{\text{new}}}{n_{\text{new}}} = \frac{1.75p}{0.5n} = \frac{1.75}{0.5} \times \frac{p}{n} = 3.5 \times \frac{p}{n}$$

Substitute the original least count:

$$\text{L.C.}_{\text{new}} = 3.5 \times 0.01 \text{ mm} = 0.035 \text{ mm}$$

Expressed in the form $\times 10^{-3}$ mm, this becomes:

$$0.035 \text{ mm} = 35 \times 10^{-3} \text{ mm}$$

Thus, the new least count is:

$$35 \times 10^{-3} \text{ mm}$$

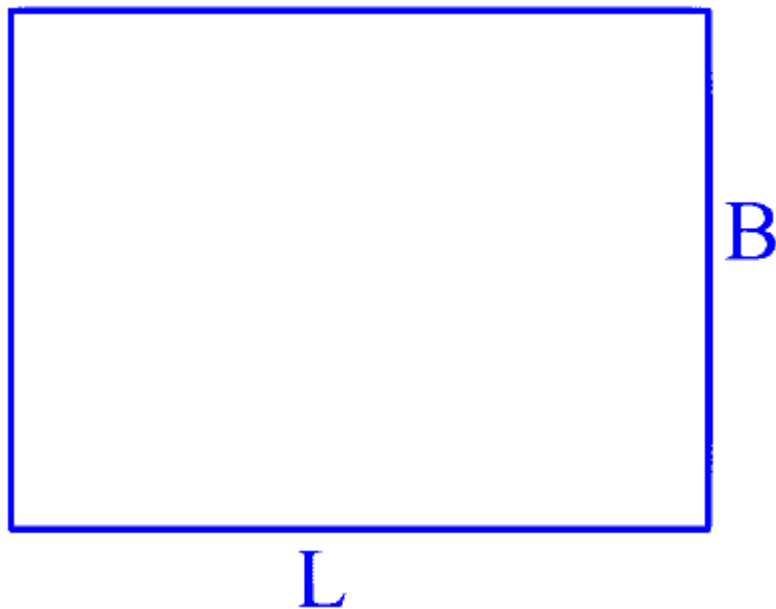
Question2

A tiny metallic rectangular sheet has length and breadth of 5 mm and 2.5 mm , respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where x is _____ .

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Answer: 3

Solution:



Given, $L = 5 \text{ mm}$, $B = 2.5 \text{ mm}$

We know, least count of a screw gauge,

$$L.C. = \frac{\text{Pitch length}}{\text{No. of division on circular scale}}$$

$$\Rightarrow L.C. = \frac{0.75}{15} = 0.05 \text{ mm}$$

We know, $A = LB$

By taking $\ln$ on both sides,

$$\Rightarrow \ln A = \ln L + \ln B$$

by differentiating both sides,

$$\Rightarrow \frac{dA}{A} = \frac{dL}{L} + \frac{dB}{B}$$

Here, $dL = dB = 0.05 \text{ mm}$ (L.C.)

So fractional error,

$$\frac{dA}{A} = \frac{0.05}{5} + \frac{0.05}{2.5}$$

$$= \frac{1}{100} + \frac{2}{100} = \frac{3}{100} = \frac{x}{100} \text{ (given)}$$

Hence, $x = 3$.

Question3

A physical quantity Q is related to four observables a, b, c, d as follows :

$$Q = \frac{ab^4}{cd}$$

where,

$a = (60 \pm 3) \text{ Pa}$; $b = (20 \pm 0.1) \text{ m}$; $c = (40 \pm 0.2) \text{ Nsm}^{-2}$ and $d = (50 \pm 0.1) \text{ m}$, then the percentage error in Q is $\frac{x}{1000}$, where $x = \underline{\hspace{2cm}}$.

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Answer: 7700

Solution:

$$\text{Given, } Q = \frac{ab^4}{cd}$$

$$a = (60 \pm 3) \text{ Pa} \Rightarrow a = 60 \text{ Pa}, \Delta a = 3 \text{ Pa}$$

$$b = (20 \pm 0.1) m \Rightarrow b = 20m, \Delta b = 0.1 m$$

$$c = (40 \pm 0.2) Nsm^{-2} \Rightarrow c = 40 Nsm^{-2}, \Delta c = 0.2 Nsm^{-2}$$

$$d = (50 \pm 0.1) m \Rightarrow d = 50 m, \Delta d = 0.1m$$

$$\text{As, } Q = \frac{ab^4}{cd}$$

by taking $\ln$ on both sides,

$$\ln Q = \ln a + 4 \ln b - \ln c - \ln d$$

Now, by differentiating,

$$\frac{dQ}{Q} = \frac{da}{a} + 4 \frac{db}{b} - \frac{dc}{c} - \frac{dd}{d}$$

So, maximum fractional error in Q is given by,

$$\frac{\Delta Q}{Q} = \frac{\Delta a}{a} + 4 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d}$$

$$\Rightarrow \frac{\Delta Q}{Q} = \frac{3}{60} + 4 \left(\frac{0.1}{20} \right) + \frac{0.2}{40} + \frac{0.1}{40}$$

$$= \frac{1}{20} + \frac{1}{50} + \frac{1}{200} + \frac{1}{500}$$

$$\Rightarrow \frac{\Delta Q}{Q} = \frac{50+20+5+2}{1000} = \frac{77}{1000}$$

$$\text{Hence the \% error in } Q = \frac{\Delta Q}{Q} \times 100\%$$

$$= \frac{7700}{1000} \% = \frac{x}{1000} \text{ (given)}$$

$$\text{So, } x = 7700$$

Question4

A physical quantity C is related to four other quantities p, q, r and s as follows

$$C = \frac{pq^2}{r^3\sqrt{s}}$$

The percentage errors in the measurement of p, q, r and s are 1%, 2%, 3% and 2%, respectively. The percentage error in the measurement of C will be _____%

Answer: 15

Solution:

To determine the percentage error in the measurement of C , which is related to p , q , r , and s as:

$$C = \frac{pq^2}{r^3\sqrt{s}}$$

we first express it in terms of powers:

$$C = p^1 q^2 r^{-3} s^{-1/2}$$

The percentage error in C can be calculated using the formula for the propagation of error, which is:

$$\left(\frac{\Delta C}{C}\right)_{\max} = \left|\frac{\Delta p}{p}\right| + 2\left|\frac{\Delta q}{q}\right| + 3\left|\frac{\Delta r}{r}\right| + \frac{1}{2}\left|\frac{\Delta s}{s}\right|$$

Given the percentage errors for p , q , r , and s are 1%, 2%, 3%, and 2% respectively, we substitute these values into the formula:

$$\left(\frac{\Delta C}{C}\right)_{\max} = 1\% + 2 \times 2\% + 3 \times 3\% + \frac{1}{2} \times 2\%$$

Calculating each part:

Contribution from p : 1%

Contribution from q : 4% (since $2 \times 2\% = 4\%$)

Contribution from r : 9% (since $3 \times 3\% = 9\%$)

Contribution from s : 1% (since $\frac{1}{2} \times 2\% = 1\%$)

Adding these contributions together:

$$1\% + 4\% + 9\% + 1\% = 15\%$$

Thus, the maximum percentage error in the measurement of C is 15%.

Question5

Given below are two statements :

Statement I: In a vernier callipers, one vernier scale division is always smaller than one main scale division.

Statement II : The vernier constant is given by one main scale division multiplied by the number of vernier scale divisions.

In the light of the above statements, choose the correct answer from the options given below.

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Options:

- A. Both Statement I and Statement II are false
- B. Statement I is true but Statement II is false
- C. Both Statement I and Statement II are true
- D. Statement I is false but Statement II is true

Answer: B

Solution:

In general, one vernier scale division is smaller than one main scale division but in some modified cases it may be not correct. Also least count is given by one main scale division divided by number of vernier scale division for normal vernier calliper.

Hence, option 2 is correct.

Question6

If B is magnetic field and μ_0 is permeability of free space, then the dimensions of (B/μ_0) is

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Options:

- A. $MT^{-2} A^{-1}$
- B. $L^{-1} A$
- C. $ML^2 T^{-2} A^{-1}$

D. $\text{LT}^{-2} \text{A}^{-1}$

Answer: B

Solution:

To determine the dimensions of $\left(\frac{B}{\mu_0}\right)$, we start with the formula for the magnetic field B inside a solenoid:

$$B = \mu_0 n I$$

where $n = \frac{N}{L}$ is the number of turns per unit length, I is the current, and μ_0 is the permeability of free space.

Rearranging the formula gives:

$$\frac{B}{\mu_0} = n I = \frac{N}{L} I$$

From the above, the dimensions of $\left(\frac{B}{\mu_0}\right)$ are:

$$\left[\frac{B}{\mu_0}\right] = [L^{-1} A]$$

Thus, the dimensional formula of $\left(\frac{B}{\mu_0}\right)$ corresponds to $L^{-1} A$.

Question7

The maximum percentage error in the measurement of density of a wire is

[Given, mass of wire = $(0.60 \pm 0.003)\text{g}$

radius of wire = $(0.50 \pm 0.01)\text{cm}$

length of wire = $(10.00 \pm 0.05)\text{cm}$]

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Options:

A. 7

B. 8

C. 5

D. 4

Answer: C

Solution:

To determine the maximum percentage error in the density of the wire, follow these steps:

The density of the wire is given by the formula for a cylinder:

$$\rho = \frac{m}{\pi r^2 l}$$

When calculating the maximum percentage (relative) error, you add the relative errors from each variable, taking into account the power to which each variable is raised. Thus, for density:

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 2 \frac{\Delta r}{r} + \frac{\Delta l}{l}$$

Here:

$\frac{\Delta m}{m}$ is the relative error in the mass.

$2 \frac{\Delta r}{r}$ is due to the radius being squared.

$\frac{\Delta l}{l}$ is the relative error in the length.

Now plug in the given values:

Mass: $m = 0.60$ g with an error $\Delta m = 0.003$ g

$$\frac{\Delta m}{m} = \frac{0.003}{0.60} = 0.005 \text{ or } 0.5\%.$$

Radius: $r = 0.50$ cm with an error $\Delta r = 0.01$ cm

$$\frac{\Delta r}{r} = \frac{0.01}{0.50} = 0.02 \text{ or } 2\%. \text{ Since the radius is squared in the formula, multiply by 2:}$$

$$2 \frac{\Delta r}{r} = 2 \times 0.02 = 0.04 \text{ or } 4\%.$$

Length: $l = 10.00$ cm with an error $\Delta l = 0.05$ cm

$$\frac{\Delta l}{l} = \frac{0.05}{10.00} = 0.005 \text{ or } 0.5\%.$$

Sum these contributions to find the overall maximum relative error:

$$\frac{\Delta \rho}{\rho} = 0.005 + 0.04 + 0.005 = 0.05$$

Converting this relative error into a percentage gives:

$$0.05 \times 100\% = 5\%$$

Thus, the maximum percentage error in the density of the wire is 5%, which corresponds to Option C.

Question8

The position of a particle moving on x -axis is given by $x(t) = A \sin t + B \cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is

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Options:

A. L

B. $L^3 T^{-2}$

C. $L^2 T^{-2}$

D. L^2

Answer: C

Solution:

Let's analyze the position function step by step.

The particle's position is given by:

$$x(t) = A \sin t + B \cos^2 t + Ct^2 + D,$$

where t represents time, which has the dimension T .

Since the overall dimensions of position $x(t)$ must be length (L), each term in the expression must also have dimensions of length.

Term $A \sin t$:

The sine function, $\sin t$, is dimensionless (its argument must be dimensionless, and by convention, if t is in an appropriate unit where the argument is dimensionless, the function itself is dimensionless).

Therefore, A must carry the dimension of length:

$$[A] = L.$$

Term $B \cos^2 t$:

Similarly, the cosine function is dimensionless, and so is its square.

Thus, B must have the dimension of length:

$$[B] = L.$$

Term Ct^2 :

Here, t^2 has the dimension T^2 .

To have the overall term with the dimension L , we require:

$$[C] \times [T^2] = L.$$

So, the dimension of C must be:

$$[C] = L T^{-2}.$$

Term D :

This is a constant term added to position, so it must also have the dimension of length:

$$[D] = L.$$

Now, we are asked to find the dimension of:

$$\frac{ABC}{D}.$$

The dimensions of A , B , and C are:

$$[A] = L, \quad [B] = L, \quad [C] = L T^{-2}.$$

Multiplying them together:

$$[ABC] = L \times L \times (L T^{-2}) = L^3 T^{-2}.$$

Since $[D] = L$, dividing by D gives:

$$\frac{[ABC]}{[D]} = \frac{L^3 T^{-2}}{L} = L^2 T^{-2}.$$

Thus, the dimension of $\frac{ABC}{D}$ is:

$$\boxed{L^2 T^{-2}}.$$

The correct answer is Option C.

Question9

Match List - I with List - II

	List - I		List - II
(A)	Permeability of free space	(I)	$[ML^2 T^{-2}]$
(B)	Magnetic field	(II)	$[MT^{-2} A^{-1}]$

	List - I		List - II
(C)	Magnetic moment	(III)	$[MLT^{-2} A^{-2}]$
(D)	Torsional constant	(IV)	$[L^2 A]$

Choose the correct answer from the options given below :

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Options:

- A. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
- B. (A) – (I), (B) – (IV), (C) – (II), (D) – (III)
- C. (A) – (II), (B) – (I), (C) – (III), (D) – (IV)
- D. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Answer: A

Solution:

The correct matching is:

(A) Permeability of free space has dimensions

$$[MLT^{-2} A^{-2}]$$

which corresponds to entry **(III)**.

(B) Magnetic field has dimensions

$$[MT^{-2} A^{-1}]$$

which corresponds to entry **(II)**.

(C) Magnetic moment in SI units is expressed in Ampere-square meters, i.e.,

$$[L^2 A]$$

which corresponds to entry **(IV)**.

(D) Torsional constant (torque per unit angular displacement) has dimensions

$$[ML^2 T^{-2}]$$

which corresponds to entry (I).

Thus, the correct answer is:

Option A: (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Question10

The energy of a system is given as $E(t) = \alpha^3 e^{-\beta t}$, where t is the time and $\beta = 0.3 \text{ s}^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5 \text{ s}$, maximum percentage error in the energy is :

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Options:

A. 6%

B. 11.6%

C. 4%

D. 8.4%

Answer: A

Solution:

We start with the energy function:

$$E(t) = \alpha^3 e^{-\beta t},$$

where $\beta = 0.3 \text{ s}^{-1}$.

Step 1. Take the Logarithm

Taking the logarithm of both sides:

$$\ln E = 3 \ln \alpha - \beta t.$$

Step 2. Differentiate to Find the Relative Error

Differentiate both sides:

$$\frac{dE}{E} = 3 \frac{d\alpha}{\alpha} - \beta dt.$$

For maximum error estimation, we consider the absolute values and sum the contributions:

$$\left| \frac{\Delta E}{E} \right| \approx 3 \left| \frac{\Delta \alpha}{\alpha} \right| + \beta |\Delta t|.$$

Step 3. Plug in the Given Errors

The percentage error in α is 1.2% (i.e., $\Delta\alpha/\alpha = 0.012$).

The percentage error in t is 1.6%, so for $t = 5$ s:

$$\Delta t = 0.016 \times 5 = 0.08 \text{ s}.$$

Now substitute these values:

$$\left| \frac{\Delta E}{E} \right| \approx 3(0.012) + 0.3(0.08).$$

Calculating each term:

$$3(0.012) = 0.036 \text{ (or 3.6\%)},$$

$$0.3(0.08) = 0.024 \text{ (or 2.4\%)}. \quad \square$$

Step 4. Compute the Total Maximum Percentage Error

Add the contributions:

$$\left| \frac{\Delta E}{E} \right| \approx 0.036 + 0.024 = 0.06,$$

which is 6%.

Thus, the maximum percentage error in the energy at $t = 5$ s is:

$$\boxed{6\%}$$

Answer: Option A (6%).

Question11

For an experimental expression $y = \frac{32.3 \times 1125}{27.4}$, where all the digits are significant. Then to report the value of y we should write

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Options:

A. $y = 1326.186$

B. $y = 1326.2$

C. $y = 1326.19$

D. $y = 1330$

Answer: D

Solution:

32.3 (3 sig figs), 1125 (4 sig figs), 27.4 (3 sig figs)

In multiplication and division, the final answer must be reported with the same number of significant figures as the factor with the fewest significant figures, which here is 3.

First, calculate the unrounded result:

$$y = \frac{32.3 \times 1125}{27.4} \approx \frac{36337.5}{27.4} \approx 1326.186$$

Expressing 1326.186 in scientific notation:

$$1326.186 \approx 1.326186 \times 10^3$$

Rounding to 3 significant figures:

The first three significant digits are 1, 3, and 2.

Considering the fourth digit (6) causes the last significant digit to round up.

Thus,

$$1.326186 \times 10^3 \approx 1.33 \times 10^3$$

In standard decimal form, this is written as:

$$1330$$

Hence, the final reported value is:

$$y = 1330$$

Question12

Match List - I with List - II.

List - I	List - II
(A) Angular Impulse	(I) $M^0 L^2 T^{-2}$
(B) Latent Heat	(II) $M L^2 T^{-3} A^{-1}$
(C) Electrical resistivity	(III) $M L^2 T^{-1}$
(D) Electromotive force	(IV) $M L^3 T^{-3} A^{-2}$

Choose the correct answer from the options given below:

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Options:

A.

(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

B.

(A)-(I), (B)-(III), (C)-(IV), (D)-(II)

C.

(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

D.

(A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Answer: C

Solution:

Matching the quantities from List - I with their correct dimensions from List - II is crucial to understanding their physical significance. Here is the correct matching:

(A) Angular Impulse: The dimensions of angular impulse are equivalent to those of angular momentum, represented as ML^2T^{-1} . Hence, (A) corresponds to (III).

(B) Latent Heat: Latent heat has dimensions of energy per unit mass, which simplifies to $M^0L^2T^{-2}$. Hence, (B) corresponds to (I).

(C) Electrical resistivity: Electrical resistivity has the dimensions of resistance multiplied by length, which can be represented as $ML^3T^{-3}A^{-2}$. Hence, (C) corresponds to (IV).

(D) Electromotive force: Electromotive force (emf) is similar to electric potential difference and has dimensions $ML^2T^{-3}A^{-1}$. Hence, (D) corresponds to (II).

Therefore, the correct answer is **Option C**:

(A)-(III)

(B)-(I)

(C)-(IV)

(D)-(II)

Question13

The expression given below shows the variation of velocity (v) with time (t),

$$v = At^2 + \frac{Bt}{C+t}.$$

The dimension of ABC is :

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Options:

A.

$$[M^0L^1T^{-2}]$$

B.

$$[M^0L^2T^{-3}]$$

C.

$$[M^0L^2T^{-2}]$$

D.

$$[M^0L^1T^{-3}]$$

Answer: B

Solution:

$$[LT^{-1}] = [A] [T^2] = \frac{[B][T]}{[C] + [T]}$$

$$[C] = [T]$$

$$[A] = [LT^{-3}]$$

$$[B] = [LT^{-1}]$$

$$[ABC] = [L^2 T^{-3}]$$

Question14

The pair of physical quantities not having the same dimensions is :

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Options:

A.

Angular momentum and Planck's constant

B.

Torque and energy

C.

Surface tension and impulse

D.

Pressure and Young's modulus

Answer: C

Solution:

$$[\tau] = [E]$$

$$[\sigma] \neq [I]$$

$$[L] = [h]$$

$$[P] = [Y]$$

Question15

Match List - I with List - II.

List - I	List - II
(A) Young's Modulus	(I) $M L^{-1} T^{-1}$
(B) Torque	(II) $M L^{-1} T^{-2}$
(C) Coefficient of Viscosity	(III) $M^{-1} L^3 T^{-2}$

(D) Gravitational Constant	(IV) $M L^2 T^{-2}$
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Choose the correct answer from the options given below :

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Options:

A. (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

B. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

C. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Answer: D

Solution:

We know, Young's modulus, $E = \frac{\sigma}{\epsilon}$

$$\Rightarrow E = \frac{\frac{F}{A}}{\frac{\Delta L}{L}} = \frac{F}{A} \left(\frac{L}{\Delta L} \right)$$

$$\Rightarrow [E] = \frac{[F]}{[A]} = \frac{[M^1 L^1 T^{-2}]}{[L^2]}$$

$$(A) \Rightarrow [E] = [M^1 L^{-1} T^{-2}]$$

(B) we know, Torque = $r \times F$

$$[\tau] = [r][F]$$

$$= [L][M^1 L^1 T^{-2}]$$

$$[\tau] = [M^1 L^2 T^{-2}]$$

(C) We know, $F = \mu A \frac{u}{y}$

where, F = force, μ = coefficient of viscosity

A = area $\frac{u}{y}$ = rate of shear deformation

$$\Rightarrow \mu = \frac{Fy}{Au}$$

$$\Rightarrow [\mu] = \frac{[M^1 L^1 T^{-2}][L]}{[L^2][LT^{-1}]}$$

$$\Rightarrow [\mu] = [M^1 L^{-1} T^{-1}]$$

(D) We know, $F = \frac{Gm_1m_2}{r^2}$ (Newton's law of gravitation)

$$\Rightarrow G = \frac{Fr^2}{m_1m_2} \Leftrightarrow G = \frac{Fr^2}{m_1m_2}$$

$$\Rightarrow [G] = \frac{[M^1L^1T^{-2}][L^2]}{[M^2]}$$

$$\Rightarrow [G] = [M^{-1}L^3T^{-2}]$$

Hence, option D is correct.

Question 16

Match List I with List II.

	List - I		List - II
(A)	Coefficient of viscosity	(I)	$[ML^0 T^{-3}]$
(B)	Intensity of wave	(II)	$[ML^{-2} T^{-2}]$
(C)	Pressure gradient	(III)	$[M^{-1}LT^2]$
(D)	Compressibility	(IV)	$[ML^{-1} T^{-1}]$

Choose the correct answer from the options given below:

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Options:

A. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

B. (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

D. (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Answer: C

Solution:

Here are the dimensional formulas:

Coefficient of viscosity, μ

$$[\mu] = \frac{\text{pressure} \times \text{time}}{[L]} = \frac{[M L^{-1} T^{-2}] [T]}{[L]} = [M L^{-1} T^{-1}] \longrightarrow \text{(IV)}$$

Intensity of a wave, I

$$[I] = \frac{\text{power}}{\text{area}} = \frac{[M L^2 T^{-3}]}{[L^2]} = [M L^0 T^{-3}] \longrightarrow \text{(I)}$$

Pressure gradient, dP/dx

$$\left[\frac{dP}{dx} \right] = \frac{[M L^{-1} T^{-2}]}{[L]} = [M L^{-2} T^{-2}] \longrightarrow \text{(II)}$$

Compressibility, κ

$$[\kappa] = \frac{1}{[\text{pressure}]} = [M^{-1} L T^2] \longrightarrow \text{(III)}$$

Matching gives

(A)–(IV), (B)–(I), (C)–(II), (D)–(III), i.e. **Option C**.

Question17

Given a charge q , current I and permeability of vacuum μ_0 . Which of the following quantity has the dimension of momentum ?

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Options:

A. qI/μ_0

B. $q^2\mu_0 I$

C. $q\mu_0/I$

D. $q\mu_0 I$

Answer: D

Solution:

$$Q = AT$$

$$I = A$$

$$\mu_0 = MLT^{-2} A^{-2}$$

$$P = Q^x \mu_0^y I^z = [AT^x [MLT^{-2} A^{-2}]^y [A]^z]$$

$$MLT^{-1} = M^y L^y T^{x-2y} A^{-2y+z+x}$$

$$\text{Now; } y = 1$$

$$x - 2y = -1$$

$$-2y + z = 0$$

$$\therefore x = y = z = 1$$

Question18

If μ_0 and ϵ_0 are the permeability and permittivity of free space, respectively, then the dimension of $\left(\frac{1}{\mu_0 \epsilon_0}\right)$ is :

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Options:

A. T^2/L

B. L^2/T^2

C. T^2/L^2

D. L/T^2

Answer: B

Solution:

The expression $\frac{1}{\mu_0 \epsilon_0}$ is related to the speed of light c , given by the equation:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Therefore, we have:

$$\frac{1}{\mu_0 \epsilon_0} = c^2$$

The speed of light c has the dimensions of $L T^{-1}$, where L is the dimension of length and T is the dimension of time. Thus, when squared, the dimensions become:

$$c^2 = (L T^{-1})^2 = L^2 T^{-2}$$

Hence, the dimensions of $\frac{1}{\mu_0\epsilon_0}$ are L^2T^{-2} .

Question19

A person measures mass of 3 different particles as 435.42 g, 226.3 g and 0.125 g . According to the rules for arithmetic operations with significant figures, the addition of the masses of 3 particles will be.

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Options:

- A. 661.845 g
- B. 661.84 g
- C. 662 g
- D. 661.8 g

Answer: D

Solution:

To calculate the total mass of the three particles, we perform the following addition:

$$m_1 + m_2 + m_3 = 435.42 \text{ g} + 226.3 \text{ g} + 0.125 \text{ g}$$

When adding numbers, the result should be reported with the same number of decimal places as the measurement with the fewest decimal places. Here, the second measurement (226.3 g) has only one decimal place, which is the fewest among the given values.

Thus, according to the rules for significant figures in addition, the sum should be rounded to one decimal place:

$$\text{Total mass} = 661.8 \text{ g}$$

Question20

Match the LIST-I with LIST-II Match the LIST-I with LIST-II

LIST-I		LIST-II	
A.	Boltzmann constant Boltzmann constant	I	ML ² T ⁻¹
B	Coefficient of viscosity Coefficient of viscosity	II	MLT ⁻³ K ⁻¹
C	Planck's constant Planck's constant	III	ML ² T ⁻² K ⁻¹
D	Thermal conductivity Thermal conductivity	IV	ML ⁻¹ T ⁻¹

Choose the correct answer from the options given below:

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Options:

A. A - III, B - IV, C - I, D - II

B. A - III, B - IV, C - II, D - I

C. A - III, B - II, C - I, D - IV

D. A - II, B - III, C - IV, D - I

Answer: A

Solution:

Explanation of Dimensional Analysis

In order to match the quantities from LIST-I with their respective dimensions in LIST-II, we need to analyze their dimensional formulas:

(A) Boltzmann Constant [k]:

The equation relating pressure (P), volume (V), number of moles (N), and temperature (T) gives us:

$$[k] = \frac{PV}{NT} = \frac{ML^2 T^{-2}}{K} = ML^2 T^{-2} K^{-1}$$

This corresponds to option III.

(B) Coefficient of Viscosity [η]:

The coefficient of viscosity can be defined using the relation:

$$[\eta] = \frac{F}{6\pi rV} = \frac{MLT^{-2}}{L^2 T^{-1}} = ML^{-1} T^{-1}$$

This corresponds to option IV.

(C) Planck's Constant [h]:

The relationship between energy (E) and frequency (f) gives the dimensional formula:

$$[h] = \frac{E}{f} = \frac{ML^2 T^{-2}}{T^{-1}} = ML^2 T^{-1}$$

This corresponds to option I.

(D) Thermal Conductivity [k]:

From the heat conduction equation, we get:

$$k = \frac{(ML^2 T^{-3})L}{L^2 \cdot K} = MLT^{-3} K^{-1}$$

This corresponds to option II.

To conclude, the correct matches are:

A - III

B - IV

C - I

D - II

This analysis ensures each physical quantity is aligned with its correct dimensional representation.

Question21

In an electromagnetic system, the quantity representing the ratio of electric flux and magnetic flux has dimension of $M^P L^Q T^R A^S$, where value of ' Q ' and ' R ' are

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Options:

A. $(3, -5)$

B. $(-2, 1)$

C. $(-2, 2)$

D. $(1, -1)$

Answer: D

Solution:

To solve this problem, we need to find the dimensions of the electric flux and the magnetic flux, and then take their ratio.

Determine the dimension of electric flux (Φ_e):

Electric flux is defined as

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{A}.$$

The electric field $\mathbf{E}$ has dimensions:

$$[E] = \frac{\text{Force}}{\text{Charge}}.$$

Since force has dimensions MLT^{-2} , and charge (in SI) has dimensions AT , we get:

$$[E] = \frac{MLT^{-2}}{AT} = MLT^{-3}A^{-1}.$$

The area element dA has dimensions L^2 .

Thus, the electric flux has dimensions:

$$[\Phi_E] = [E][A] = MLT^{-3}A^{-1} \times L^2 = ML^3T^{-3}A^{-1}.$$

Determine the dimension of magnetic flux (Φ_B):

Magnetic flux is defined as

$$\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A}.$$

The magnetic field $\mathbf{B}$ has dimensions determined via the Lorentz force law (using $F = qvB$), giving:

$$[B] = \frac{F}{qv} = \frac{MLT^{-2}}{(AT)(LT^{-1})} = MT^{-2}A^{-1}.$$

Again, the area element has dimensions L^2 .

So the magnetic flux has dimensions:

$$[\Phi_B] = [B][A] = (MT^{-2}A^{-1})(L^2) = ML^2T^{-2}A^{-1}.$$

Find the ratio of electric flux to magnetic flux:

Taking the ratio:

$$\frac{\Phi_E}{\Phi_B} = \frac{ML^3T^{-3}A^{-1}}{ML^2T^{-2}A^{-1}}.$$

Canceling like terms:

Mass (M) cancels.

Current (A^{-1}) cancels.

For length, $L^{3-2} = L^1$.

For time, $T^{-3-(-2)} = T^{-1}$.

Therefore, the ratio has dimensions:

$$L^1T^{-1}.$$

This means in the expression

$$M^P L^Q T^R A^S,$$

the exponents for length and time are $Q = 1$ and $R = -1$.

Looking at the options provided, this corresponds to Option D: $(1, -1)$.

Thus, the answer is Option D.

Question22

For the determination of refractive index of glass slab, a travelling microscope is used whose main scale contains 300 equal divisions equals to 15 cm . The vernier scale attached to the microscope has 25 divisions equals to 24 divisions of main scale. The least count (LC) of the travelling microscope is (in cm) :

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Options:

A. 0.002

B. 0.0025

C. 0.0005

D. 0.001

Answer: A

Solution:

Determine one main scale division (msd):

$$300 \text{ msd} = 15 \text{ cm}$$

Therefore,

$$1 \text{ msd} = \frac{15}{300} \text{ cm} = 0.05 \text{ cm}$$

Determine one vernier scale division (vsd):

$$25 \text{ vsd} = 24 \text{ msd}$$

Therefore,

$$1 \text{ vsd} = \frac{24}{25} \text{ msd}$$

Calculate the least count (LC):

The least count is given by the difference between one main scale division and one vernier scale division:

$$\text{LC} = 1 \text{ msd} - 1 \text{ vsd}$$

Substitute the expression for 1 vsd:

$$\text{LC} = 1 \text{ msd} - \frac{24}{25} \times 1 \text{ msd}$$

This simplifies to:

$$\text{LC} = \frac{1}{25} \times 1 \text{ msd}$$

Calculate the LC in cm:

Substituting the value of 1 msd:

$$\text{LC} = \frac{1}{25} \times 0.05 \text{ cm} = 0.002 \text{ cm}$$

Hence, the least count of the traveling microscope is 0.002 cm.

Question23

In an electromagnetic system, a quantity defined as the ratio of electric dipole moment and magnetic dipole moment has dimension of $[M^P L^Q T^R A^S]$. The value of P and Q are :

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Options:

A. $-1, 0$

B. $0, -1$

C. $-1, 1$

D. $1, -1$

Answer: B

Solution:

Electric dipole moment ($\vec{P}$) = $q \times 2\ell$

Magnetic dipole moment ($\vec{M}$) = IA

$$\left[\frac{P}{M} \right] = \left[\frac{LTA}{L^2 A} \right] = L^{-1} T = M^0 L^{-1} T^1 A^0$$

After comparing values of P&Q are 0, -1

Correct Answer : Option 4

Question24

Match List - I with List - II.

List - I	List - II
(A) Mass density	(I) $[ML^2T^{-3}]$
(B) Impulse	(II) $[MLT^{-1}]$
(C) Power	(III) $[ML^2T^0]$
(D) Moment of inertia	(IV) $[ML^{-3}T^0]$

Choose the correct answer from the options given below :

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Options:

A.

(A)-(IV), (B)-(II), (C)-(III), (D)-(I)

B.

(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C.

(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

D.

(A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Answer: D

Solution:

(A) Mass density:

Mass density is defined as mass per unit volume.

The dimension of mass is $[M]$.

The dimension of volume is $[L^3]$.

Therefore, the dimension of mass density is $\frac{[M]}{[L^3]} = [ML^{-3}T^0]$.

Matches with (IV).

(B) Impulse:

Impulse is defined as the change in momentum, or force multiplied by time.

Impulse = Force x Time

The dimension of force is $[MLT^{-2}]$.

The dimension of time is $[T]$.

Therefore, the dimension of impulse is $[MLT^{-2}] \cdot [T] = [MLT^{-1}]$.

Matches with (II).

(C) Power:

Power is defined as the rate of doing work or energy per unit time.

Power = Work / Time

The dimension of work (or energy) is $[ML^2T^{-2}]$.

The dimension of time is $[T]$.

Therefore, the dimension of power is $\frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$.

Matches with (I).

(D) Moment of inertia:

Moment of inertia is defined as $I = mr^2$, where m is mass and r is the distance from the axis of rotation.

The dimension of mass is $[M]$.

The dimension of distance squared is $[L^2]$.

Therefore, the dimension of moment of inertia is $[ML^2T^0]$.

Matches with (III).

So, the correct matches are:

(A) - (IV)

(B) - (II)

(C) - (I)

(D) - (III)

Therefore, the correct option is **D**.

Question 25

The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equal to that of:

(μ_0 = Vacuum permeability and ϵ_0 = Vacuum permittivity)

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Options:

A.

Voltage

B.

Inductance

C.

Resistance

D.

Capacitance

Answer: C

Solution:

The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ can be understood as follows:

Vacuum permeability (μ_0) and vacuum permittivity (ϵ_0) relate to the properties of inductance and capacitance, respectively, in a vacuum. We start by considering the formulas for inductance (L) and capacitance (C):

Inductance (L) is given by:

$$L = \frac{\mu_0 \cdot \text{Number of turns}^2 \cdot \text{Area}}{\text{Length}}$$

Capacitance (C) is given by:

$$C = \frac{\text{Area} \cdot \epsilon_0}{\text{Distance}}$$

From these, the ratio $\frac{L}{C}$ can be expressed as $\frac{\mu_0}{\epsilon_0}$:

$$\frac{L}{C} \propto \frac{\mu_0}{\epsilon_0}$$

Taking the square root of this ratio, we have:

$$\sqrt{\frac{\mu_0}{\epsilon_0}} \propto \sqrt{\frac{L}{C}}$$

This simplifies further to considering the relationship between time constant (τ) and resistance (R):

$$\frac{L}{C} = \frac{\tau R}{(\tau/R)} = R^2$$

Thus, by taking the square root:

$$\sqrt{\frac{\mu_0}{\epsilon_0}} = R$$

In conclusion, the dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equivalent to the dimension of resistance, R .

Question 26

A quantity Q is formulated as $X^{-2}Y^{+\frac{3}{2}}Z^{-\frac{2}{5}}$. X , Y , and Z are independent parameters which have fractional errors of 0.1, 0.2, and 0.5, respectively in measurement. The maximum fractional error of Q is

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Options:

A.

0.6

B.

0.8

C.

0.7

D.

0.1

Answer: C

Solution:

The quantity Q is expressed as $Q = X^{-2}Y^{+\frac{3}{2}}Z^{-\frac{2}{5}}$. The variables X , Y , and Z are independent parameters with fractional errors of 0.1, 0.2, and 0.5, respectively. To determine the maximum fractional error in Q , we can use the following method:

$$\text{Fractional error in } Q = |-2| \frac{\Delta X}{X} + \left| \frac{3}{2} \right| \frac{\Delta Y}{Y} + \left| -\frac{2}{5} \right| \frac{\Delta Z}{Z}$$

Substituting the fractional errors into the equation:

$$= 2 \times 0.1 + \frac{3}{2} \times 0.2 + \frac{2}{5} \times 0.5$$

Calculating each term gives:

$$= 0.2 + 0.3 + 0.2$$

Summing these values results in:

$$= 0.7$$

Thus, the maximum fractional error in Q is 0.7.

Question27

Identify the physical quantity that cannot be measured using spherometer :

[27-Jan-2024 Shift 1]

Options:

A.

Radius of curvature of concave surface

B.

Specific rotation of liquids

C.

Thickness of thin plates

D.

Radius of curvature of convex surface

Answer: B

Solution:

Spherometer can be used to measure curvature of surface.

Question28

Given below are two statements: one is labelled as Assertion(A) and the other is labelled as Reason (R).

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below :

[27-Jan-2024 Shift 2]

Options:

A.

Both (A) and (R) are correct and (R) is the correct explanation of (A)

B.

Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C.

(A) is true but (R) is false

D.

(A) is false but (R) is true

Answer: B

Solution:

Assertion & Reason both are correct Theory

Question29

If 50 Vernier divisions are equal to 49 main scale divisions of a travelling microscope and one smallest reading of main scale is 0.5 mm, the Vernier constant of travelling microscope is:

[30-Jan-2024 Shift 2]

Options:

A.

0.1 mm

B.

0.1 cm

C.

0.01 cm

D.

0.01 mm

Answer: D

Solution:

$$50V + S = 49S + S$$

$$S = 50(S - V)$$

$$.5 = 50(S - V)$$

$$S - V = \frac{0.5}{50} = \frac{1}{100} = 0.01 \text{ mm}$$

Question30

The measured value of the length of a simple pendulum is 20cm with 2mm accuracy. The time for 50 oscillations was measured to be 40 seconds with 1 second resolution. From these measurements, the accuracy in the measurement of acceleration due to gravity is N%. The value of N is:

[31-Jan-2024 Shift 2]

Options:

A.

4

B.

8

C.

6

D.

5

Answer: C

Solution:

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2 \Delta T}{T}$$

$$= \frac{0.2}{20} + 2 \left(\frac{1}{40} \right)$$

$$= \frac{0.3}{20}$$

$$\text{Percentage change} = \frac{0.3}{20} \times 100 = 6\%$$

Question31

The resistance $R = V/I$ where $V = (200 \pm 5)V$ and $I = (20 \pm 0.2)A$, the percentage error in the measurement of R is :

[29-Jan-2024 Shift 1]

Options:

A.

3.5%

B.

7%

C.

3%

D.

5.5%

Answer: A

Solution:

$$R = \frac{V}{I}$$

According to error analysis

$$\frac{dR}{R} = \frac{dV}{V} + \frac{dI}{I}$$

$$\frac{dR}{R} = \frac{5}{200} + \frac{0.2}{20}$$

$$\frac{dR}{R} = \frac{7}{200}$$

$$\% \text{ error } \frac{dR}{R} \times 100 = \frac{7}{200} \times 100 = 3.5\%$$

Question32

A physical quantity Q is found to depend on quantities a, b, c by the relation $Q = a^4 b^3 / c^2$. The percentage error in a, b and c are 3%, 4% and 5% respectively. Then, the percentage error in Q is :

[29-Jan-2024 Shift 2]

Options:

A.

66%

B.

43%

C.

34%

D.

14%

Answer: C

Solution:

$$\text{Sol. } Q = \frac{a^4 b^3}{c^2}$$

$$\frac{\Delta Q}{Q} = 4 \frac{\Delta a}{a} + 3 \frac{\Delta b}{b} + 2 \frac{\Delta c}{c}$$

$$\frac{\Delta Q}{Q} \times 100 = 4 \left(\frac{\Delta a}{a} \times 100 \right) + 3 \left(\frac{\Delta b}{b} \times 100 \right) + 2 \left(\frac{\Delta c}{c} \times 100 \right)$$

$$\% \text{ error in } Q = 4 \times 3\% + 3 \times 4\% + 2 \times 5\%$$

$$= 12\% + 12\% + 10\%$$

$$= 34\%$$

Question33

A vector has magnitude same as that of $\vec{A} = 3\hat{j} + 4\hat{j}$ and is parallel to $\vec{B} = 4\hat{i} + 3\hat{j}$.

The x and y components of this vector in first quadrant are x and 3 respectively where x = _____

[30-Jan-2024 Shift 2]

Answer: 4

Solution:

$$\vec{N} = |\vec{A}| \hat{B} = \frac{5(4\hat{i} + 3\hat{j})}{5} = 4\hat{i} + 3\hat{j}$$

$$\therefore x = 4$$

Question34

If the percentage errors in measuring the length and the diameter of a wire are 0.1% each. The percentage error in measuring its resistance will be:

[31-Jan-2024 Shift 1]

Options:

A.

0.2%

B.

0.3%

C.

0.1%

D.

0.144%

Answer: B

Solution:

$$R = \frac{\rho L}{\pi \frac{d^2}{4}}$$

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + \frac{2 \Delta d}{d}$$

$$\frac{\Delta L}{L} = 0.1\% \text{ and } \frac{\Delta d}{d} = 0.1\%$$

$$\frac{\Delta R}{R} = 0.3\%$$

Question35

If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ , then

[31-Jan-2024 Shift 2]

Options:

A.

$$|\vec{A} - \vec{B}| = \sqrt{2}R \sin\left(\frac{\theta}{2}\right)$$

B.

$$|\vec{A} + \vec{B}| = 2R \sin\left(\frac{\theta}{2}\right)$$

C.

$$|\vec{A} + \vec{B}| = 2R \cos\left(\frac{\theta}{2}\right)$$

D.

$$|\vec{A} - \vec{B}| = 2R \cos\left(\frac{\theta}{2}\right)$$

Answer: C

Solution:

The magnitude of resultant vector

$$R' = \sqrt{a^2 + b^2 + 2ab \cos \theta}$$

Here $a = b = R$

$$\text{Then } R' = \sqrt{R^2 + R^2 + 2R^2 \cos \theta}$$

$$= R\sqrt{2}\sqrt{1 + \cos \theta}$$

$$= \sqrt{2}R \sqrt{2\cos^2 \frac{\theta}{2}}$$

$$= 2R \cos \frac{\theta}{2}$$

Question36

Given below are two statements :

Statement (I) : Planck's constant and angular momentum have same dimensions.

Statement (II) : Linear momentum and moment of force have same dimensions.

In the light of the above statements, choose the correct answer from the options given below :

[27-Jan-2024 Shift 1]

Options:

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are false

C.

Both Statement I and Statement II are true

D.

Statement I is false but Statement II is true

Answer: A

Solution:

$$[h] = ML^2T^{-1}$$

$$[L] = ML^2T^{-1}$$

$$[P] = MLT^{-1}$$

$$[\tau] = ML^2T^{-2}$$

(Here h is Planck's constant, L is angular momentum, P is linear momentum and τ is moment of force)

Question37

The equation of state of a real gas is given by $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, where P, V and T are pressure, volume and temperature respectively and R is the universal gas constant. The dimensions of a/b^2 is similar to that of :

[27-Jan-2024 Shift 2]

Options:

A.

PV

B.

P

C.

RT

D.

R

Answer: B

Solution:

$$[P] = \left[\frac{a}{V^2} \right] \Rightarrow [a] = [PV^2]$$

$$\text{And } [V] = [b]$$

$$\frac{[a]}{[b^2]} = \frac{[PV^2]}{[V^2]} = [P]$$

Question38

Match List-I with List-II.

	List-I		List-II
A.	Coefficient of viscosity	I.	$\{[ML^2T^{-2}]\}$
B.	Surface Tension	II.	$\{[ML^2T^{-1}]\}$
C.	Angular momentum	III.	$\{[ML^{-1}T^{-1}]\}$
D.	Rotational kinetic energy	IV.	$\{[ML^0T^{-2}]\}$

[30-Jan-2024 Shift 1]

Options:

A.

A-II, B-I, C-IV, D-III

B.

A-I, B-II, C-III, D-IV

C.

A-III, B-IV, C-II, D-I

D.

A-IV B-III C-II. D-I

Answer: C

Solution:

$$F = \eta A \frac{dv}{dy}$$

$$[MLT^{-2}] = \eta [L^2][T^{-1}]$$

$$\eta = [ML^{-1}T^{-1}]$$

$$S.T = \frac{F}{\ell} = \frac{[MLT^{-2}]}{[L]} = [ML^0T^{-2}]$$

$$L = mvr = [ML^2T^{-1}]$$

$$K.E = \frac{1}{2}I\omega^2 = [ML^2T^{-2}]$$

Question39

If mass is written as $m = kc^P G^{-1/2} h^{1/2}$ then the value of P will be :
(Constants have their usual meaning with k a dimensionless constant)

[30-Jan-2024 Shift 2]

Options:

A.

$1/2$

B.

$1/3$

C.

2

D.

$1/4$

Answer: A

Solution:

$$m = kc^P G^{-1/2} h^{1/2}$$

$$M^1 L^0 T^0 = [L T^{-1}]^P [M^{-1} L^3 T^{-2}]^{-1/2} [M L^2 T^{-1}]^{1/2}$$

By comparing $P = 1/2$

Question40

A force is represented by $F = ax^2 + bt^{1/2}$

Where x = distance and t = time. The dimensions of b^2/a are :

[31-Jan-2024 Shift 1]

Options:

A.

$$[ML^3T^{-3}]$$

B.

$$[MLT^{-2}]$$

C.

$$[ML^{-1}T^{-1}]$$

D.

$$[ML^2T^{-3}]$$

Answer: A

Solution:

$$F = ax^2 + bt^{1/2}$$

$$[a] = \frac{[F]}{[x^2]} = [M^1L^{-1}T^{-2}]$$

$$[b] = \frac{[F]}{[t^{1/2}]} = [M^1L^1T^{-5/2}]$$

$$\left[\frac{b^2}{a} \right] = \frac{[M^2L^2T^{-5}]}{[M^1L^{-1}T^{-2}]} = [M^1L^3T^{-3}]$$

Question41

Consider two physical quantities A and B related to each other as $E = \frac{B - x^2}{At}$ where E, x and t have dimensions of energy,

[31-Jan-2024 Shift 2]

Options:

A.

$$L^{-2}M^1T^0$$

B.

$$L^2M^{-1}T^1$$

C.

$$L^{-2}M^{-1}T^1$$

D.

$$L^0M^{-1}T^1$$

Answer: B

Solution:

$$[B] = L^2$$

$$A = \frac{x^2}{tE} = \frac{L^2}{TML^2T^{-2}} = \frac{1}{MT^{-1}}$$

$$[A] = M^{-1}T$$

$$[AB] = [L^2M^{-1}T^1]$$

Question42

The dimensional formula of angular impulse is :

[1-Feb-2024 Shift 1]

Options:

A.

$$[ML^{-2}T^{-1}]$$

B.

$$[ML^2T^{-2}]$$

C.

$$[MLT^{-1}]$$

D.

$$[ML^2T^{-1}]$$

Answer: D

Solution:

Angular impulse = change in angular momentum.

$$\begin{aligned} [\text{Angular impulse}] &= [\text{Angular momentum}] = [mvr] \\ &= [ML^2T^{-1}] \end{aligned}$$

Question43

10 divisions on the main scale of a Vernier calliper coincide with 11 divisions on the Vernier scale. If each division on the main scale is of 5 units, the least count of the instrument is :

[1-Feb-2024 Shift 1]

Options:

A.

$$1/2$$

B.

$$10/11$$

C.

$$50/11$$

D.

5/11

Answer: D

Solution:

$$10 \text{ MSD} = 11 \text{ VSD}$$

$$1 \text{ VSD} = \frac{10}{11} \text{ MSD}$$

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \frac{10}{11} \text{ MSD}$$

$$= \frac{1 \text{ MSD}}{11}$$

$$= \frac{5}{11} \text{ units}$$

Question44

The radius (r), length (l) and resistance (R) of a metal wire was measured in the laboratory as

$$r = (0.35 \pm 0.05) \text{ cm}$$

$$R = (100 \pm 10) \text{ ohm}$$

$$l = (15 \pm 0.2) \text{ cm}$$

The percentage error in resistivity of the material of the wire is :

[1-Feb-2024 Shift 1]

Options:

A.

25.6%

B.

39.9%

C.

37.3%

D.

35.6%

Answer: B

Solution:

$$\rho = R \frac{\rho}{\ell}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta R}{R} + 2 \frac{\Delta r}{r} + \frac{\Delta \ell}{\ell}$$

$$= \frac{10}{100} + 2 \times \frac{0.05}{0.35} + \frac{0.2}{15}$$

$$= \frac{1}{10} + \frac{2}{7} + \frac{1}{75}$$

$$\frac{\Delta \rho}{\rho} = 39.9\%$$

Question45

Choose the correct answer from the options given below :

Match List - I with List - II.

List - I (Number)	List - II (Significant figure)
(A) 1001	(I) 3
(B) 010.1	(II) 4
(C) 100.100	(III) 5
(D) 0.0010010	(IV) 6

Options:

A.

(A)-(III), (B)-(IV), (C)-(II), (D)-(I)

B.

(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

C.

(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D.

(A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Answer: C

Solution:

Theoretical

Question46

Match List I with List II

	LIST I		LIST II
A.	Planck's constant (h)	I.	$[M^1L^2T^{-2}]$
B.	Stopping potential (Vs)	II.	$[M^1L^1T^{-1}]$
C.	Work function (ϕ)	III.	$[M^1L^2T^{-1}]$
D.	Momentum (p)	IV.	$[M^1L^2T^{-3}A^{-1}]$

[24-Jan-2023 Shift 1]

Options:

A. A-III, B-I, C-II, D-IV

B. A-III, B-IV, C-I, D-II

C. A-II, B-IV, C-III, D-I

D. A-I, B-III, C-IV, D-II

Answer: B

Solution:

Solution:

(A) Planck's constant

$$h\nu = E$$

$$h = \frac{E}{\nu} = \frac{M^1 L^2 T^{-2}}{T^{-1}} = M^1 L^2 T^{-1}$$

(B) $E = qV$

$$V = \frac{E}{q} = \frac{M^1 L^2 T^{-2}}{A^1 T^1} = M^1 L^2 T^{-3} A^{-1} (IV)$$

(C) ϕ (work function) = energy

$$= M^1 L^2 T^{-2}$$

(D) Momentum (p) = F.t

$$= M^1 L^1 T^{-2} T^1$$

$$= M^1 L^1 T^{-1}$$

Question47

The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as : $\nu = r^a \rho^b s^c$. The values of a, b and c respectively are

[24-Jan-2023 Shift 2]

Options:

A. $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

B. $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

C. $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$

D. $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Answer: A

Solution:

Solution:

$$[T^{-1}] = [L^1]^a [M^1 L^{-3}]^b \left[\frac{MLT^{-2}}{L} \right]^c$$

$$\Rightarrow T^{-1} = M^{b+c} \cdot L^{a-3b} \cdot T^{-2c}$$

$$c = \frac{1}{2}, b = -\frac{1}{2}, a - 3b = 0$$

$$a + \frac{3}{2} = 0 \Rightarrow a = -\frac{3}{2}$$

Question48

Match List I with List II

List - I	List - II
A. Surface tension	I. $\text{Kg m}^{-1} \text{s}^{-1}$
B. Pressure	II. Kg ms^{-1}
C. Viscosity	III. $\text{Kg m}^{-1} \text{s}^{-2}$
D. Impulse	IV. Kg s^{-2}

**Choose the correct answer from the options given below :
[25-Jan-2023 Shift 1]**

Options:

A. A-IV, B-III, C- II, D- I

B. A-IV, B-III, C-I, D-II

C. A-III, B-IV, C-I, D-II

D. A-II, B-I, C-III, D-IV

Answer: B

Solution:

Solution:

$$(A) \text{ Surface Tension} = \frac{F}{\ell} = \frac{MLT^{-2}}{L} = ML^{-1}T^{-2}$$

$$= Kgs^{-2} \text{ (IV)}$$

$$(B) \text{ Pressure} = \frac{F}{A} = \frac{MLT^{-2}}{L^2}$$

$$= kg\,m^{-1}\,s^{-2} \text{ (III)}$$

$$(C) \text{ Viscosity} = \frac{F}{A \left(\frac{dV}{dz} \right)} = \frac{MLT^{-2}}{L^2 \left(\frac{LT^{-1}}{L} \right)}$$

$$= ML^{-1}T^{-1} = kg\,m^{-1}\,s^{-1} \text{ (I)}$$

$$(D) \text{ Impulse} = \int Fdt = MLT^{-2} \times T$$

$$= MLT^{-1} = Kgms^{-1} \text{ (II)}$$

So A-(IV), B-(III), C-(I), D- (II)

Question49

Match List I with List II

List	List II
A. Young's Modulus (Y)	I. $[ML^{-1}T^{-1}]$
B. Co-efficient of Viscosity (η)	II. $[ML^2T^{-1}]$
C. Planck's Constant (h)	III. $[ML^{-1}T^{-2}]$
D. Work Function (ϕ)	IV. $[ML^2T^{-2}]$

Choose the correct answer from the options given below:
[25-Jan-2023 Shift 2]

Options:

A. A-II, B-III, C-IV, D-I

B. A-III, B-I, C-II, D-IV

C. A-I, B-III, C-IV, D-II

D. A-I, B-II, C-III, D-IV

Answer: B

Solution:

Solution:

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta \ell / \ell} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$$

$$F = 6\pi\eta rv \Rightarrow \eta = \frac{F}{6\pi rv}$$

$$[\eta] = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1}T^{-1}]$$

$$E = hv \Rightarrow h = \frac{E}{v} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Work function has same dimension as that of energy, so $[\phi] = [ML^2T^{-2}]$

Question 50

Match List I with List II :

List-I (Physical Quantity)	List-II (Dimensional Formula)
A Pressure gradient	I $[M^0L^2T^{-2}]$
B Energy density	II $[M^1L^{-1}T^{-2}]$
C Electric Field	III $[M^1L^{-2}T^{-2}]$
D Latent heat	IV $[M^1L^1T^{-3}A^{-1}]$

Choose the correct answer from the options given below:
[29-Jan-2023 Shift 1]

Options:

A. A-III, B-II, C-I, D-IV

B. A-II, B-III, C-IV, D-I

C. A-III, B-II, C-IV, D-I

D. A-II, B-III, C-I, D-IV

Answer: C

Solution:

Solution:

$$\begin{aligned}\text{Pressure gradient} &= \frac{dp}{dx} = \frac{[ML^{-1}T^{-2}]}{[L]} \\ &= [M^1L^{-2}T^{-2}] \\ \text{Energy density} &= \frac{\text{energy}}{\text{volume}} = \frac{[ML^2T^{-2}]}{[L^3]} \\ &= [M^1L^{-1}T^{-2}] \\ \text{Electric field} &= \frac{\text{Force}}{\text{charge } e} = \frac{[MLT^{-2}]}{[A \cdot T]} \\ &= [M^1L^1T^{-3}A^{-1}] \\ \text{Latent heat} &= \frac{\text{heat}}{\text{mass}} = \frac{[ML^2T^{-2}]}{[M]} \\ &= [M^0L^2T^{-2}]\end{aligned}$$

Question51

The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than $(0, 0)$, then find out the correct dimensions of A and B in a new equation:

$$(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2.$$

The dimensions of t is given as $[T^{-1}]$.

[29-Jan-2023 Shift 2]

Options:

A. $A = [L^{-1}T]$, $B = [LT^{-1}]$

B. $A = [LT]$, $B = [L^{-1}T^{-1}]$

C. $A = [L^{-1}T^{-1}]$, $B = [LT^{-1}]$

D. $A = [L^{-1}T^{-1}]$, $B = [LT]$

Answer: B

Solution:

Solution:

$$(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$$

$$[At] = A \times \frac{1}{T} = L$$

$$\therefore [A] = T^1 L^1$$

$\frac{t}{B}$ is in meters

$$\therefore \frac{1}{T[B]} = L$$

$$\therefore [B] = T^{-1} L^{-1}$$

$\therefore$ Correct Ans. (2)

Question 52

Electric field in a certain region is given by $\vec{E} = \left(\frac{A}{x^2} \hat{i} + \frac{B}{y^3} \hat{j} \right)$. The SI

unit of A and B are :

[30-Jan-2023 Shift 1]

Options:

A. Nm^3C^{-1} ; Nm^2C^{-1}

B. Nm^2C^{-1} ; Nm^3C^{-1}

C. Nm^3C ; Nm^2C

D. Nm^2C ; Nm^3C

Answer: B

Solution:

Solution:

$$\vec{E} = \frac{A}{x^2} \hat{i} + \frac{B}{y^3} \hat{j}$$

$$\left[\frac{A}{x^2} \right] = \text{NC}^{-1} \Rightarrow [A] = \text{Nm}^2\text{C}^{-1}$$

$$\left[\frac{B}{y^3} \right] = \text{NC}^{-1} \Rightarrow [B] = \text{Nm}^3\text{C}^{-1}$$

Question53

Match List I with List II.

List I	List II
A. Torque	I. $\text{kg m}^{-1} \text{s}^{-2}$
B. Energy density	II. kg ms^{-1}
C. Pressure gradient	III. $\text{kg m}^{-2} \text{s}^{-2}$
D. Impulse	IV. $\text{kg m}^2 \text{s}^{-2}$

Choose the correct answer from the options given below :
[30-Jan-2023 Shift 2]

Options:

- A. A-IV, B-III, C-I, D-II
- B. A-I, B-IV, C-III, D-II
- C. A-IV, B-I, C-II, D-III
- D. A-IV, B-I, C-III, D-II

Answer: D

Solution:

Solution:

Question54

If R , X_L and X_C represent resistance, inductive reactance and capacitive reactance. Then which of the following is dimensionless:
[31-Jan-2023 Shift 1]

Options:

A. RX_LX_C

B. $\frac{R}{\sqrt{X_LX_C}}$

C. $\frac{R}{X_LX_C}$

D. $R \frac{X_L}{X_C}$

Answer: B

Solution:

Solution:

All three have same dimension therefore $\frac{R}{\sqrt{X_LX_C}}$ is dimensionless.

Option 2

Question55

Match List-I with List-II.

List-I	List-II
A. Angular momentum	I. $[ML^2T^{-2}]$.
B. Torque	II. $[ML^{-2}T^{-2}]$
C. Stress	III. $[ML^2T^{-1}]$
D. Pressure gradient	IV. $[ML^{-1}T^{-2}]$.

Choose the correct answer from the options given below:
[31-Jan-2023 Shift 2]

Options:

A. A-I, B-IV, C-III, D-II

B. A-III, B-I, C-IV, D-II

C. A-II, B-III, C-IV, D-I

D. A-IV, B-II, C-I, D-III

Answer: B

Solution:

Solution:

$$\vec{L} = \vec{r} \times \vec{p} \Rightarrow [L] = [M^0 L^1 T^0][M^1 L^1 T^{-1}]$$

$$= [M^1 L^2 T^{-1}]$$

$$\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow [\tau] = [L^1][MLT^{-2}]$$

$$= [ML^2 T^{-2}]$$

$$\text{Stress} \equiv \text{Pressure} = \frac{F}{A} \Rightarrow [\text{Stress}] = [ML^{-1} T^{-2}]$$

$$\text{Pressure Gradient} = \frac{dP}{dx} \Rightarrow [\text{Pressure Gradient}] = [ML^{-2} T^{-2}]$$

Question 56

Vectors $a\hat{i} + b\hat{j} + k\hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is _____
[24-Jan-2023 Shift 1]

Answer: 1

Solution:

Solution:

For two perpendicular vectors

$$(a\hat{i} + b\hat{j} + k\hat{k}) \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 0$$

$$2a - 3b + 4 = 0$$

On solving, $2a - 3b = -4$

Also given

$$3a + 2b = 7$$

We get $a = 1, b = 2$

$$\frac{a}{b} = \frac{x}{2} \Rightarrow x = \frac{2a}{b} = \frac{2 \times 1}{2}$$

$$\Rightarrow x = 1$$

Question57

If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other.

Then, the value of m will be :

[24-Jan-2023 Shift 2]

Options:

A. 1

B. -1

C. -3

D. 2

Answer: D

Solution:

Solution:

$$\vec{P} \cdot \vec{Q} = 0$$

$$(\hat{i} + 2m\hat{j} + m\hat{k}) \cdot (4\hat{i} - 2\hat{j} + m\hat{k}) = 0$$

$$\Rightarrow 4 - 4m + m^2 = 0$$

$$\Rightarrow (m - 2)^2 = 0 \Rightarrow m = 2$$

Question58

If $\vec{P} = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $\vec{Q} = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$ then, The unit vector in the direction of $\vec{P} \times \vec{Q}$ is $\frac{1}{x}(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k})$. The value of x is

[25-Jan-2023 Shift 1]

Answer: 4

Solution:

Solution:

$$\begin{aligned}\vec{P} \times \vec{Q} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & \sqrt{3} & 2 \\ 4 & \sqrt{3} & 2.5 \end{vmatrix} = \sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k} \\ \Rightarrow \frac{|\vec{P} \times \vec{Q}|}{|\vec{P}| |\vec{Q}|} &= \frac{1}{2} \left(\sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k} \right) \\ &= \frac{1}{4} (\sqrt{3} \hat{i} + \hat{j} - 2\sqrt{3} \hat{k}) \quad x = 4\end{aligned}$$

Question 59

In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab as 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is

[29-Jan-2023 Shift 2]

Answer: 41

Solution:

Solution:

$$u = \frac{h}{h'} = \frac{5.25}{5.00}$$

$$\text{Least count} = \frac{1}{20} \text{ cm} - \frac{49}{50} \cdot \frac{1}{20} \text{ cm}$$

$$= \frac{1}{50} \times \frac{1}{20} \text{ cm} = 0.01 \text{ mm}$$

$$\ln u = \ln h - \ln h'$$

$$\frac{du}{u} = \frac{dh}{h} - \frac{dh'}{h'}$$

$$du = \left[\frac{0.01}{5.25} + \frac{0.01}{5.00} \right] \frac{5.25}{5.00}$$

$$= \frac{41}{10} \times 10^{-3}$$

$$\text{Ans.} = 41$$

Question60

In a metre bridge experiment the balance point is obtained if the gaps are closed by 2Ω and 3Ω . A shunt of $X \Omega$ is added to 3Ω resistor to shift the balancing point by 22.5 cm. The value of X is _____
[29-Jan-2023 Shift 1]

Answer: 2

Solution:

Solution:

$$\frac{2}{\left(\frac{3x}{3+x} \right)} = \frac{40 + 22.5}{60 - 22.5} = \frac{62.5}{37.5} = \frac{5}{3}$$

$$\frac{6}{5} = \frac{3x}{3+x}$$

$$6 + 2x = 5x \Rightarrow x = 2$$

Question61

A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω . When a resistance of 15Ω is used for shunting null point moves to 300 cm. The internal resistance of the cell is _____ Ω .

[29-Jan-2023 Shift 2]

Answer: 5

Solution:

Solution:

$$\frac{\varepsilon}{r+5} \times 5 = 200 \times \dots (1)$$

$$\frac{\varepsilon \times 15}{r+15} = 300 \times \dots (2)$$

$$\Rightarrow r = 5\Omega$$

Ans. 5

Question62

In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by 0.5 mm on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46th division the circular scale coincide with the reference line. The diameter of the wire is _____ $\times 10^{-2}$ mm.

[30-Jan-2023 Shift 1]

Answer: 220

Solution:

Solution:

$$\text{Least count} = \frac{\text{Pitch}}{\text{No. of circular divisions}}$$

$$= \frac{0.5 \text{ mm}}{100}$$

$$\text{Least count} = 5 \times 10^{-3} \text{ mm}$$

$$\text{Positive Error} = \text{MSR} + \text{CSR(LC)}$$

$$= 0 \text{ mm} + 6(5 \times 10^{-3} \text{ mm})$$

Reading of Diameter = MSR + CSR (LC) – Positive zero error

$$= 4 \times 0.5 \text{ mm} + (46(5 \times 10^{-3})) - 6(5 \times 10^{-3}) \text{ mm}$$

$$= 2 \text{ mm} + 40 \times 5 \times 10^{-3} \text{ mm} = 2.2 \text{ mm (Ans.)}$$

Question63

$\left(P + \frac{a}{V^2}\right)(V - b) = RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a, b, R are the constants. The physical quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will be :

[1-Feb-2023 Shift 1]

Options:

- A. Bulk modulus
- B. Modulus of rigidity
- C. Compressibility
- D. Energy density

Answer: C

Solution:

Solution:

$$[b] = [V]$$

$$\left[\frac{a}{b^2}\right] = [P] \quad \therefore \quad \left[\frac{b^2}{a}\right] = \frac{1}{[P]} = \frac{1}{[B]} = [K]$$

Question64

If the velocity of light c, universal gravitational constant G and planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is:

[1-Feb-2023 Shift 2]

Options:

A. $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^1 \right]$

B. $[h^1 c^1 G^{-1}]$

C. $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^{\frac{1}{2}} \right]$

D. $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^{-\frac{1}{2}} \right]$

Answer: D

Solution:

Solution:

Say dimensional formale of mass is $H^x C^y G^z$

$$M^1 = (ML^2T^{-1})^x (LT^{-1})(M^{-1}L^3T^{-2})^z$$

$$M^1 L^0 T^0 = M^{x-z} L^{2x+y+3z} T^{-x-y-2z}$$

on comparing both side

$$x - z = 1$$

$$2x + y + 3z = 0$$

$$-x - y - 2z = 0$$

On solving above equations we get

$$x = \frac{1}{2} \quad y = \frac{1}{2} \quad z = \frac{-1}{2}$$

Question65

In an experiment to find emf of a cell using potentiometer, the length of null point for a cell of emf 1.5V is found to be 60 cm. If this cell is replaced by another cell of emf E . the length-of null point increases by 40 cm. The value of E is $\frac{x}{10}$ V. The value of x is _____.

[1-Feb-2023 Shift 1]

Answer: 25

Solution:

Solution:

$$\frac{E_1}{E_2} = \frac{I_1}{I_2}$$

$$\frac{1.5}{E_2} = \frac{60}{60+40} = \frac{6}{10} = \frac{3}{5}$$

$$E_2 = \frac{5}{2} = \frac{x}{10}$$

$$x = 25$$

Question66

Dimension of $\frac{1}{\mu_0 \epsilon_0}$ should be equal to
[8-Apr-2023 shift 1]

Options:

A. T/L

B. T²/L²

C. L/T

D. L²/T²

Answer: D

Solution:

Solution:

Dimension of $\frac{1}{\mu_0 \epsilon_0}$

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \Rightarrow \frac{1}{\mu_0 \epsilon_0} = c^2$$

$$\left[\frac{1}{\mu_0 \epsilon_0} \right] = [c^2]$$

$$= \left[\frac{L^2}{T^2} \right]$$

Question67

Match List I with List II

LIST - I	LIST - II
A. Torque	I. $ML^{-2}T^{-2}$
B. Stress	II. ML^2T^{-2}
C. Pressure gradient	III. $MLL^{-1}t^{-1}$
D. Coefficient of viscosity	IV. $ML^{-1}T^{-2}$

Choose the correct answer from the options given below:
[8-Apr-2023 shift 2]

Options:

A. A-III, B-IV, C-I, D-II

B. A-II, B-I, C-IV, D-III

C. A-IV, B-II, C-III, D-I

D. A-II, B-IV, C-I, D-III

Answer: D

Solution:

Solution:

$$[\text{Torque}] = F \cdot L$$
$$MLT^{-2} \cdot L = ML^2T^{-2}$$

$$[\text{Stress}] = \frac{F}{A}$$

$$\frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

$$[\text{Pressure gradient}] = \frac{\Delta P}{\Delta L} = \frac{F}{A^2 \cdot L}$$

$$= \frac{MLT^{-2}}{L^3}$$

$$= ML^{-2}T^{-2}$$

$$F = nA \frac{dv}{dy}$$

$$\eta = ML^{-1}T^{-1}$$

Question68

Given below are two statements :

Statements I : Astronomical unit (Au), Parsec (Pc) and Light year (ly) are units for measuring astronomical distances.

Statements II : $Au < \text{Parsec (Pc)} < ly$

In the light of the above statements, choose the most appropriate answer from the options given below :

[11-Apr-2023 shift 1]

Options:

- A. Both Statements I and Statements II are incorrect.
- B. Both Statements I and Statements II are correct.
- C. Statements I is incorrect but Statements II are correct.
- D. Statements I is correct but Statements II are incorrect.

Answer: D

Solution:

Solution:

A.V., Par sec and light year are the unit of distance

Light year $\rightarrow$ distance travelled by light in one year

$$1 \text{ ly} = 9.5 \times 10^{15} \text{ m}$$

$$\text{parsec} = 3.262 \text{ light year}$$

$$A.V. = 1.58 \times 10^{-5} \text{ light year}$$

$$A.V. < 1y < \text{Parsec}.$$

Statement I correct and statement II incorrect.

Question69

If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be:

[11-Apr-2023 shift 2]

Options:

A. $FV^{-2}T^2$

B. FV^4T^{-6}

C. $FV^{-4}T^{-2}$

D. $F^2V^{-2}T^6$

Answer: C

Solution:

Solution:

$$\rho = F^x V^y T^z$$

$$[ML^{-3}] = [MLT^{-2}]^x [LT^{-1}]^y [T]^z$$

$$1 = x \quad x = 1$$

$$-3 = x + y \quad y = -4$$

$$0 = -2x - y + z \quad z = 2x + 4 = 2 - 4 = -2$$

$$\rho = F V^{-4} T^{-2} \quad \text{Ans. (3)}$$

Question 70

LIST I	LIST II
A. Spring constant	I. $[T^{-1}]$
B. Angular speed	II. $[MT^{-2}]$
C. Angular momentum	III. $[ML^2]$
D. Moment of Inertia	IV. $[ML^2T^{-1}]$

**Choose the correct answer from the options given below :
[12-Apr-2023 shift 1]**

Options:

A. A-II, B-III, C-I, D-IV

B. A-IV, B-I, C-III, D-II

C. A-I, B-III, C-II, D-IV

D. A-II, B-I, C-IV, D-III

Answer: D

Solution:

Solution:

(i) Spring constant

$$F = Kx$$

$$K = \frac{F}{x} = \frac{\text{kgm} \cdot \text{sec}}{\text{m}} = \text{kg} \cdot \text{sec}^{-2} = [\text{ML}^0\text{T}^{-2}]$$

(ii) Angular speed : (ω)

$$\omega = \frac{2\pi}{T}$$

$$\omega = [\text{M}^0\text{L}^0\text{T}^{-1}]$$

(iii) Angular momentum

$$L = MVR \Rightarrow \text{Kg} \cdot \text{m} \cdot \text{sec}^{-1} \cdot \text{m} = \text{kgm}^2 \cdot \text{sec}^{-1} \\ = [\text{ML}^2\text{T}^{-1}]$$

(iv) Moment of Inertia

$$I = MR^2 = \text{Kg} \cdot \text{m}^2 \\ = [\text{ML}^2\text{T}^0]$$

Question 71

In the equation $\left[x + \frac{a}{y^2} \right] [Y - b] = RT$, X is pressure, Y is volume, R is universal gas constant and T is temperature. The physical quantity equivalent to the ratio $\frac{a}{b}$ is :

[13-Apr-2023 shift 2]

Options:

A. Coefficient of viscosity

B. Energy

C. Impulse

D. Pressure gradient

Answer: B

Solution:

Solution:

x and $\frac{a}{y^2}$ have same dimensions

y and b have same dimensions

$$[a] = [ML^5T^{-2}]$$

$$[b] = [L^3]$$

$$\frac{[a]}{[b]} = ML^2T^{-2} \text{ has dimension of energy}$$

Question72

The speed of a wave produced in water is given by $v = \lambda^a g^b \rho^c$. Where λ , g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a, b and c respectively, are :

[15-Apr-2023 shift 1]

Options:

A. $\frac{1}{2}, 0, \frac{1}{2}$

B. $1, -1, 0$

C. $\frac{1}{2}, \frac{1}{2}, 0$

D. $1, 1, 0$

Answer: C

Solution:

Solution:

By dimensional analysis,

$$V = \lambda^a g^b \rho^c$$

$$[L^1T^{-1}] = [L^1]^a [L^1T^{-2}]^b [M^1L^{-3}]^c$$

$$[L^1T^{-1}] = [M^c L^{a+b-3c} T^{-2b}]$$

On comparing respective powers,

$$c = 0, -2b = -1, a + b - 3c = 1,$$

$$b = \frac{1}{2}, a = \frac{1}{2}$$

Question73

Two resistances are given as $R_1 = (10 \pm 0.5)\Omega$ and $R_2 = (15 \pm 0.5)\Omega$.

The percentage error in the measurement of equivalent resistance when they are connected in parallel is -

[6-Apr-2023 shift 1]

Options:

A. 2.33

B. 4.33

C. 5.33

D. 6.33

Answer: B

Solution:

Solution:

In parallel combination, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{15} = \frac{5}{30} = \frac{1}{6}$$

Now, for error calculation,

$$\frac{dR_{eq}}{R_{eq}^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2}$$

$$\Rightarrow \frac{dR_{eq}}{36} = \frac{0.5}{100} + \frac{0.5}{225}$$

$$dR_{eq} = 36 \times 0.5 \times \left(\frac{13}{900} \right) = 18 \times \frac{13}{900} = \frac{26}{100} = 0.26$$

$$\text{Now, } \frac{dR_{eq}}{R_{eq}} \times 100 = \frac{0.26}{6} \times 100 = \frac{26}{6} = 4.33$$

Question74

The length of a metallic wire is increased by 20% and its area of cross section is reduced by 4%. The percentage change in resistance of the metallic wire is _____

[6-Apr-2023 shift 1]

Answer: 25

Solution:

Solution:

$$R = \frac{\rho l}{A}$$

$$R' = \frac{\rho \times 1.2l}{0.96A} = \frac{10}{8} \times R = 1.25R$$

It means 25% increase in Resistance.

Question 75

Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is:
[8-Apr-2023 shift 1]

Options:

A. $\frac{5A}{2}$

B. $\frac{\sqrt{5}A^2}{2}$

C. $\frac{\sqrt{5}A}{4}$

D. $\frac{\sqrt{5}A}{2}$

Answer: D

Solution:

Solution:

$$|\vec{F}_1| = A, |\vec{F}_2| = \frac{A}{2} \quad \theta = \frac{\pi}{2}$$

$$|\vec{F}_{\text{net}}| = \sqrt{F_1^2 + F_2^2}$$

$$= \sqrt{A^2 + \left(\frac{A}{2}\right)^2}$$

$$|\vec{F}_{\text{net}}| = \frac{\sqrt{5}A}{2}$$

Question76

A physical quantity P is given as

$$P = \frac{a^2 b^3}{c \sqrt{d}}$$

The percentage error in the measurement of a, b, c and d are 1%, 2%, 3% and 4% respectively. The percentage error in the measurement of quantity P will be
[10-Apr-2023 shift 1]

Options:

A. 14%

B. 13%

C. 16%

D. 12%

Answer: B

Solution:

Solution:

$$\frac{dP}{P} \times 100 = \left(2 \frac{da}{a} + 3 \frac{db}{b} + \frac{dc}{c} + \frac{1}{2} \frac{d(d)}{d} \right) \times 100$$

$$= 2 \times 1 + 3 \times 2 + 3 + \frac{1}{2} \times 4$$

$$= 2 + 6 + 3 + 2$$

$$= 13\%$$

Question77

Three forces $F_1 = 10\text{N}$, $F_2 = 8\text{N}$, $F_3 = 6\text{N}$ are acting on a particle of mass 5 kg. The forces F_2 and F_3 are applied perpendicularly so that

particle remains at rest. If the force F_1 is removed, then the acceleration of the particle is :
[12-Apr-2023 shift 1]

Options:

A. 4.8ms^{-2}

B. 0.5ms^{-2}

C. 7ms^{-2}

D. 2ms^{-2}

Answer: D

Solution:

Solution:

$$F_1 = 10\text{N}$$

$$F_2 = 8\text{N}$$

$$F_3 = 6\text{N}$$

$$\vec{a} = \frac{F_2\hat{i} + F_3\hat{j}}{5}$$

$$\vec{a} = \frac{8\hat{i} + 6\hat{j}}{5}$$

$$|\vec{a}| = \frac{\sqrt{64 + 36}}{5}$$
$$= \frac{10}{5} = 2\text{m/sec}^2$$

OR

$$F_1^2 = F_2^2 + F_3^2$$

Now F_1 is removed then net force

$$F_{\text{net}} = \sqrt{F_2^2 + F_3^2}$$
$$= \sqrt{64 + 36}$$

Question78

A vector in $x - y$ plane makes an angle of 30° with y -axis. The magnitude of y -component of vector is $2\sqrt{3}$. The magnitude of x -

component of the vector will be :
[15-Apr-2023 shift 1]

Options:

A. 2

B. $\sqrt{3}$

C. $\frac{1}{\sqrt{3}}$

D. 6

Answer: A

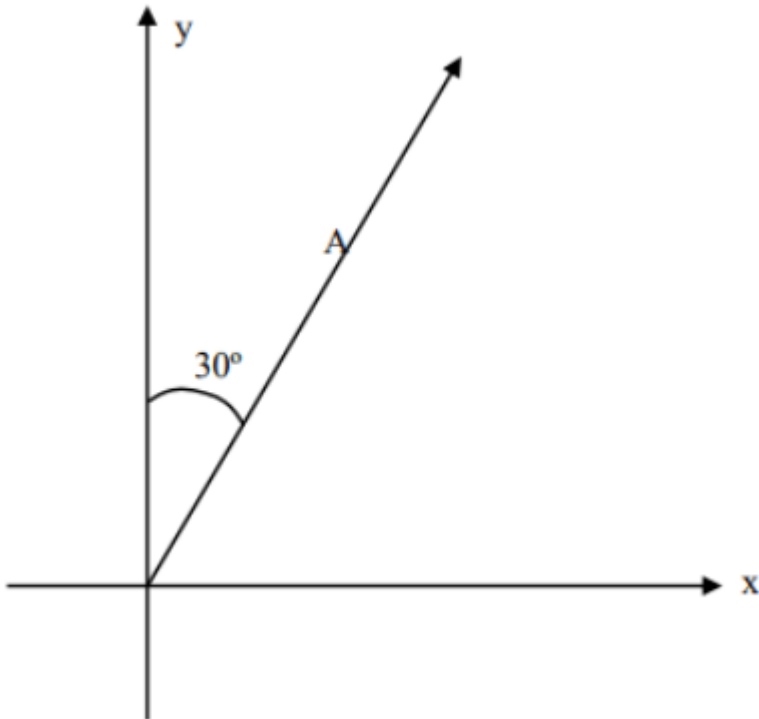
Solution:

Solution:

$$A_y = A \cos 30^\circ = 2\sqrt{3}$$

$$A = 4$$

$$A_x = A \sin 30^\circ = 2$$



Question79

In an experiment with vernier calipers of least count 0.1 mm, when two jaws are joined together the zero of vernier scale lies right to the

zero of the main scale and 6th division of vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 cm and 3.3 cm marks, and 4th division of vernier scale coincides with the main scale division. The diameter of bob is measured as
[10-Apr-2023 shift 2]

Options:

- A. 3.25 cm
- B. 3.22 cm
- C. 3.18 cm
- D. 3.26 cm

Answer: C

Solution:

Solution:

The zero error in vernier scale is $= 6 \times 0.1 \text{ mm} = 0.6 \text{ mm}$ (+ve zero error)

Note: +ve zero error will have to be subtracted

From the reading of the object.

Now, the diameter measured with the help of Vernier scale is

Given by $\rightarrow \text{M.S.D} + \text{V.S.D} \times \text{L.S}$

$$\Rightarrow 3.2 \text{ cm} + 0.1 \text{ mm} \times 4$$

$$= 3.24 \text{ cm}$$

The actual diameter is $\Rightarrow 3.24 \text{ mm} - (\text{zero error}) = 3.24 - 0.6 = 3.18 \text{ cm}$

Question80

Identify the pair of physical quantities which have different dimensions:

[24-Jun-2022-Shift-1]

Options:

- A. Wave number and Rydberg's constant
- B. Stress and Coefficient of elasticity
- C. Coercivity and Magnetisation

D. Specific heat capacity and Latent heat

Answer: D

Solution:

Solution:

$$[S] = \frac{[C]}{[m] \times [\Delta T]}$$

$$\text{and, } [L] = \frac{[Q]}{[m]}$$

⇒ They have different dimensions.

Question81

**Identify the pair of physical quantities that have same dimensions:
[24-Jun-2022-Shift-2]**

Options:

- A. velocity gradient and decay constant
- B. Wien's constant and Stefan constant
- C. angular frequency and angular momentum
- D. wave number and Avogadro number

Answer: A

Solution:

Solution:

$$\text{Velocity gradient} = \frac{dv}{dx}$$

$$\Rightarrow \text{Dimensions are } \frac{[LT^{-1}]}{[L]} = [T^{-1}]$$

Decay constant λ has dimensions of $[T^{-1}]$ because of the relation $\frac{dN}{dt} = -\lambda N$

⇒ Velocity gradient and decay constant have same dimensions.

Question82

An expression for a dimensionless quantity P is given by

$$P = \frac{\alpha}{\beta} \log_e \left(\frac{kt}{\beta x} \right); \text{ where } \alpha \text{ and } \beta \text{ are constants, } x \text{ is distance; } k \text{ is}$$

Boltzmann constant and t is the temperature. Then the dimensions of α will be :

[26-Jun-2022-Shift-1]

Options:

A. $[M^0 L^{-1} T^0]$

B. $[ML^0 T^{-2}]$

C. $[MLT^{-2}]$

D. $[ML^2 T^{-2}]$

Answer: C

Solution:

Solution:

$$[P] = \frac{[\alpha]}{[\beta]} \Rightarrow 1 \frac{[\alpha]}{[\beta]}$$

$$\Rightarrow [\alpha] = [\beta] \dots \dots (1)$$

$$\text{Now, } \left[\frac{kt}{\beta x} \right] = 1$$

$$\text{We know, } E = \frac{3}{2} kT \text{ (K T G)}$$

$$\therefore [\text{Energy}] = [KT]$$

$$\therefore [\beta x] = [kT]$$

$$[\beta] = \frac{ML^2 T^{-2}}{[x]} = \frac{ML^2 T^{-2}}{L}$$

$$\therefore [\alpha] = [\beta] = MLT^{-2}$$

Question83

The dimension of mutual inductance is :

[26-Jun-2022-Shift-2]

Options:

A. $[ML^2 T^{-2} A^{-1}]$

B. $[M L^2 T^{-3} A^{-1}]$

C. $[M L^2 T^{-2} A^{-2}]$

D. $[M L^2 T^{-3} A^{-2}]$

Answer: C

Solution:

Solution:

$$\because U = \frac{1}{2} M i^2$$

$$\Rightarrow [M] = \frac{[U]}{[i^2]} = \frac{M L^2 T^{-2}}{A^2}$$
$$= [M L^2 T^{-2} A^{-2}]$$

Question84

The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be :

[27-Jun-2022-Shift-2]

Options:

A. $[M L^{-1} T^{-1}]$

B. $[M L^{-1} T^{-2}]$

C. $[M L^2 T^{-1}]$

D. $[M^{-1} L^3 T^0]$

Answer: A

Solution:

Solution:

$$[\text{pascal-second}] = \frac{M L T^{-2}}{L^2} \times T$$
$$= M L^{-1} T^{-1}$$

Question85

If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?

[27-Jun-2022-Shift-2]

Options:

A. RC

B. $\frac{L}{R}$

C. $\sqrt{LC}$

D. $\frac{L}{C}$

Answer: D

Solution:

Solution:

$\left(\frac{L}{C}\right)$ does not have dimension of time.

RC , $\frac{L}{R}$ are time constant while $\sqrt{LC}$ is reciprocal of angular frequency or having dimension of time.

$$U = \frac{1}{2}Li^2 = \frac{1}{2}CV^2$$

So, $\left[\frac{L}{C}\right] = \frac{V^2}{i^2} = R^2$ is not the dimension of time.

Question86

In van der Waal equation $\left[P + \frac{a}{V^2}\right][V - b] = RT$; P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to:

[29-Jun-2022-Shift-1]

Options:

A. $\frac{P}{V}$

B. $\frac{V}{P}$

C. PV

D. PV^3

Answer: C

Solution:

Solution:

From the equation

$$[a] \equiv [PV^2]$$

$$[b] \equiv [V]$$

$$\Rightarrow \left[\frac{a}{b} \right] \equiv [PV]$$

Question87

If $Z = \frac{A^2 B^3}{C^4}$, then the relative error in Z will be :

[25-Jun-2022-Shift-1]

Options:

A. $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$

B. $\frac{2 \Delta A}{A} + \frac{3 \Delta B}{B} - \frac{4 \Delta C}{C}$

C. $\frac{2 \Delta A}{A} + \frac{3 \Delta B}{B} + \frac{4 \Delta C}{C}$

D. $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Answer: C

Solution:

Solution:

$$Z = \frac{A^2 B^3}{C^4}$$

$$\therefore \frac{\Delta Z}{Z} = \frac{2 \Delta A}{A} + 3 \times \frac{\Delta B}{B} + \frac{4 \Delta C}{C}$$

Question88

$\vec{A}$ is a vector quantity such that $|\vec{A}| = \text{non-zero constant}$. Which of the following expression is true for $\vec{A}$?
[25-Jun-2022-Shift-1]

Options:

A. $\vec{A} \cdot \vec{A} = 0$

B. $\vec{A} \times \vec{A} < 0$

C. $\vec{A} \times \vec{A} = 0$

D. $\vec{A} \times \vec{A} > 0$

Answer: C

Solution:

Solution:

$$\begin{aligned}\vec{A} \times \vec{A} &= A \times A \times \sin 0^\circ \\ &= 0\end{aligned}$$

Question89

Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other?
[25-Jun-2022-Shift-1]

Options:

A. $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan \frac{\theta}{2}$

B. $\left| \hat{A} - \hat{B} \right| = \left| \hat{A} + \hat{B} \right| \tan \frac{\theta}{2}$

C. $\left| \hat{A} + \hat{B} \right| = \left| \hat{A} - \hat{B} \right| \cos \frac{\theta}{2}$

D. $\left| \hat{A} - \hat{B} \right| = \left| \hat{A} + \hat{B} \right| \cos \frac{\theta}{2}$

Answer: B

Solution:

Solution:

$$\therefore \left| \hat{A} - \hat{B} \right| = 2 \sin \left(\frac{\theta}{2} \right)$$

$$\text{and, } \left| \hat{A} + \hat{B} \right| = 2 \cos \left(\frac{\theta}{2} \right)$$

$$\Rightarrow \frac{\left| \hat{A} - \hat{B} \right|}{\left| \hat{A} + \hat{B} \right|} = \tan \left(\frac{\theta}{2} \right)$$

Question90

For $z = a^2 x^3 y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12% respectively, then the percentage error for 'z' will be _____ %
[25-Jun-2022-Shift-2]

Answer: 18

Solution:

Solution:

$$\begin{aligned} \% \text{ error in } z &= 3 \times 4 + \frac{1}{2} \times 12 \\ &= 12 + 6 = 18\% \end{aligned}$$

Question91

A silver wire has a mass $(0.6 \pm 0.006)\text{g}$, radius $(0.5 \pm 0.005)\text{ mm}$ and length $(4 \pm 0.04)\text{ cm}$. The maximum percentage error in the measurement of its density will be :

[27-Jun-2022-Shift-1]

Options:

A. 4%

B. 3%

C. 6%

D. 7%

Answer: A

Solution:

Solution:

$$\rho = \frac{m}{V} = \frac{m}{\pi r^2 \times l}$$

$$\therefore \% \text{ error in } \rho = \left(\frac{0.006}{0.6} + 2 \times \frac{0.005}{0.5} + \frac{0.04}{4} \right) \times 100$$
$$= 4\%$$

Question92

The distance of the Sun from earth is $1.5 \times 10^{11}\text{ m}$ and its angular diameter is (2000) s when observed from the earth. The diameter of the Sun will be :

[27-Jun-2022-Shift-2]

Options:

A. $2.45 \times 10^{10}\text{ m}$

B. $1.45 \times 10^{10}\text{ m}$

C. $1.45 \times 10^9\text{ m}$

D. $0.14 \times 10^9 \text{m}$

Answer: C

Solution:

Solution:

$$\text{Diameter} = r \times \delta$$

$$= 1.5 \times 10^{11} \times (2000) \times \left(\frac{1}{3600} \right) \times \left(\frac{\pi}{180} \right)$$

$$= 1.45 \times 10^9 \text{m}$$

Question93

Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to two times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

[29-Jun-2022-Shift-1]

Options:

A. $\sin^{-1} \left(\frac{3}{5} \right)$

B. $\sin^{-1} \left(\frac{1}{3} \right)$

C. $\cos^{-1} \left(\frac{3}{5} \right)$

D. $\cos^{-1} \left(\frac{1}{3} \right)$

Answer: C

Solution:

Solution:

$$\sqrt{A^2 + A^2 + 2A^2 \cos \theta} = 2 \sqrt{A^2 + A^2 + 2A^2(-\cos \theta)}$$

$$\Rightarrow 2A^2 + 2A^2 \cos \theta = 8A^2 + 8A^2(-\cos \theta)$$

$$\Rightarrow 5 \cos \theta = 3$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{3}{5} \right)$$

Question94

In a vernier callipers, each cm on the main scale is divided into 20 equal parts. If tenth vernier scale division coincides with ninth main scale division. Then the value of vernier constant will be

$$\underline{\hspace{2cm}} \times 10^{-2} \text{ mm}$$

[26-Jun-2022-Shift-1]

Answer: 5

Solution:

Solution:

$$1 \text{ MSD} = \frac{1}{20} \text{ cm}$$

$$\because 10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ VSD} = \frac{9}{10} \times \frac{1}{20} \text{ cm} = \frac{9}{200} \times 10 \text{ mm} = 0.45 \text{ mm}$$

$$\text{Now, } 1 \text{ MSD} = \frac{1}{20} \times 10 \text{ mm} = 0.50 \text{ mm}$$

$$\text{LC} = (0.50 - 0.45) \text{ mm} = 0.05 \text{ mm}$$

$$= 5 \times 10^{-2} \text{ mm}$$

Question95

A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then

least count of the travelling microscope is $\underline{\hspace{2cm}} \times 10^{-6} \text{ m}$.

[26-Jun-2022-Shift-2]

Answer: 5

Solution:

Solution:

$$40M = 1 \text{ cm}$$

$$\Rightarrow M = 0.025 \text{ cm} \dots \dots \dots (1)$$

$$\text{Also, } 50V = 49M$$

$$\Rightarrow \text{Least count} = M - V = M - \frac{49}{50}M = \frac{M}{50}$$

$$\Rightarrow LC = \frac{0.025}{50} \text{ cm}$$

$$= \frac{250}{50} \times 10^{-6} \text{ m}$$

$$\Rightarrow LC = 5 \times 10^{-6} \text{ m}$$

Question96

A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.19 mm and 1.20 mm.

The percentage error is $\frac{x}{121}\%$. The value of x is

[28-Jun-2022-Shift-2]

Answer: 150

Solution:

Solution:

$$I_{\text{mean}} = \frac{1.22 + 1.23 + 1.19 + 1.20}{4} = 1.21$$

$$\Delta I_{\text{mean}} = \frac{0.01 + 0.02 + 0.02 + 0.01}{4} = 0.015$$

$$\text{So } \%I = \frac{\Delta I_{\text{mean}}}{I_{\text{mean}}} \times 100 = \frac{0.015}{1.21} \times 100 = \frac{150}{121}\%$$

$$x = 150$$

Question97

The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (-0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5 . The corrected diameter will be _____ $\times 10^{-2}$ cm
[29-Jun-2022-Shift-2]

Answer: 180

Solution:

Solution:

Since zero error is negative, we will add 0.05 cm.

$$\begin{aligned}\Rightarrow \text{Corrected reading} &= 1.7 \text{ cm} + 5 \times 0.1 \text{ mm} + 0.05 \text{ cm} \\ &= 180 \times 10^{-2} \text{ cm}\end{aligned}$$

Question98

If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is:

[25-Jul-2022-Shift-1]

Options:

A. $[PA^{-1}T^0]$

B. $[PAT^{-1}]$

C. $[PA^{-1}T]$

D. $[PA^{-1}T^{-1}]$

Answer: A

Solution:

Solution:

$$[\eta] = [M L^{-1} T^{-1}]$$

Now if $[\eta] = [P]^a [A]^b [T]^c$

$$\Rightarrow [M L^{-1} T^{-1}] = [M L^1 T^{-1}]^a [L^2]^b [T]^c$$

$$\Rightarrow a = 1, a + 2b = -1, -a + c = -1$$

$$\Rightarrow a = 1, b = -1, c = 0$$

$$\Rightarrow [\eta] = [P][A]^{-1}[T]^0$$

$$= [P A^{-1} T^0]$$

Question99

A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of :
[27-Jul-2022-Shift-1]

Options:

A. 15%

B. 25%

C. 75%

D. 5%

Answer: B

Solution:

Solution:

Dimensional formula for Torque

$$[\tau] = [M L^2 T^{-2}]$$

Now

$$\text{Percentage error in torque} = \% \tau = \% M + 2\% L$$

$$2\% T$$

$$\% \tau = 25\%$$

Question100

An expression of energy density is given by $u = \frac{\alpha}{\beta} \sin\left(\frac{\alpha x}{kt}\right)$, where α, β are constants, x is displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be :
[27-Jul-2022-Shift-2]

Options:

A. $[ML^2T^{-2}\theta^{-1}]$

B. $[M^0L^2T^{-2}]$

C. $[M^0L^0T^0]$

D. $[M^0L^2T^0]$

Answer: D

Solution:

Solution:

$$u = \frac{\alpha}{\beta} \sin\left(\frac{\alpha x}{kt}\right)$$

$$[\alpha] = \left[\frac{kt}{x} \right] = \frac{[\text{Energy}]}{[\text{Dis tance}]}$$

$$[\beta] = \frac{[\alpha]}{[u]}$$

$$= \frac{[\text{Energy}]/[\text{Distan ce}]}{[\text{Energy}]/ \text{Volume}]}$$

$$= [L^2]$$

Question101

The dimensions of $\left(\frac{B^2}{\mu_0} \right)$ will be :

(if μ_0 : permeability of free space and B : magnetic field)

[28-Jul-2022-Shift-1]

Options:

A. $[ML^2T^{-2}]$

B. $[MLT^{-2}]$

C. $[ML^{-1}T^{-2}]$

D. $[ML^2T^{-2}A^{-1}]$

Answer: C

Solution:

Solution:

$$\left[\frac{B^2}{\mu_0} \right] = [\text{Energy density}] = \frac{ML^2T^{-2}}{L^3} = ML^{-1}T^{-2}$$

Question102

Consider the efficiency of carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta} \log e^{\frac{\beta x}{kT}}$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the incorrect option:
[28-Jul-2022-Shift-2]

Options:

A. Dimensions of β is same as that of force.

B. Dimensions of $\alpha^{-1}x$ is same as that of energy.

C. Dimensions of $\eta^{-1} \sin \theta$ is same as that of $\alpha\beta$

D. Dimensions of α is same as that of β .

Answer: D

Solution:

Solution:

$$(A) [\beta] = \left[\frac{kT}{x} \right] = \left[\frac{E}{x} \right] = [MLT^{-2}] = [F]$$

$$(B) [\alpha\beta] = [M^0L^0T^0]$$

$$[\alpha]^{-1} = [\beta] = \left[\frac{kT}{x} \right]$$

So $[\alpha]^{-1}[\chi] = [kT] = [ML^2T^{-2}]$

(C) $\eta \sin\theta = \alpha\beta$

So $[\eta \sin\theta] = [\alpha\beta]$

$[\eta] = [M^0L^0T^0]$ it is dimensionless quantity

(D) $[\alpha] \neq [\beta]$

Question103

Match List I with List II.

List I	List II
A.Torque	I. Nms^{-1}
B.Stress	II. Jkg^{-1}
C.Latent Heat	III.Nm
D.Power	IV. Nm^{-2}

**Choose the correct answer from the options given below :
[29-Jul-2022-Shift-2]**

Options:

A. A-III, B-II, C-I, D-IV

B. A-III, B-IV, C-II, D-I

C. A-IV, B-I, C-III, D-II

D. A-II, B-III, C-I, D-IV

Answer: B

Solution:

Solution:

Torque $\rightarrow Nm$

Stress $\rightarrow N/m^2$

Latent heat $\rightarrow J/kg$

Power $\rightarrow Nm/s$

A-III, B-IV, C-II, D-I

Question104

The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be :

[25-Jul-2022-Shift-2]

Options:

A. 2

B. 4

C. 6

D. 8

Answer: D

Solution:

Solution:

$$\therefore H = i^2 R t$$

$$\therefore \% \text{ error in } H = 2 \times 2\% + 1\% + 3\% = 8\%$$

Question105

If $\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})\text{m}$ and $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})\text{m}$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be m.

[26-Jul-2022-Shift-2]

Answer: 2

Solution:

Solution:

$$\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k}) \text{ m and } \vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k}) \text{ m}$$

Component of $\vec{A}$ along $\vec{B} = \vec{A} \cdot \hat{B}$

$$= \vec{A} \cdot \frac{\vec{B}}{|\vec{B}|} = \frac{2 + 6 - 2}{\sqrt{1^2 + 2^2 + 2^2}}$$
$$= \frac{6}{3} = 2$$

Question106

The one division of main scale of Vernier callipers reads 1 mm and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other, the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between 4.1 cm and 4.2 cm and 6th Vernier division coincides scale division. The diameter of the bob will be _____ $\times 10^{-2}$ cm. [27-Jul-2022-Shift-1]

Answer: 412

Solution:

Solution:

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ VSD} = .9 \text{ MSD}$$

$$\text{L.C.} = .1 \text{ mm} = .01 \text{ cm}$$

$$+ \text{ve zero error} = .4 \text{ mm}$$

$$= 0.04 \text{ cm}$$

$$\text{Negative zero error} = 4.1 \text{ cm} + 6 \times .01 = 4.12 \text{ cm}$$

$$= 412 \times 10^{-2} \text{ cm}$$

Question107

If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be _____.
 [28-Jul-2022-Shift-1]

Answer: 5

Solution:

Solution:

$$\vec{a} \cdot \vec{b} = 0$$

$$\therefore \vec{a} \cdot \vec{b} = 0$$

$$\therefore 2 \times 1 + 4 \times 2 - 2 \times \alpha = 0$$

$$\therefore \alpha = 5$$

Question108

In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1 s resolution. If measured value of length is 10 cm known to 1 mm accuracy, The accuracy in the determination of g is found to be x%. The value of x is _____.
 [28-Jul-2022-Shift-2]

Answer: 5

Solution:

Solution:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\frac{dg}{g} \times 100 = \frac{2dT}{T} \times 100 + \frac{dl}{l} \times 100$$

$$= 2 \times \frac{1}{50} \times 100 + \frac{1}{100} \times 100 = 5\%$$

Question109

A screw gauge of pitch 0.5 mm is used to measure the diameter of uniform wire of length 6.8 cm, the main scale reading is 1.5 mm and circular scale reading is 7 . The calculated curved surface area of wire to appropriate significant figures is :
[Screw gauge has 50 divisions on its circular scale]
[26-Jul-2022-Shift-1]

Options:

A. 6.8cm^2

B. 3.4cm^2

C. 3.9cm^2

D. 2.4cm^2

Answer: B

Solution:

Solution:

$$\text{Least count} = \frac{0.5}{50} \text{ mm} = 0.01 \text{ mm}$$

$$\therefore \text{Diameter, } d = 1.5 \text{ mm} + 7 \times 0.01 \\ = 1.57 \text{ mm}$$

$$\therefore \text{Surface area} = (2\pi r) \times l \\ = \pi d l$$

$$= 3.142 \times \frac{1.57}{10} \times 6.8 \text{ cm}^2$$

$$= 3.354 \text{ cm}^2 = 3.4 \text{ cm}^2$$

Question110

In a Vernier Calipers, 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4th Vernier scale division exactly coincides with

the main scale reading. One main scale division is equal to 1 mm. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6th Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be :

[26-Jul-2022-Shift-2]

Options:

A. 3.02 cm

B. 3.06 cm

C. 3.10 cm

D. 3.20 cm

Answer: C

Solution:

Solution:

$$1 \text{ M.S.D} = 1 \text{ mm}$$

$$9 \text{ M.S.D} = 10 \text{ V.S.D}$$

$$1 \text{ V.S.D} = 0.9 \text{ M.S.D} = 0.9 \text{ mm}$$

$$\text{L.C of vernier caliper} = 1 - 0.9 = 0.1 \text{ mm} = 0.01 \text{ cm}$$

$$\text{zero error} = (10 - 4) \times 0.1 \text{ mm} = -0.6 \text{ mm}$$

$$\text{Reading} = \text{M.S.R} + \text{V.S.R} - \text{Zero error}$$

$$= 3 \text{ cm} + 6 \times 0.01 - [-0.06]$$

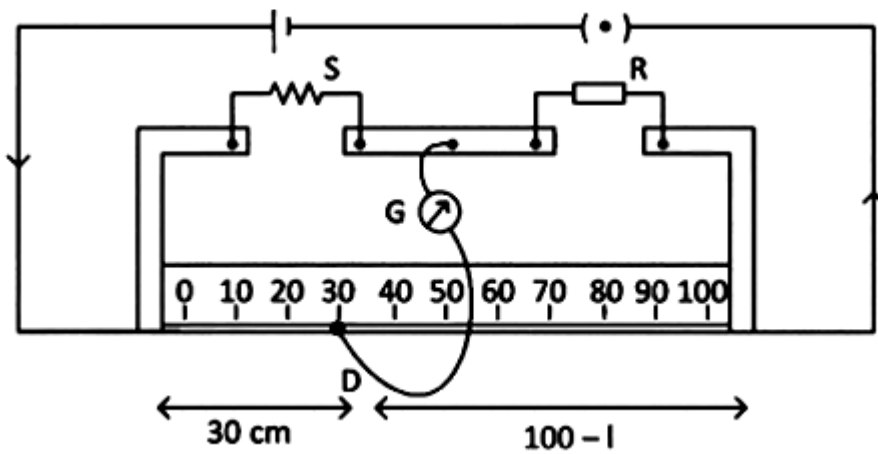
$$= 3 + 0.06 + 0.06$$

$$= 3.12 \text{ cm}$$

Nearest given answer in the options is 3.10

Question111

In a meter bridge experiment, for measuring unknown resistance 'S', the null point is obtained at a distance 30 cm from the left side as shown at point D. If R is 5.6kΩ, then the value of unknown resistance 'S' will be _____ Ω.



[27-Jul-2022-Shift-1]

Answer: 2400

Solution:

Solution:

$$\frac{S}{30} = \frac{5.6 \times 10^3}{70}$$

$$S = \frac{3}{7} \times 5.6 \times 10^3 = 2400$$

Question112

In an experiment with a convex lens, The plot of the image distance (v') against the object distance (u') measured from the focus gives a curve $v'u' = 225$. If all the distances are measured in cm. The magnitude of the focal length of the lens is ____ cm

[28-Jul-2022-Shift-2]

Answer: 15

Solution:

Solution:

$$ru = f^2 \text{ (by Newton's formula)}$$

$$f^2 = 225$$

$$f = 15 \text{ cm}$$

Question113

A travelling microscope has 20 divisions per cm on the main scale while its vernier scale has total 50 divisions and 25 vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope?

[29-Jul-2022-Shift-1]

Options:

A. 0.001 cm

B. 0.002 mm

C. 0.002 cm

D. 0.005 cm

Answer: C

Solution:

Solution:

$$1 \text{ MSD} = \frac{1}{20} \text{ cm}$$

$$1 \text{ VSD} = \frac{24}{25} \text{ MSD} = \frac{24}{25} \times \frac{1}{20} \text{ cm}$$

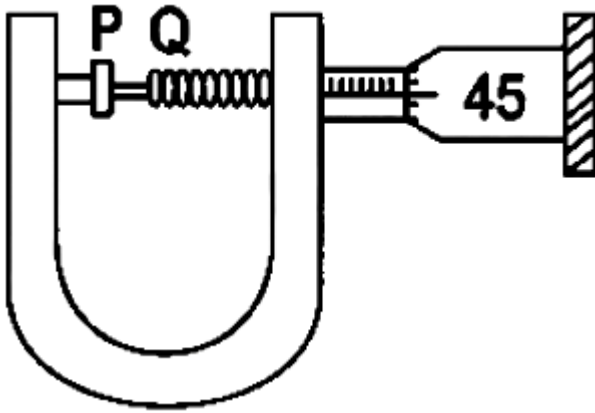
$$\therefore \text{Least count} = \frac{1}{20} \left(1 - \frac{24}{25} \right) \text{ cm}$$

$$= \frac{1}{20} \times \frac{1}{25} = \frac{1}{500} \text{ cm}$$

$$= 0.002 \text{ cm}$$

Question114

In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves 0.5 mm on main scale in one complete rotation

(B) Total divisions on circular scale = 50

(C) Main scale reading is 2.5 mm

(D) 45th division of circular scale is in the pitch line

(E) Instrument has 0.03 mm negative error

Then the diameter of wire is :

[29-Jul-2022-Shift-1]

Options:

A. 2.92 mm

B. 2.54 mm

C. 2.98 mm

D. 3.45 mm

Answer: C

Solution:

Solution:

MSR = 2.5 mm

CSR = $45 \times \frac{0.5}{50}$ mm

= 0.45 mm

Diameter reading = MSR + CSR - zero error

= 2.5 + 0.45 - (-0.03)

= 2.98 mm

Question115

The pitch of the screw gauge is 1mm and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lies 8 divisions below the reference line. When a wire is placed between the jaws, the first linear scale division is clearly visible while 72 nd division on circular scale coincides with the reference line. The radius of the wire is
[25 Feb 2021 Shift 1]

Options:

- A. 1.64mm
- B. 0.82mm
- C. 1.80mm
- D. 0.90mm

Answer: B

Solution:

Solution:

Given, pitch of screw gauge, $P = 1\text{mm}$

Number of division, $n = 100$

$$\therefore \text{Least count (LC)} = \frac{P}{n} = 1/100 = 0.01\text{mm}$$

As, zero of circular division lies 8 divisions below.

$$\therefore \text{Zero error} = 8 \times \text{LC} = 8 \times 0.01 = 0.08\text{mm}$$

Since, 1st linear scale division coincide with 72nd circular scale division.

$$\begin{aligned} \therefore \text{Radius}(r) &= \frac{[P + (72 \times \text{LC}) - \text{Zero error}]}{2} \\ &= [1 + (72 \times 0.01) - 0.08]/2 \\ &= (1.72 - 0.08)/2 = 1.64/2 \\ &= 0.82\text{mm} \end{aligned}$$

Question116

A large number of water drops, each of radius r , combine to have a drop of radius R . If the surface tension is T and mechanical equivalent of heat is J , the rise in heat energy per unit volume will be
[26 Feb 2021 Shift 1]

Options:

A. $\frac{2T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$

B. $\frac{2T}{r}$

C. $\frac{3T}{rJ}$

D. $\frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$

Answer: D

Solution:

Solution:

Given, radius of small drop = r

Radius of big drop = R

Surface tension = T

and mechanical equivalent of heat = J

As, small drops combine to form big drop.

$\therefore$ Volume of big drop (V_B) = $n \times$ Volume of small drop (V_S)

$$\Rightarrow \frac{4}{3}\pi R^3 = n \cdot \frac{4}{3}\pi r^3$$

$$\Rightarrow nr^3 = R^3$$

$$\Rightarrow r = \frac{R}{n^{1/3}} \dots (i)$$

Surface energy of small drop (E_S) = Surface tension (T) $\times$ Area (A)

$$\Rightarrow E_S = n \times 4\pi r^2 T \text{ and } E_B = 4\pi R^2 T$$

Now, change in energy will be

$$\Delta E = E_B - E_S = 4\pi T (nr^2 - R^2)$$

$$\therefore \text{Heat energy per unit volume} = \frac{\Delta E}{V} = \frac{4\pi T (nr^2 - R^2)}{J \times \frac{4}{3}\pi R^3}$$

$$= \frac{3T}{J} \left(\frac{nr^2}{R^3} - \frac{1}{R} \right) = \frac{3T}{J} \left(n \frac{R^2}{n^{2/3}R^3} - \frac{1}{R} \right)$$

$$= \frac{3T}{J} \left[\frac{n^{1/3}}{R} - \frac{1}{R} \right] \text{ [from Eq. (i)]}$$

$$= \frac{3T}{J} \left[\frac{1}{r} - \frac{1}{R} \right]$$

Question117

The period of oscillation of a simple pendulum is $T = 2\pi \sqrt{\frac{L}{g}}$.

Measured value of L is 1.0m from metre scale having a minimum division of 1mm and time of one complete oscillation is 1.95s measured from stopwatch of 0.01s resolution. The percentage error in the determination of g will be
[24 Feb 2021 Shift 2]

Options:

A. 1.13%

B. 1.03%

C. 1.33%

D. 1.30%

Answer: A

Solution:

Solution:

$$\text{Given, } T = 2\pi \sqrt{\frac{L}{g}} \dots (i)$$

where, time period, $T = 1.95\text{s}$

Length of string, $L = 1\text{m}$

Acceleration due to gravity = g

Error in time period, $\Delta T = 0.01\text{s} = 10^{-2}\text{s}$

Error in length, $\Delta L = 1\text{mm} = 1 \times 10^{-3}\text{m}$

Squaring Eq. (i) on both sides, we get

$$T^2 = 4\pi^2 \frac{L}{g}$$

$$g = 4\pi^2 \frac{L}{T^2}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2 \Delta T}{T} = \frac{10^{-3}}{1} + \frac{2 \times 10^{-2}}{1.95}$$

$$= 10^{-3} + 1.025 \times 10^{-2}$$

$$= 10^{-3} + 10.25 \times 10^{-3}$$

$$= 11.25 \times 10^{-3}$$

$$\therefore \Delta g/g \times 100 = 11.25 \times 10^{-3} \times 10^2 \\ = 1.125\% = 1.13\%$$

Question118

If C and V represent capacity and voltage respectively, then what are the dimensions of λ , where $\frac{C}{V} = \lambda$?

[26 Feb 2021 Shift 2]

Options:

A. $[M^{-2}L^{-3}I^2T^6]$

B. $[M^{-3}L^{-4}I^3T^7]$

C. $[M^{-1}L^{-3}I^{-2}T^{-1}]$

D. $[M^{-2}L^{-4}I^3T^7]$

Answer: D

Solution:

Solution:

Given, C and V represent capacity and voltage, respectively.

Dimensional formula of $[C] = [M^{-1}L^{-2}T^4I^2]$

Dimensional formula of $[V] = [ML^2T^{-3}I^{-1}]$

Therefore, dimensional formula of $[C/V] = \frac{[M^{-1}L^{-2}T^4I^2]}{[ML^2T^{-3}I^{-1}]}$

$= [M^{-2}L^{-4}T^7I^3]$

Question119

In a typical combustion engine, the work done by a gas molecule is

given $W = \alpha^2 \beta e^{\frac{-\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be

[26 Feb 2021 Shift 1]

Options:

A. $[MLT^{-2}]$

B. $[M^0LT^0]$

C. $[M^2 L T^{-2}]$

D. $[M L T^{-1}]$

Answer: B

Solution:

Solution:

Given, work done by gas molecule,

$$W = \alpha^2 \beta e^{-\beta x^2 / kT}$$

Here, x is displacement, k is Boltzmann constant, α and β are constants and T is temperature.

Dimensional formula of $[W] = [M L^2 T^{-2}]$

$\therefore$ Dimensions of $[\alpha^2 \beta] = [M L^2 T^{-2}]$

$$\Rightarrow \alpha = \left[\frac{M L^2 T^{-2}}{\beta} \right]^{1/2} \dots (i)$$

The term $[e^{-\beta x^2 / kT}]$ should be dimensionless, i.e. $[M^0 L^0 T^0]$.

$$\Rightarrow \left[\frac{\beta x^2}{kT} \right] = [M^0 L^0 T^0]$$

$$\Rightarrow [\beta] = \frac{[k][T]}{[x^2]} \dots (ii)$$

Energy of gaseous molecule $(E) = \frac{7}{2} kT$

$$[k] = [E] / [T] = [M L^2 T^{-2} K^{-1}]$$

Substituting the value of k in Eq. (ii), we get

$$[\beta] = \frac{[M L^2 T^{-2} K^{-1}][K]}{[L^2]} = [M T^{-2}]$$

Substituting the value of β in Eq. (i), we get

$$[\alpha] = \left\{ \frac{[M L^2 T^{-2}]}{[M T^{-2}]} \right\}^{1/2} = [M L^1 T^0]$$

Question120

Match List-I with List-II

List-I	List-II
A. h (Planck's constant)	1. $[MLT^{-1}]$
B. E (kinetic energy)	2. $[ML^2T^{-1}]$
C. V (electric potential)	3. $[ML^2T^{-2}]$
D. P (linear momentum)	4. $[ML^2I^{-1}T^{-3}]$

Choose the correct answer from the options given below.
[25 Feb 2021 Shift 1]

Options:

- A. (A-3), (B-4), (C-2), (D-1)
- B. (A-2), (B-3), (C-4), (D-1)
- C. (A-1), (B-2), (C-4), (D-3)
- D. (A-3), (B-2), (C-4), (D-1)

Answer: B

Solution:

Solution:

The dimensional formulae of given terms are

Planck's constant (h) = $[ML^2T^{-1}]$

Kinetic energy (E) = $[ML^2T^{-2}]$

Electric potential (V) = $[ML^{2-11}T^{-3}]$

Linear momentum (p) = $[MLT^{-1}]$

So, the correct match is

$A \rightarrow 2, B \rightarrow 3, C \rightarrow 4, D \rightarrow 1.$

Question121

The work done by a gas molecule in an isolated system is given by,

$W = \alpha\beta^2 e^{-\frac{x^2}{\alpha kT}}$, where x is the displacement k is the Boltzmann constant and T is the temperature, α and β are constants. Then the dimension of β will be :

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Options:

A. $[M L^2 T^{-2}]$

B. $[M L T^{-2}]$

C. $[M^2 L T^2]$

D. $[M^0 L T^0]$

Answer: B

Solution:

Solution:

$$\frac{x^2}{\alpha kT} \rightarrow \text{dimensionless}$$

$$\Rightarrow [\alpha] = \frac{[x^2]}{[kT]} = \frac{L^2}{M L^2 T^{-2}} = M^{-1} T^2$$

$$\text{Now, } [W] = [\alpha][\beta]^2$$

$$[\beta]^2 = \frac{[W]}{[\alpha]}$$

$$[\beta] = \left(\frac{M L^2 T^{-2}}{M^{-1} T^2} \right)^{\frac{1}{2}} = M L T^{-2}$$

Question122

Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm^3 of oleic acid per cm^3 of the solution. Then, you make a thin film of this solution (monomolecular thickness) of area 4 cm^2 by considering 100 spherical drops of radius

$\left(\frac{3}{40\pi} \right)^{\frac{1}{3}} \times 10^{-3} \text{ cm}$. Then, the thickness of oleic acid layer will be

$x \times 10^{-14}$ m, where x is
[17 Mar 2021 Shift 2]

Answer: 25

Solution:

Solution:

Given, the radius of the spherical drop

$$r = \left(\frac{3}{40\pi} \right)^{1/3} \times 10^{-3} \text{ cm}$$

The volume of 100 spherical drops, $V = 100 \times \frac{4\pi r^3}{3}$

Substituting the value of the radius in the above equation, we get

$$V = 100 \times \frac{4\pi}{3} \times \frac{3}{40\pi} \times 10^{-9}$$

$$V = 10^{-8} \text{ cm}^3$$

Now, Volume of the film = Area of the film $\times$ Thickness of the film

$$\Rightarrow \text{Volume of the film} = 4 \times t$$

The film is made up of 100 spherical drops.

$\therefore$ The volume of the film = Volume of 100 spherical drops

$$4t = 10^{-8} \Rightarrow t = 25 \times 10^{-12} \text{ m}$$

The thickness of the oleic layer,

$$t_0 = 0.01 \times t = 25 \times 10^{-14} \text{ m}$$

Comparing with $x \times 10^{-14}$, we get $x = 25$

Question123

In order to determine the Young's modulus of a wire of radius 0.2cm (measured using a scale of least count = 0.001cm) and length 1m (measured using a scale of least count = 1mm), a weight of mass 1kg (measured using a scale of least count = 1g) was hanged to get the elongation of 0.5cm (measured using a scale of least count 0.001cm). What will be the fractional error in the value of Young's modulus determined by this experiment?

[16 Mar 2021 Shift 2]

Options:

A. 0.14%

B. 0.9%

C. 9%

D. 1.4%

Answer: D

Solution:

Solution:

$$\text{Young's modulus, } Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F L}{A l}$$

$$\Rightarrow Y = \frac{mgL}{\pi R^2 l} \dots (i)$$

$$[\because F = mg \text{ and } A = \pi R^2]$$

To determine the fractional errors, we can write Eq. (i) as follows

$$\frac{\Delta Y}{Y} = \frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{2 \cdot \Delta R}{R} + \frac{\Delta l}{l}$$

$$\Rightarrow \frac{\Delta Y}{Y} \times 100 = \frac{\Delta m}{m} \times 100 + \frac{\Delta L}{L} \times 100 + \frac{2 \Delta R}{R} \times 100 + \frac{\Delta l}{l} \times 100$$

$$\Rightarrow \frac{\Delta Y}{Y} \times 100 = 100 \left[\frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{2 \Delta R}{R} + \frac{\Delta l}{l} \right]$$

$$= 100 \left[\frac{1}{1000} + \frac{1}{1000} + 2 \left(\frac{0.001}{0.2} \right) + \frac{0.001}{0.5} \right]$$

$$= \frac{1}{10} + \frac{1}{10} + 1 + \frac{1}{5} = \frac{14}{10} = 1.4\%$$

Question124

**One main scale division of a vernier callipers is a cm and n th division of the vernier scale coincide with $(n - 1)$ th division of the main scale
[16 Mar 2021 Shift 1]**

Options:

A. The least count of the callipers (in mm) is

B. $\frac{10na}{(n-1)}$

C. $\frac{10a}{(n-1)}$

D. $\left(\frac{n-1}{10n}\right)a$

E. $\frac{10a}{n}$

Answer: E

Solution:

Solution:

According to the question,

One division of main scale reading = a cm

n th vernier scale division = $(n-1)$ th main scale division

$$\therefore \text{One division of vernier scale reading} = \frac{(n-1) \times a}{n} \dots (i)$$

We know that,

Least count (LC) =

[1 main scale division – 1 vernier scale division] cm

$$= a - \frac{(n-1)a}{n} \quad [\text{using Eq. (i)}]$$

$$= \frac{a(n - n + 1)}{n} = \frac{a}{n} \text{cm} = \frac{a}{n} \times 10\text{mm}$$

$$\Rightarrow LC = \frac{10a}{n} \text{mm}$$

Question125

The resistance $R = \frac{V}{I}$, where $V = (50 \pm 2)V$ and $I = (20 \pm 0.2)A$. The percentage error in R is $x\%$. The value of x to the nearest integer is.....

[16 Mar 2021 Shift 1]

Answer: 5

Solution:

Solution:

$$\text{Given, } R = \frac{V}{I} \dots (i)$$

where, $V = (50 \pm 2)V$

$I = (20 \pm 0.2)A$

From Eq. (i)

$$\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$$

$$\begin{aligned} \% \text{ error in } R &= \left[\frac{2}{50} \times 100 + \frac{0.2}{20} \times 100 \right] \% \\ &= [2 \times 2 + 0.2 \times 5] \% \\ &= 5\% \end{aligned}$$

Question126

A person is swimming with a speed of 10m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is xm/s. The value of x to the nearest integer is

[18 Mar 2021 Shift 1]

Answer: 5

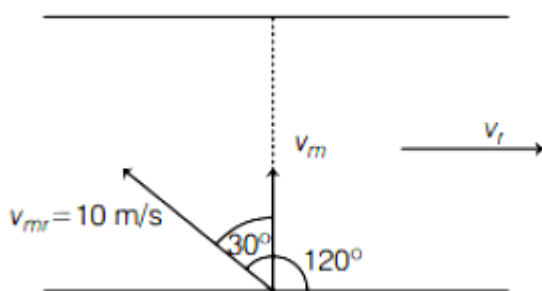
Solution:

Solution:

Given,

A person's swimming speed with respect to the river, $v_{mr} = 10\text{m/s}$

As shown in the figure,



Let's determine the speed of the river flow, v_r

In X-direction,

$$v_r = v_{mr} \sin 30^\circ$$

$$v_r = 10 \times \sin 30^\circ$$

$$v_r = 5\text{m/s}$$

Hence, the speed of the river flow is 5m/s. So, the value of the x to the nearest integer is 5 .

Question127

The vernier scale used for measurement has a positive zero error of 0.2mm. If while taking a measurement, it was noted that ' 0 ' on the vernier scale lies between 8.5cm and 8.6cm, vernier coincidence is 6 , then the correct value of measurement is cm.

[17 Mar 2021 Shift 1]

Options:

A. 8.36

B. 8.54

C. 8.58

D. 8.56

Answer: B

Solution:

Solution:

Given, positive zero error = 0.2mm = 0.02cm

[$\therefore$ Least count LC = 0.01cm]

Main scale reading = 8.5cm

Vernier scale reading

= Vernier scale coincidence $\times$ Least count

= $6 \times 0.01 = 0.06\text{cm}$

Final reading = Main scale reading + Vernier scale reading - Zero error

= $8.5 + 0.06 - 0.02$

= 8.54cm

Question128

Assertion A : If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.

Reason R :

Least Count = $\frac{\text{Pitch}}{\text{Total divisions on circular scale}}$

In the light of the above statements, choose the most appropriate answer from the options given below :

[27 Jul 2021 Shift 1]

Options:

- A. A is not correct but R is correct.
- B. Both A and R are correct and R is the correct explanation of A.
- C. A is correct but R is not correct.
- D. Both A and R are correct and R is NOT the correct explanation of A.

Answer: A**Solution:**

Solution:

$$\text{Least Count} = \frac{\text{Pitch}}{\text{Total divisions on circular scale}}$$

In 5 revolution, distance travel, 5 mm

In 1 revolution, it will travel 1 mm.

$$\text{So least count} = \frac{1}{50} = 0.02$$

Question 129

The force is given in terms of time t and displacement x by the equation

$$F = A \cos Bx + C \sin Dt$$

The dimensional formula of $\frac{AD}{B}$ is :

[25 Jul 2021 Shift 2]**Options:**

- A. $[M^0 L T^{-1}]$
- B. $[M L^2 T^{-3}]$
- C. $[M^1 L^1 T^{-2}]$
- D. $[M^2 L^2 T^{-3}]$

Answer: B**Solution:**

Solution:

$$[A] = [M L T^{-2}]$$

$$[B] = [L^{-1}]$$

$$[D] = [T^{-1}]$$

$$\left[\frac{AD}{B} \right] = \frac{[M L T^{-2}][T^{-1}]}{[L^{-1}]}$$

$$\left[\frac{D}{B} \right] = [M L^2 T^{-3}]$$

Question130

If time (t), velocity (v), and angular momentum (l) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and l is :

[20 Jul 2021 Shift 2]

Options:

A. $[t^{-1} v^1 l^{-2}]$

B. $[t^1 v^2 l^{-1}]$

C. $[t^{-2} v^{-1} l^1]$

D. $[t^{-1} v^{-2} l^1]$

Answer: D

Solution:

Solution:

$$m \propto t^a v^b l^c$$

$$m \propto [T]^a [L T^{-1}]^b [M L^2 T^{-1}]^c$$

$$M^1 L^0 T^0 = M^c L^{b+2c} T^{a-b-c}$$

comparing powers

$$c = 1, b = -2, a = -1$$

$$m \propto t^{-1} v^{-2} l^1$$

Question131

The entropy of any system is given by

$$S = \alpha^2 \beta \ln \left[\frac{\mu k R}{J \beta^2} + 3 \right]$$

where α and β are the constants. μ , J , k and R are no. of moles, mechanical equivalent of heat, Boltzmann constant and gas constant respectively.

$$\left[\text{Take } S = \frac{dQ}{T} \right]$$

**Choose the incorrect option from the following :
[20 Jul 2021 Shift 1]**

Options:

- A. α and J have the same dimensions.
- B. S , β , k and μR have the same dimensions.
- C. S and α have different dimensions.
- D. α and k have the same dimensions.

Answer: D

Solution:

Solution:

$$S = \alpha^2 \beta \ln \left(\frac{\mu K R}{J \beta^2} + 3 \right)$$

$$S = \frac{Q}{T} = \text{joule} / K$$

$$[\alpha^2 \beta] = \text{Joule} / K$$

$$PV = nRT \quad \left[\frac{\mu K R}{J \beta^2} \right] = 1$$

$$R = \frac{\text{Joule}}{K}$$

$$\Rightarrow R = \frac{\text{Joule}}{K}, K = \frac{\text{Joule}}{R}$$

$$\Rightarrow \beta = \left(\frac{\text{Joule}}{K} \right)$$

$$\alpha^2 \beta = \left(\frac{\text{Joule}}{K} \right)$$

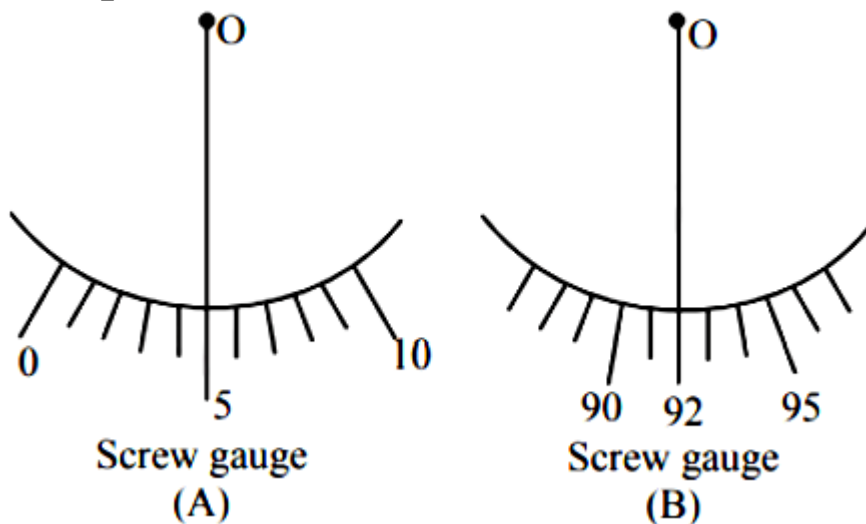
$$\Rightarrow \alpha = \text{dimensionless}$$

Question132

Student A and Student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____.

[Figure shows position of reference 'O' when jaws of screw gauge are closed]

Given pitch = 0.1 cm.



[25 Jul 2021 Shift 1]

Answer: 13

Solution:

Solution:

For (A)

$$\text{Reading} = \text{MSR} + \text{CSR} + \text{Error}$$

$$0.322 = 0.300 + \text{CSR} + 5 \times \text{LC}$$

$$0.322 = 0.300 + \text{CSR} + 0.005$$

$$\text{CSR} = 0.017$$

For B

$$\text{Reading} = \text{MSR} + \text{CSR} + \text{Error}$$

$$0.322 = 0.200 + \text{CSR} + 0.092$$

$$\text{CSR} = 0.030$$

$$\text{Difference} = 0.030 - 0.017 = 0.013 \text{ cm}$$

$$\text{Division on circular scale} = \frac{0.013}{0.001} = 13$$

Question133

Three students S_1 , S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different number of oscillations. The observation same as shown in the table.

Student No.	Length of pendulum (cm)	No. of oscillations (n)	Total time for n oscillations	Time period (s)
1.	64.0	8	128.0	16.0
2.	64.0	4	64.0	16.0
3.	20.0	4	36.0	9.0

(Least count of length = 0.1 m

least count for time = 0.1 s)

If E_1 , E_2 and E_3 are the percentage errors in ' g ' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by student no. _____.

[22 Jul 2021 Shift 2]

Answer: 1

Solution:

Solution:

$$T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

$$\Delta T = \frac{\text{least count of time } (\Delta T_0)}{\text{number of oscillations}(n)}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T_0}{nT}$$

As Δl and ΔT_0 are same for all observations so $\frac{\Delta g}{g}$ is minimum for highest value of l , n and T

$\Rightarrow$ Minimum percentage error in g is for student number-1

Question134

The diameter of a spherical bob is measured using a Vernier callipers. 9 divisions of the main scale, in the vernier calipers, are equal to 10 divisions of vernier scale. One main scale division is 1 mm. The main scale reading is 10 mm and 8th division of vernier scale was found to coincide exactly with one of the main scale division. If the given vernier callipers has positive zero error of 0.04 cm, then the radius of the bob is $\times 10^{-2}$ cm.
[31 Aug 2021 Shift 2]

Answer: 52

Solution:

Solution:

Given, 9 divisions of main scale are equal to 10 divisions of Vernier scale.

i.e. 9 MSD = 10 VSD

$$\Rightarrow \text{VSD} = \frac{9}{10} \text{MSD} \dots (i)$$

Size of 1 main scale division,

$$1 \text{ MSD} = 1 \text{ mm}$$

Now, least count, $\text{LC} = 1\text{MSD} - 1\text{VSD} \dots (ii)$

Using Eqs. (i) and (ii), we get

$$\text{LC} = 1 \text{ MSD} - \frac{9}{10} \text{MSD} = \frac{1}{10} \text{MSD} = \frac{1}{10} \text{mm}$$

While measuring the diameter of bob.

Main Scale Reading, $\text{MSR} = 10 \text{ mm}$

Vernier Scale Reading, $\text{VSR} = 8$

Zero error, $e = 0.04 \text{ cm}$

Now, diameter, $d = [\text{MSR} + \text{LC} \times \text{VSR}] - e$

$$= \left(10 \text{ mm} + \frac{1}{10} \times 8 \text{ mm} \right) - 0.04 \text{ cm}$$

$$= (10.8) \text{ mm} - 0.04 \text{ cm}$$

$$= 1.08 \text{ cm} - 0.04 \text{ cm} = 1.04 \text{ cm}$$

$$\text{Radius, } r = \frac{d}{2} = \frac{1.04}{2} \text{ cm} = 0.52 \text{ cm} = 52 \times 10^{-2} \text{ cm}$$

$\therefore$ Correct answer is 52.

Question135

If the length of the pendulum in pendulum clock increases by 0.1%, then the error in time per day is
[26 Aug 2021 Shift 2]

Options:

A. 86.4s

B. 4.32s

C. 43.2s

D. 8.64s

Answer: C

Solution:

Solution:

Increase in length of pendulum is 0.1%.

$$\text{i.e. } \frac{\Delta L}{L} \times 100 = 0.1$$

Time period of pendulum is given by

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Here, 2π and g are constant.

$$\frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta L}{L} \times 100$$

$$\Rightarrow \frac{\Delta T}{T} \times 100 = \frac{1}{2} \times 0.1$$

$$\Delta T = 0.05 \times \frac{T}{100}$$

In one single day, the time in seconds is

$$T = 24 \times 60 \times 60 = 86400\text{s}$$

$$\therefore \Delta T = 0.05 \times \frac{86400}{100} = 43.2\text{s}$$

Thus, the error in time per day is 43.2 s.

Question136

In a screw gauge, 5th division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale, and the main scale moves by 0.5 mm on a complete rotation. For a particular observation the reading on the main scale is 5 mm and the 20th division of the circular scale

coincides with reference line. Calculate the true reading.
[26 Aug 2021 Shift 1]

Options:

A. 5.00 mm

B. 5.25 mm

C. 5.15 mm

D. 5.20 mm

Answer: C

Solution:

Solution:

Least count of a screw gauge is given by

$$\text{Least count (LC)} = \frac{\text{Pitch}}{\text{Number of circular scale divisions}}$$

$$\Rightarrow \text{LC} = \frac{0.5}{50} \dots (i)$$

The actual reading (true reading)

= Main scale reading + Least count $\times$ Circular division – Zero error

$$= 5 + \left(\frac{0.5}{50} \right) \times 20 - \frac{0.5}{50} \times 5$$

[$\therefore$ Zero error = Least count $\times$ Coinciding division]

$$= 5 + \frac{0.5}{50}(20 - 5)$$

$$= 5 + \frac{0.5}{50}(15)$$

$$= 5 + \frac{15}{100}$$

$$= \frac{515}{100} \text{ mm} = 5.15 \text{ mm}$$

Question137

Which of the following equations is dimensionally incorrect? Where, t = time, h = height, s = surface tension, θ = angle, ρ = density, a, r = radius, g = acceleration due to gravity, V = volume, p = pressure, W = work done, τ = torque, ϵ = permittivity, E = electric field, J = current density, L = length.

[31 Aug 2021 Shift 1]

Options:

$$A. V = \frac{\pi p a^4}{8 \eta L}$$

$$B. h = \frac{2s \cos \theta}{\rho g r}$$

$$C. J = \epsilon \frac{\partial E}{\partial t}$$

$$D. W = \tau \theta$$

Answer: A

Solution:

Solution:

As we know that,

Dimensional formula of volume = $[M^0 L^3 T^0]$... (i)

Since, $F = 6\pi\eta r v$

$$\therefore \eta \propto \frac{F}{rv}$$

where, η is viscosity and F is force.

$$\therefore [\eta] = \frac{[ML T^{-2}]}{[L \cdot LT^{-1}]} = [ML^{-1} T^{-1}]$$

$$\text{So, } \left[\frac{\pi p a^4}{8 \eta L} \right] = \frac{[ML^{-1} T^{-2}][L^4]}{[ML^{-1} T^{-1}][L^1]} \\ = [M^0 L^3 T^0] \dots (ii)$$

Since, Eq. (i) is not equal to Eq (ii), so option (a) is wrong.

Now, since formula of capillary rise in tube, $h = \frac{2s \cos \theta}{\rho g r}$

Dimensional formula of LHS part,

$$\therefore [h] = [L]$$

Dimensional formula of RHS part

$$= \frac{[s]}{[\rho][g][r]} = \frac{[MT^{-2}]}{[ML^{-3}][LT^{-2}][L]} = [L]$$

Hence, $h = \frac{2s \cos \theta}{\rho g r}$ is dimensionally correct.

So, option (b) will also be dimensionally correct.

In option (c),

$$J = \epsilon \frac{dE}{dt} \dots (iii)$$

$$\Rightarrow J = \epsilon \frac{E}{t}$$

Dimension of current density J is calculated as

$$\text{Since, } J = \frac{1}{A}$$

$$\therefore [J] = \frac{[I]}{[A]} = \frac{[A]}{[L^2]}$$

$$\Rightarrow [J] = [AL^{-2}] \dots (iv)$$

Again, we know that

$$E = \frac{1}{4\pi\epsilon} \cdot \frac{q}{r^2}$$

$$\Rightarrow \epsilon E = \frac{1}{4\pi} \cdot \frac{q}{r^2}$$

$$\Rightarrow \frac{\epsilon E}{t} = \frac{1}{4\pi} \cdot \frac{q}{tr^2}$$

$$\Rightarrow \left[\frac{\epsilon E}{t} \right] = \frac{[q]}{[t][r^2]} = \frac{[AT]}{[T][L^2]}$$

$$\left[\frac{\epsilon E}{t} \right] = [AL^{-2}] \dots (v)$$

Form Eqs. (iv) and (v), we see that Eq. (iii) is dimensionally correct.

In option (d) $W = \tau\theta$

and $\tau = r \times F$

So, dimensional formula of

$$[\tau][\theta] = [r][F] = [L][MLT^{-2}] = [ML^2T^{-2}] = [W]$$

Question138

If velocity [v], time [T] and force [F] are choosen as the base quantities, the dimensions of the mass will be [31 Aug 2021 Shift 2]

Options:

A. $[FT^{-1}v^{-1}]$

B. $[FTv^{-1}]$

C. $[FT^2v]$

D. $[FvT^{-1}]$

Answer: B

Solution:

Solution:

When the velocity (v), time (T) and force (F) are chosen as base quantities. Then, mass is given by

$$m \propto v^x T^y F^z \dots (i)$$

Using dimensional formula of all quantities,

$$[ML^0T^0] = [LT^{-1}]^x [T]^y [MLT^{-2}]^z$$

$$[M^1L^0T^0] = [M^z L^{x+z} T^{-x+y-2z}]$$

Comparing the powers of dimensions on both sides, we get

$$z = 1$$

$$x + z = 0 \text{ and } -x + y - 2z = 0$$

$$\Rightarrow x + 1 = 0$$

$$\Rightarrow x = -1$$

$$\text{and } -(-1) + y - 2(1) = 0$$

$$\Rightarrow 1 + y - 2 = 0$$

$$\Rightarrow y = 1$$

Substituting these values in Eq. (i), we get

$$m \propto v^{-1} T^1 F^1$$

$$\Rightarrow m = [FTv^{-1}]$$

Question 139

**If force (F), length (L) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density:
[27 Aug 2021 Shift 2]**

Options:

A. $[FL^{-4}T^2]$

B. $[FL^{-3}T^2]$

C. $[FL^{-5}T^2]$

D. $[FL^{-3}T^3]$

Answer: A

Solution:

Solution:

As we know that, the dimensional formula of

$$\text{density } [D] = [ML^{-3}T^0]$$

Since, dimensional formula of force

$$[F] = [MLT^{-2}]$$

Dimensional formula of length

$$[L] = [M^0L^1T^0]$$

Dimensional formula of time

$$[T] = [M^0L^0T^1]$$

$$\therefore [D] = [F]^a [L]^b [T]^c$$

$$\Rightarrow [ML^{-3}T^0] = [ML^{-2}]^a$$

$$[M^0L^1T^0]^b [M^0L^0T^1]^c$$

$$= [M^a L^{a+b} T^{-2a+c}]$$

Comparing powers of dimensions on both sides, we get

$$ca = 1$$

$$a + b = -3$$

$$\Rightarrow b = -3 - 1 = -4$$

$$\text{and } -2a + c = 0$$

$$c = 2a$$

$$\Rightarrow c = 2 \times 1 = 2$$

$\therefore$ Dimensional formula of density will be $[F^1 L^{-4} T^2]$.

Question 140

Which of the following is not a dimensionless quantity?
[27 Aug 2021 Shift 1]

Options:

- A. Relative magnetic permeability (μ_r)
- B. Power factor
- C. Permeability of free space (μ_0)
- D. Quality factor

Answer: C

Solution:

Solution:

Relative magnetic permeability (μ_r) It is the ratio of permeability of medium to the permeability of free space i.e., $\mu_r = \frac{\mu_m}{\mu_0}$

As, it is a ratio of permeabilities. So, it is unitless and dimensionless quantity.

Power factor ($\cos \phi$) It is cosine of phase difference between alternating current and alternating voltage. Hence, it has only a numerical value, so it is also a unitless and dimensionless quantity.

Permeability of free space (μ_0) It is the ratio of magnetising field induction (B) to magnetising field intensity (H)

$$\text{i.e. } \mu_0 = \frac{B}{H}$$

$$\Rightarrow [\mu_0] = \frac{[B]}{[H]} \dots (i)$$

Since, $H = nI$

where, n = number of turns per unit length

and I = current

$$\therefore [H] = [n][I]$$

$$= [L^{-1}][A] = [AL^{-1}]$$

We know that, force on a current carrying conductor in magnetic field B is given as

$$F = IBl$$

where, l = length of the conductor,

B = magnetic field

and I = current.

$$\therefore B = \frac{F}{Il}$$

$$\Rightarrow [B] = \frac{[F]}{[I][l]} = \frac{[MLT^{-2}]}{[A][L]} = [MT^{-2}A^{-1}]$$

$\therefore$ From Eq. (i), we have

$$[\mu_0] = \frac{[MT^{-2}A^{-1}]}{[AL^{-1}]} = [MLT^{-2}A^{-2}]$$

Hence, permeability of free space is not a dimensionless quantity.

Quality factor It is the ratio of energy stored to the energy dissipated per cycle.

$$\text{Quality factor} = \frac{\text{Energy stored}}{\text{Energy dissipated per cycle}}$$

As, it is ratio of energies so, it will be unitless and dimensionless quantity.

Question141

Match List-I with List-II.

List - I	List - II
A. Magnetic induction	1. $[ML^2T^{-2}A^{-1}]$
B. Magnetic flux	2. $[ML^{-1}A]$
C. Magnetic permeability	3. $[MT^{-2}A^{-1}]$
D. Magnetisation	4. $[MLT^{-2}A^{-2}]$

Choose the most appropriate answer from the options given below.
[26 Aug 2021 Shift 2]

Options:

A. A-2 B-4 C-1 D-3

B. A-2 B-1 C-4 D-3

C. A-3 B-2 C-4 D-1

D. None of the above

Answer: D

Solution:

Solution:

Magnetic induction

Magnetic force of induction can be given as

$$F = qvB$$

$$\Rightarrow B = \frac{F}{qv}$$

$$[B] = \frac{[F]}{[q][v]} = \frac{[MLT^{-2}]}{[AT][LT^{-1}]} = [M^1T^{-2}A^{-1}]$$

Magnetic Flux Magnetic flux is the product of magnetic induction and area.

$$\phi = B \cdot A$$

$$[\phi] = [B][A] = [M^1T^{-2}A^{-1}][L^2] \\ = [ML^2T^{-2}A^{-1}]$$

Magnetic permeability

We know that, magnetic field inside the solenoid.

$$B = \mu_0 nI$$

$$\Rightarrow \mu_0 = \frac{B}{nI} \text{ (where, } n \text{ is number of turns per unit length)}$$

$$\Rightarrow [\mu_0] = \frac{[B]}{[n][I]} = \frac{[MT^{-2}A^{-1}]}{[L^{-1}][A]} = [MLT^{-2}A^{-2}]$$

Magnetisation The magnetic dipole moment acquired per unit volume is known as magnetisation.

$$I = \frac{M}{V}$$

$$[I] = \frac{[L^2A]}{[L^3]} = [L^{-1}A]$$

Hence, no option is correct.

Question 142

If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimension of P in the formula $P = EL^2M^{-5}G^{-2}$ is
[26 Aug 2021 Shift 1]

Options:

A. $[M^0L^1T^0]$

B. $[M^{-1}L^{-1}T^2]$

C. $[M^1L^1T^{-2}]$

D. $[M^0L^0T^0]$

Answer: D

Solution:

Solution:

Dimension of energy, $E = [ML^2T^{-2}]$

Angular momentum, $L = [ML^2T^{-1}]$

Mass, $M = [M]$

and gravitational constant, $G = [M^{-1}L^3T^{-2}]$

The dimension of P in formula,

$P = EL^2M^{-5}G^{-2}$ is given as follows

$$\begin{aligned}[P] &= \frac{[E][L^2]}{[M^5][G^2]} \\ &= \frac{[ML^2T^{-2}] \times [M^2L^4T^{-2}]}{[M^5][M^{-2}L^6T^{-4}]} \\ &= [M^0L^0T^0]\end{aligned}$$

Question143

If E and H represent the intensity of electric field and magnetising field respectively, then the unit of E / H will be
[27 Aug 2021 Shift 1]

Options:

A. ohm

B. mho

C. joule

D. newton

Answer: A

Solution:

Solution:

Unit of intensity of electric field E is Vm^{-1} .

Unit of intensity of magnetising field H is Am^{-1} .

Unit of E / H can be calculated as

$$\frac{\text{Unit of E}}{\text{Unit of H}} = \frac{Vm^{-1}}{Am^{-1}} = VA = \text{ohm}.$$

Thus, the unit of E / H will be ohm.

Question144

Match List-I with List-II.

List-I	List-II
A . R_H (Rydberg constant)	1. $\text{kg m}^{-1} \text{s}^{-1}$
B. h (Planck's constant)	2. $\text{kg m}^2 \text{s}^{-1}$
C . μ_B (Magnetic field energy density)	3. m^{-1}
D. η (Coefficient of viscosity)	4. $\text{kg m}^{-1} \text{s}^{-2}$

Choose the most appropriate answer from the options given below.
[27 Aug 2021 Shift 2]

Options:

A. A-2 B-3 C-4 D-1

B. A-3 B-2 C-4 D-1

C. A-4 B-2 C-1 D-3

D. A-3 B-2 C-1 D-4

Answer: B

Solution:

Solution:

As we know that, SI unit of following terms are

(a) R_H (Rydberg constant) = m^{-1}

(b) h (Planck's constant)

= $\text{J} \cdot \text{s} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{s} = \text{kg} \cdot \text{m}^2 \text{s}^{-1}$

(c) Magnetic field energy density

$(\mu_B) = \frac{\text{Energy}}{\text{Volume}}$
= $\frac{\text{kg} \cdot \text{m}^2 \text{s}^{-2}}{\text{m}^3}$

$$= \text{kgm}^{-1}\text{s}^{-2}$$

(d) η (coefficient of viscosity)

$$\therefore F = \eta A \frac{dv}{dx}$$

$$\therefore \eta = \frac{F dx}{A dv} = \frac{\text{kg} \cdot \text{ms}^{-2} \cdot \text{m}}{\text{m}^2 \cdot \text{ms}^{-1}} = \text{kgm}^{-1}\text{s}^{-1}$$

So, the correct match is A – 3, B – 2, C – 4 and D-1.

Question145

Two resistors $R_1 = (4 \pm 0.8)\Omega$ and $R_2 = (4 \pm 0.4)\Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be

[1 Sep 2021 Shift 2]

Options:

A. $(4 \pm 0.4)\Omega$

B. $(2 \pm 0.4)\Omega$

C. $(2 \pm 0.3)\Omega$

D. $(4 \pm 0.3)\Omega$

Answer: C

Solution:

Solution:

Given,

$$R_1 = (4 \pm 0.8)\Omega$$

$$R_2 = (4 \pm 0.4)\Omega$$

Equivalent resistance when the resistors are connected in parallel is given by

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{4}$$

$$R_{eq} = 2\Omega$$

Now,

$$\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

Substituting the values in the above equation, we get

$$\frac{\Delta R_{eq}}{4} = \frac{0.8}{16} + \frac{0.4}{16}$$

$$\Delta R_{eq} = 0.3\Omega$$

∴ The equivalent resistance in parallel combination is

$$R_{eq} = (2 \pm 0.3)\Omega.$$

Question 146

A student determined Young's modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$.

The value of g is taken to be 9.8m/s^2 , without any significant error, his observations are as following.

Physical quantity	Least count of the equipment used for measurement	Observed value
Mass (M)	1 g	2 kg
Length of bar (L)	1mm	1m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

Then, the fractional error in the measurement of Y is
[1 Sep 2021 Shift 2]

Options:

- A. 0.0083
- B. 0.0155
- C. 0.155
- D. 0.083

Answer: B

Solution:

Solution:

The given formula of Young's modulus of elasticity,

$$Y = \frac{mgL^3}{4bd^3\delta}$$

where, Y = Young's modulus of elasticity,

m = mass of the bar,

L = length of the bar

b = breadth of the bar
d = thickness of the bar

and δ = depression of the bar.

There is no error in the value of the g.

The fractional error in the measurement of Y,

$$\frac{\Delta Y}{Y} = \frac{\Delta M}{M} + 3 \frac{\Delta L}{L} + \frac{\Delta b}{b} + 3 \frac{\Delta d}{d} + \frac{\Delta \delta}{\delta}$$

Substituting the values in the above expression, we get

$$\frac{\Delta Y}{Y} = \frac{10^{-3}}{2} + 3 \frac{(1 \times 10^{-3})}{1} + \frac{0.1 \times 10^{-3}}{4 \times 10^{-2}} + 3 \frac{(0.01 \times 10^{-3})}{0.4 \times 10^{-2}} + \frac{(0.01 \times 10^{-3})}{5 \times 10^{-3}}$$

$$\frac{\Delta Y}{Y} = 0.0155$$

The fractional error in the measurement of the Young's modulus is 0.0155.

Question147

If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is:

[9 Jan. 2020 I]

Options:

- A. 0.001 cm
- B. 0.02 mm
- C. 0.01 cm
- D. 0.001 mm

Answer: A

Solution:

Solution:

When screw on a screw-gauge is given six rotations, it moves by 3mm on the main scale

$$\therefore \text{Pitch} = \frac{3}{6} = 0.5 \text{ mm}$$

$$\therefore \text{Least count L.C.} = \frac{\text{Pitch}}{\text{CSD}} = \frac{0.5 \text{ mm}}{50}$$

$$= \frac{1}{100} \text{ mm} = 0.01 \text{ mm} = 0.001 \text{ cm}$$

Question148

For the four sets of three measured physical quantities as given below. Which of the following options is correct?

(A) $A_1 = 24.36$, $B_1 = 0.0724$, $C_1 = 256.2$

(B) $A_2 = 24.44$, $B_2 = 16.082$, $C_2 = 240.2$

(C) $A_3 = 25.2$, $B_3 = 19.2812$, $C_3 = 236.183$

(D) $A_A = 25$, $B_A = 236.191$, $C_A = 19.5$

[9 Jan. 2020 II]

Options:

A.

$$A_4 + B_4 + C_4 < A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2$$

B.

$$A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$$

C.

$$A_4 + B_4 + C_4 < A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3$$

D. $A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2 < A_4 + B_4 + C_4$

E. None of the above

Answer: E

Solution:

Solution:

$$D_1 = A_1 + B_1 + C_1 = 24.36 + 0.0724 + 256.2 = 280.6$$

$$D_2 = A_2 + B_2 + C_2 = 24.44 + 16.082 + 240.2 = 280.7$$

$$D_3 = A_3 + B_3 + C_3 = 25.2 + 19.2812 + 236.183 = 280.7$$

$$D_4 = A_4 + B_4 + C_4 = 25 + 236.191 + 19.5 = 281$$

Question149

A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the

pendulum is 25.0 cm and a stop watch with 1 s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is:
[8 Jan. 2020 II]

Options:

A. 5.40%

B. 3.40%

C. 4.40%

D. 2.40%

Answer: C

Solution:

Solution:

Given, Length of simple pendulum, $l = 25.0$ cm

Time of 40 oscillation, $T = 50$ s

Time period of pendulum $T = 2\pi \sqrt{\frac{l}{g}}$

$$\Rightarrow T^2 = \frac{4\pi^2 l}{g} \Rightarrow g = \frac{4\pi^2 l}{T^2}$$

$$\Rightarrow \text{Fractional error in } g = \frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

$$\Rightarrow \frac{\Delta g}{g} = \left(\frac{0.1}{25.0} \right) + 2 \left(\frac{1}{50} \right) = 0.044$$

$$\therefore \text{Percentage error in } g = \frac{\Delta g}{g} \times 100 = 4.4\%$$

Question150

Quantities The quantities $x = l\sqrt{\mu_0 \epsilon_0}$, $y = \frac{E}{B}$ and $z = \frac{1}{CR}$ are defined where C -capacitance, R -Resistance, l -length, E -Electric field, B -magnetic field and ϵ_0 , μ_0 , – free space permittivity and permeability respectively. Then :
[Sep. 05,2020 (II)]

Options:

A. x, y and z have the same dimension.

- B. Only x and z have the same dimension.
- C. Only x and y have the same dimension.
- D. Only y and z have the same dimension.

Answer: A

Solution:

Solution:

(a) We know that

$$\text{Speed of light, } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = x$$

$$\text{Also, } c = \frac{E}{B} = y$$

$$\text{Time constant, } \tau = Rc = t$$

$$\therefore z = \frac{1}{Rc} = \frac{1}{t} = \text{Speed}$$

Thus, x, y, z will have the same dimension of speed.

Question151

Dimensional formula for thermal conductivity is (here K denotes the temperature :

[Sep. 04, 2020 (I)]

Options:

- A. $MLT^{-2}K$
- B. $MLT^{-2}K^{-2}$
- C. $MLT^{-3}K$
- D. $MLT^{-3}K^{-1}$

Answer: D

Solution:

Solution:

From formula, $\frac{dQ}{dt} = kA \frac{dT}{dx}$

$$\Rightarrow k = \frac{\left(\frac{dQ}{dt}\right)}{A \left(\frac{dT}{dx}\right)}$$

$$[k] = \frac{[ML^2T^{-3}]}{[L^2][KL^{-1}]} = [MLT^{-3}K^{-1}]$$

Question152

**A quantity x is given by $\left(\frac{IFv^2}{WL^4}\right)$ in terms of moment of inertia I , force F , velocity v, work W and Length L. The dimensional formula for x is same as that of :
[Sep. 04, 2020 (II)]**

Options:

- A. planck's constant
- B. force constant
- C. energy density
- D. coefficient of viscosity

Answer: C

Solution:

Solution:

Dimension of Force $F = M^1L^1T^{-2}$

Dimension of velocity $V = L^1T^{-1}$

Dimension of work $= M^1L^2T^{-2}$

Dimension of length $= L$

Moment of inertia $= ML^2$

$$\therefore x = \frac{IFv^2}{WL^4}$$

$$= \frac{(M^1L^2)(M^1L^1T^{-2})(L^1T^{-1})^2}{(M^1L^2T^{-2})(L^4)}$$

$$= \frac{M^1L^{-2}T^{-2}}{L^3} = M^1L^{-1}T^{-2} = \text{Energy density}$$

Question153

Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is :

[Sep. 03, 2020 (II)]

Options:

A. $M L^2 T^{-2}$

B. $M L^0 T^{-3}$

C. $M^2 L^0 T^{-1}$

D. $M L T^{-2}$

Answer: B

Solution:

Solution:

$$\text{Solar constant} = \frac{\text{Energy}}{\text{Time Area}}$$

$$\text{Dimension of Energy, } E = M L^2 T^{-2}$$

$$\text{Dimension of Time} = T$$

$$\text{Dimension of Area} = L^2$$

$$\therefore \text{Dimension of Solar constant} = \frac{M^1 L^2 T^{-2}}{T L^2} = M^1 L^0 T^{-3}$$

Question154

If speed V, area A and force F are chosen as fundamental units, then the dimension of Young's modulus will be :

[Sep. 02, 2020 (I)]

Options:

A. $F A^2 V^{-1}$

B. $F A^2 V^{-3}$

C. $F A^2 V^{-2}$

D. $F A^{-1} V^0$

Answer: D

Solution:

Solution:

Young's modulus, $Y = \frac{\text{stress}}{\text{strain}}$

$$\Rightarrow Y = \frac{F/A}{\Delta l_0/l_0} = F A^{-1} V^0$$

Question 155

If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is : [Sep. 02, 2020 (II)]

Options:

A. $[P^2 A T^{-2}]$

B. $[P A^{-1} T^{-2}]$

C. $\left[P A^{\frac{1}{2}} T^{-1} \right]$

D. $\left[P^{\frac{1}{2}} A T^{-1} \right]$

Answer: C

Solution:

Solution:

Energy, $E \propto A^a T^b P^c$

Or, $E = k A^a T^b P^c \dots (i)$

where k is a dimensionless constant and a, b and c are the exponents.

Dimension of momentum, $P = M^1 L^1 T^{-1}$

Dimension of area, $A = L^2$

Dimension of time, $T = T^1$

Putting these value in equation (i), we get

$$M^1 L^2 T^{-2} = M^c L^{2a+c} T^{b-c}$$

by comparison

$$c = 1$$

$$2a + c = 2$$

$$b - c = -2$$

$$c = 1, a = \frac{1}{2}, b = -1$$

$$\therefore E = A^{\frac{1}{2}} T^{-1} P^1$$

Question156

A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale. The nature of zero error involved, and the least count of the screw gauge, are respectively :

[Sep. 06, 2020 (I)]

Options:

A. Negative, 2 mm

B. Positive, 10 mm

C. Positive, 0.1 mm

D. Positive, 0.1 mm

Answer: B

Solution:

Solution:

Given : No. of division on circular scale of screw gauge = 50

Pitch = 0.5 mm

$$\text{Least count of screw gauge} = \frac{\text{Pitch}}{\text{No. of division on circular scale}}$$

$$= \frac{0.5}{50} \text{ mm} = 1 \times 10^{-5} \text{ m} = 10 \mu\text{m}$$

And nature of zero error is positive.

Question157

The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is _____.

[NA Sep. 06, 2020 (I)]

Answer: 1050

Solution:

Solution:

$$\text{Density, } \rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi\left(\frac{D}{2}\right)^3} \Rightarrow \rho = \frac{6}{\pi} M D^{-3}$$

$$\therefore \% \left(\frac{\Delta \rho}{\rho} \right) = \frac{\Delta m}{m} + 3 \left(\frac{\Delta D}{D} \right) = 6 + 3 \times 1.5 = 10.5\%$$

$$\% \left(\frac{\Delta \rho}{\rho} \right) = \frac{1050}{100} \% = \left(\frac{x}{100} \right) \%$$

$$\therefore x = 1050.00$$

Question158

A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, 5.65 mm, The average of these four reading is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as :

[Sep. 06, 2020 (II)]

Options:

A. $(5.5375 \pm 0.0739) \text{ mm}$

B. (5.5375 ± 0.0740) mm

C. (5.538 ± 0.074) mm

D. (5.54 ± 0.07) mm

Answer: D

Solution:

Solution:

Average diameter, $d_{av} = 5.5375$ mm

Deviation of data, $\Delta d = 0.07395$ mm

As the measured data are upto two digits after decimal, therefore answer should be in two digits after decimal.

$\therefore d = (5.54 \pm 0.07)$ mm

Question159

A physical quantity z depends on four observables a , b , c and d , as

$z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c} d^3}$. The percentages of error in the measurement of a , b , c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is :

[Sep. 05, 2020 (I)]

Options:

A. 12.25%

B. 16.5%

C. 13.5%

D. 14.5%

Answer: D

Solution:

Solution:

$$\text{Given : } z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c} d^3}$$

$$\begin{aligned} \text{Percentage error in } Z, &= \frac{\Delta Z}{Z} = \frac{2\Delta a}{a} + \frac{2}{3} \frac{\Delta b}{b} + \frac{1}{2} \frac{\Delta c}{c} + \frac{3\Delta d}{d} \\ &= 2 \times 2 + \frac{2}{3} \times 1.5 + \frac{1}{2} \times 4 + 3 \times 2.5 = 14.5\% \end{aligned}$$

Question160

Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as :

[Sep. 03, 2020 (I)]

Options:

- A. 2.121 cm
- B. 2.124 cm
- C. 2.125 cm
- D. 2.123 cm

Answer: A

Solution:

Solution:

Thickness = M.S. Reading + Circular Scale Reading (L.C.)

$$\text{Here LC} = \frac{\text{Pitch}}{\text{Circular scale division}} = \frac{0.1}{50} = 0.002 \text{ cm per division}$$

So, correct measurement is measurement of integral multiple of L.C.

Question161

The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main

scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division)
[Sep. 02, 2020 (I)]

Options:

- A. 3.2 cm
- B. 3.21 cm
- C. 3.07 cm
- D. 2.99 cm

Answer: C

Solution:

Solution:

$$\text{L.C. of vernier callipers} = 1 \text{ MSD} - 1 \text{ VSD} = \left(1 - \frac{9}{10}\right) \times 1 = 0.1 \text{ mm} = 0.01 \text{ cm}$$

Here 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale.

Zero error = $7 \times 0.1 = 0.7 \text{ mm} = 0.07 \text{ cm}$.

Length of the cylinder = measured value – zero error = $(3.1 + 4 \times 0.01) - 0.07 = 3.07 \text{ cm}$.

Question162

The density of a material in SI unit is 128 kg m^{-3} . In certain units in which the unit of length is 25 cm and the unit of mass is 50g, the numerical value of density of the material is:
[10 Jan. 2019 I]

Options:

- A. 40
- B. 16
- C. 640
- D. 410

Answer: A

Solution:

Solution:

Density of material in SI unit, = 128 kg/m^3

Density of material in new system = $\frac{128(50\text{g})(20)}{(25 \text{ cm})^3(4)^3} = \frac{128}{64}(20) = 40 \text{ units}$

Question163

Let l , r , c and v represent inductance, resistance, capacitance and voltage, respectively. The dimension of $\frac{1}{rcv}$ in SI units will be:

[12 Jan. 2019 II]

Options:

A. $[LA^{-2}]$

B. $[A^{-1}]$

C. $[LT A]$

D. $[LT^2]$

Answer: B

Solution:

Solution:

As we know, $\left[\frac{1}{r}\right] = [T]$ and $[cv] = [AT]$

$\therefore \left[\frac{1}{rcv}\right] = \left[\frac{T}{AT}\right] = [A^{-1}]$

Question164

The force of interaction between two atoms is given by

$F = \alpha\beta \exp\left(-\frac{x^2}{\alpha kT}\right)$; where x is the distance, k is the Boltzmann

constant and T is temperature and α and β are two constants.
 The dimensions of beta is:
 [11 Jan. 2019 I]

Options:

A. $M^0 L^2 T^{-4}$

B. $M^2 L T^{-4}$

C. $M L T^{-2}$

D. $M^2 L^2 T^{-2}$

Answer: B

Solution:

Solution:

Force of interaction between two atoms, $F = \alpha \beta e^{\left(\frac{-x^2}{\alpha k T}\right)}$
 Since exponential terms are dimensionless

$$\therefore \left[\frac{x^2}{\alpha k T} \right] = M^0 L^0 T^0$$

$$\Rightarrow \frac{L^2}{[\alpha] M L^2 T^{-2}} = M^0 L^0 T^0$$

$$\Rightarrow [\alpha] = M^{-1} T^2$$

$$[F] = [\alpha][\beta]$$

$$M L T^{-2} = M^{-1} T^2 [\beta]$$

$$\Rightarrow [\beta] = M^2 L T^{-4}$$

Question165

If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be:
 [11 Jan. 2019 II]

Options:

A. $V^{-2} A^2 F^{-2}$

B. $V^{-2} A^2 F^2$

C. $V^{-4}A^{-2}F$

D. $V^{-4}A^2F$

Answer: D

Solution:

Solution:

$$\text{Let } [Y] = [V]^a [F]^b [A]^c$$

$$[M L^{-1} T^{-2}] = [L T^{-1}]^a [M L T^{-2}]^b [L T^{-2}]^c$$

$$[M L^{-1} T^{-2}] = [M^b L^{a+b+c} T^{-a-2b-2c}]$$

Comparing power both side of similar terms we get,

$$b = 1, a + b + c = -1, -a - 2b - 2c = -2$$

solving above equations we get:

$$a = -4, b = 1, c = 2$$

$$\text{so } [Y] = [V^{-4} F A^2] = [V^{-4} A^2 F]$$

Question 166

A quantity f is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed of light, G

universal gravitational constant and h is the Planck's constant.

Dimension of f is that of :

[9 Jan. 2019 I]

Options:

A. area

B. energy

C. momentum

D. volume

Answer: B

Solution:

Solution:

$$\text{Dimension of } [h] = [M L^2 T^{-1}]$$

$$[G] = [L T^{-2}]$$

$$[G] = [M^{-1}L^3T^{-2}]$$

$$\begin{aligned} \text{Hence dimension of } \left[\sqrt{\frac{hC^5}{G}} \right] &= \frac{[ML^2T^{-1}] \cdot [L^5T^{-5}]}{[M^{-1}L^3T^{-2}]} \\ &= [ML^2T^{-2}] = \text{energy} \end{aligned}$$

Question 167

Expression for time in terms of G (universal gravitational constant), h (Planck's constant) and c (speed of light) is proportional to:
[9 Jan. 2019 II]

Options:

A. $\sqrt{\frac{hc^5}{G}}$

B. $\sqrt{\frac{c^3}{Gh}}$

C. $\sqrt{\frac{Gh}{c^5}}$

D. $\sqrt{\frac{Gh}{c^3}}$

Answer: C

Solution:

Solution:

$$\text{Let } t \propto G^x h^y C^z$$

$$\text{Dimensions of } G = [M^{-1}L^3T^{-2}]$$

$$h = [ML^2T^{-1}] \text{ and } C = [LT^{-1}]$$

$$[T] = [M^{-1}L^3T^{-2}]^x [ML^2T^{-1}]^y [LT^{-1}]^z$$

$$[M^0L^0T^1] = [M^{-x+y}L^{3x+2y+z}T^{-2x-y-z}]$$

By comparing the powers of M, L, T both the sides

$$-x + y = 0 \Rightarrow x = y$$

$$3x + 2y + z = 0 \Rightarrow 5x + z = 0 \dots\dots(i)$$

$$-2x - y - z = 1 \Rightarrow 3x + z = -1 \dots\dots(ii)$$

Solving eqns. (i) and (ii),

$$x = y = \frac{1}{2}, z = -\frac{5}{2}$$

$$\therefore t \propto \sqrt{\frac{Gh}{C^5}}$$

Question 168

The dimensions of stopping potential V_0 in photoelectric effect in units of Planck's constant 'h', speed of light 'c' and Gravitational constant 'G' and ampere A is:

[8 Jan. 2019 I]

Options:

A. $h^{\frac{1}{3}} G^{\frac{2}{3}} c^{\frac{1}{3}} A^{-1}$

B. $h^{\frac{2}{3}} c^{\frac{5}{3}} G^{\frac{1}{3}} A^{-1}$

C. $h^{-\frac{2}{3}} e^{-\frac{1}{3}} G^{\frac{4}{3}} A^{-1}$

D. $h^0 A^{-1} G^{-1} C^5$

Answer: D

Solution:

Solution:

$$\text{Stopping potential } (V_0) \propto h^x A^y G^z C^r$$

$$\text{Here, } h = \text{Planck's constant} = [ML^2T^{-1}]$$

$$I = \text{current} = [A]$$

$$G = \text{Gravitational constant} = [M^{-1}L^3T^{-2}]$$

$$\text{and } c = \text{speed of light} = [LT^{-1}]$$

$$V_0 = \text{potential} = [ML^2T^{-3}A^{-1}]$$

$$\therefore [ML^2T^{-3}A^{-1}] = [ML^2T^{-1}]^x [A]^y [M^{-1}L^3T^{-2}]^z [LT^{-1}]^r$$

$$M^{x-z}; L^{2x+3z+r}; T^{-x-2z-r}; A^y$$

Comparing dimension of M, L, T, A, we get

$$y = -1, x = 0, z = -1, r = 5$$

$$\therefore V_0 \propto h^0 A^{-1} G^{-1} C^5$$

Question169

The dimensions of $\frac{B^2}{2\mu_0}$, where B is magnetic field and μ_0 is the magnetic permeability of vacuum, is:
[8 Jan. 2019 II]

Options:

- A. $M L T^{-2}$
- B. $M L^2 T^{-1}$
- C. $M L^2 T^{-2}$
- D. $M L^{-1} T^{-2}$

Answer: D

Solution:

Solution:

The quantity $\frac{B^2}{2\mu_0}$ is the energy density of magnetic field

$$\Rightarrow \left[\frac{B^2}{2\mu_0} \right] = \frac{\text{Energy}}{\text{Volume}} = \frac{\text{Force} \times \text{displacement}}{(\text{displacement})^3} = \left[\frac{M L^2 T^{-2}}{L^3} \right] = M L^{-1} T^{-2}$$

Question170

The least count of the main scale of a screw gauge is 1 mm. The minimum number of divisions on its circular scale required to measure 5 μm diameter of a wire is:
[12 Jan. 2019 I]

Options:

- A. 50
- B. 200
- C. 100

D. 500

Answer: B

Solution:

Solution:

Least count of main scale of screw gauge = 1 mm

Least count of screw gauge = $\frac{\text{Pitch}}{\text{Number of division on circular scale}}$

$$5 \times 10^{-6} = \frac{10^{-3}}{N}$$

$$\Rightarrow N = 200$$

Question171

**The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figures?
[10 Jan. 2019 II]**

Options:

A. $4264 \pm 81 \text{ cm}^3$

B. $4264.4 \pm 81.0 \text{ cm}^3$

C. $4260 \pm 80 \text{ cm}^3$

D. $4300 \pm 80 \text{ cm}^3$

Answer: C

Solution:

Solution:

$$v = \frac{\pi d^2}{4} h = 4260 \text{ cm}^3$$

$$\frac{\Delta v}{v} = \frac{2 \Delta d}{d} + \frac{\Delta h}{h}$$

$$\Delta v = 2 \times \frac{0.1v}{12.6} + \frac{0.1v}{34.2} = \frac{0.2}{126} \times 4260 + \frac{0.1 \times 4260}{34.2} = 80$$

$$\therefore \text{Volume} = 4260 \pm 80 \text{ cm}^3.$$

Question172

The pitch and the number of divisions, on the circular scale for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 division below the mean line.

The readings of the main scale and the circular scale, for a thin sheet, are 5.5 mm and 48 respectively, the thickness of the sheet is:
[9 Jan. 2019 II]

Options:

A. 5.755 mm

B. 5.950 mm

C. 5.725 mm

D. 5.740 mm

Answer: C

Solution:

Solution:

Least count of screw gauge,

$$LC = \frac{\text{Pitch}}{\text{No. of division}}$$

$$= 0.5 \times 10^{-3} = 0.5 \times 10^{-2} \text{ mm} + \text{ve error}$$

$$= 3 \times 0.5 \times 10^{-2} \text{ mm} = 1.5 \times 10^{-2} \text{ mm} = 0.015 \text{ mm}$$

$$\begin{aligned} \text{Reading} &= \text{MSR} + \text{CSR} - (+\text{ve error}) = 5.5 \text{ mm} + (48 \times 0.5 \times 10^{-2}) - 0.015 \\ &= 5.5 + 0.24 - 0.015 = 5.725 \text{ mm} \end{aligned}$$

Question173

Which of the following combinations has the dimension of electrical resistance (ϵ_0 is the permittivity of vacuum and μ_0 is the permeability of vacuum)?

[12 April 2019 I]

Options:

A. $\sqrt{\frac{\mu_0}{\epsilon_0}}$

B. $\frac{\mu_0}{\epsilon_0}$

C. $\sqrt{\frac{\epsilon_0}{\mu_0}}$

D. $\frac{\epsilon_0}{\mu_0}$

Answer: A

Solution:

Solution:

$$\sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{\mu_0^2}{\epsilon_0 \mu_0}} = \mu_0 c \left(\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c \right)$$

$$\mu_0 c \rightarrow \text{MLT}^{-2}\text{A}^{-2} \times \text{LT}^{-1}$$

$$\text{ML}^2\text{T}^{-3}\text{A}^{-2}$$

Dimensions of resistance

Question174

In the formula $X = 5Y Z^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units ?

[10 April 2019 II]

Options:

A. $[\text{M}^{-3}\text{L}^{-2}\text{T}^8\text{A}^4]$

B. $[\text{M}^{-1}\text{L}^{-2}\text{T}^4\text{A}^2]$

C. $[\text{M}^{-2}\text{L}^0\text{T}^{-4}\text{A}^{-2}]$

D. $[\text{M}^{-2}\text{L}^{-2}\text{T}^6\text{A}^3]$

Answer: A

Solution:

Solution:

$$X = 5Y Z^2$$

$$\Rightarrow Y \propto \frac{X}{Z^2} \dots (i)$$

$$X = \text{Capacitance} = \frac{Q}{V} = \frac{Q^2}{W} = \frac{[A^2 T^2]}{[M L^2 T^{-2}]}$$

$$X = [M^{-1} L^{-2} T^4 A^2]$$

$$Z = B = \frac{F}{IL} [\because F = I L B]$$

$$Z = [M T^{-2} A^{-1}]$$

$$Y = \frac{[M^{-1} L^{-2} T^4 A^2]}{[M T^{-2} A^{-1}]^2}$$

$$Y = [M^{-3} L^{-2} T^8 A^4] \text{ (Using (i))}$$

Question 175

In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is:

[8 April 2019 I]

Options:

A. $A^{-1} T M L^3$

B. $A T^2 M^{-1} L^{-1}$

C. $A T^{-3} M L^{3/2}$

D. $A^2 T^3 M^{-1} L^{-2}$

Answer: D

Solution:

Solution:

$$\left[\sqrt{\frac{\epsilon_0}{\mu_0}} \right] = \sqrt{\frac{\epsilon_0^2}{\mu_0 \epsilon_0}} = \left[\frac{\epsilon_0}{\sqrt{\mu_0 \epsilon_0}} \right] = \epsilon_0 C [L T^{-1}] \times [\epsilon_0] \left[\because \frac{1}{\sqrt{\mu_0 \epsilon_0}} = C \right]$$

$$\because F = \frac{q^2}{4\pi\epsilon_0 r^2}$$

$$\Rightarrow [\varepsilon_0] = \frac{[AT]^2}{[MLT^{-2}] \times [L^2]} = [A^2 M^{-1} L^{-3} T^4]$$

$$\therefore \left[\sqrt{\frac{\varepsilon_0}{\mu_0}} \right] = [LT^{-1}] \times [A^2 M^{-1} L^{-3} T^4]$$

$$= [M^{-1} L^{-2} T^3 A^2]$$

Question176

In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is:

[9 April 2019 I]

Options:

- A. 0.01 kg/m^3
- B. 0.10 kg/m^3
- C. 0.013 kg/m^3
- D. 0.31 kg/m^3

Answer: D

Solution:

Solution:

$$\delta = \frac{M}{V} = \frac{M}{l^3} = M l^{-3}$$

$$\frac{\Delta \delta}{\delta} = \frac{\Delta M}{M} + 3 \frac{\Delta l}{l} = \frac{0.10}{10.00} + 3 \left(\frac{0.01}{0.10} \right) = 0.31 \text{ kg/m}^3$$

Question177

The area of a square is 5.29 cm^2 . The area of 7 such squares taking into account the significant figures is:

[9 April 2019 II]

Options:

A. 37 cm^2

B. 37.030 cm^2

C. 37.03 cm^2

D. 37.0 cm^2

Answer: D

Solution:

Solution:

$$A = 7 \times 5.29 = 37.03 \text{ cm}^2$$

The result should have three significant figures, so $A = 37.0 \text{ cm}^2$

Question178

In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to :

[8 April 2019 II]

Options:

A. 0.7%

B. 0.2%

C. 3.5%

D. 6.8%

Answer: D

Solution:

Solution:

We have

$$T = 2\pi \sqrt{\frac{l}{g}} \text{ or } g = 4\pi^2 \frac{l}{T^2}$$

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta R}{R} \times 100 + 2 \frac{\Delta T}{T} \times 100$$

$$= \frac{0.1}{55} \times 100 + 2 \left(\frac{1}{30} \right) \times 100$$

$$= 0.18 + 6.67 = 6.8\%$$

Question179

The characteristic distance at which quantum gravitational effects are significant, the Planck length, can be determined from a suitable combination of the fundamental physical constants G, h and c.

Which of the following correctly gives the Planck length?

[Online April 15, 2018]

Options:

A. G^2hc

B. $\left(\frac{Gh}{c^3} \right)^{\frac{1}{2}}$

C. $G^{\frac{1}{2}}h^2c$

D. Gh^2c^3

Answer: B

Solution:

Solution:

Planck length is a unit of length, $l_p = 1.616229 \times 10^{-35} \text{m}$

$$l_p = \sqrt{\frac{hG}{c^3}}$$

Question180

The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is:
[2018]

Options:

- A. 2.5%
- B. 3.5%
- C. 4.5%
- D. 6%

Answer: C

Solution:

Solution:
 $1.5\% + 3(1\%) = 4.5\%$

Question181

The percentage errors in quantities P, Q, R and S are 0.5% 1%, 3% and 1.5% respectively in the measurement of a physical quantity

$$A = \frac{P^3 Q^2}{\sqrt{RS}}.$$

The maximum percentage error in the value of A will be
[Online April 16, 2018]

Options:

- A. 8.5%
- B. 6.0%
- C. 7.5%
- D. 6.5%

Answer: D

Solution:

Solution:

Maximum percentage error in A = + 3(% error in P) 2(% error in Q) + $\frac{1}{2}$ (% error in R) + 1(% error in S)

$$= 3 \times 0.5 + 2 \times 1 + \frac{1}{2} \times 3 + 1 \times 1.5$$
$$= 1.5 + 2 + 1.5 + 1.5 = 6.5\%$$

Question182

The relative uncertainty in the period of a satellite orbiting around the earth is 10^{-2} . If the relative uncertainty in the radius of the orbit is negligible, the relative uncertainty in the mass of the earth is [Online April 16, 2018]

Options:

A. 3×10^{-2}

B. 10^{-2}

C. 2×10^{-2}

D. 6×10^{-2}

Answer: C

Solution:

Solution:

From Kepler's law, time period of a satellite,

$$T = 2\pi \sqrt{\frac{r^3}{Gm}} \Rightarrow T^2 = \frac{4\pi^2}{GM} r^3$$

Relative uncertainty in the mass of the earth

$$\left| \frac{\Delta M}{M} \right| = 2 \frac{\Delta T}{T} = 2 \times 10^{-2}$$

($\because 4\pi$ & G constant and relative uncertainty in radius $\frac{\Delta r}{r}$ negligible)

Question183

The relative error in the determination of the surface area of a sphere is α . Then the relative error in the determination of its volume is

[Online April 15, 2018]

Options:

A. $\frac{2}{3}\alpha$

B. 2α

C. $\frac{3}{2}\alpha$

D. α

Answer: C

Solution:

Solution:

Relative error in Surface area, $\frac{\Delta s}{s} = 2 \times \frac{\Delta r}{r} = \alpha$ and relative error in volume, $\frac{\Delta v}{v} = 3 \times \frac{\Delta r}{r}$

$\therefore$ Relative error in volume w.r.t. relative error in area, $\frac{\Delta v}{v} = \frac{3}{2}\alpha$

Question184

In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm. There are 100 circular scale divisions.

The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions.

Assuming negligible zero error, the thickness of the wire is:

[Online April 15, 2018]

Options:

A. 0.0430 cm

B. 0.3150 cm

C. 0.4300 cm

D. 0.2150 cm

Answer: D

Solution:

Solution:

$$\begin{aligned}\text{Least count} &= \frac{\text{Value of 1 part on main scale}}{\text{Number of parts on vernier scale}} \\ &= \frac{0.25}{5 \times 100} \text{ cm} = 5 \times 10^{-4} \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Reading} &= 4 \times 0.05 \text{ cm} + 30 \times 5 \times 10^{-4} \text{ cm} \\ &= (0.2 + 0.0150) \text{ cm} = 0.2150 \text{ cm (Thickness of wire)}\end{aligned}$$

Question185

**Time (T), velocity (C) and angular momentum (h) are chosen as fundamental quantities instead of mass, length and time. In terms of these, the dimensions of mass would be :
[Online April 8, 2017]**

Options:

A. $[M] = [T^{-1}C^{-2}h]$

B. $[M] = [T^{-1}C^2h]$

C. $[M] = [T^{-1}C^{-2}h^{-1}]$

D. $[M] = [T C^{-2}h]$

Answer: A

Solution:

Solution:

$$\begin{aligned}\text{Let mass, related as } M &\propto T^x C^y h^z \\ M^1 L^0 T^0 &= (T^1)^x (L^1 T^{-1})^y (M^1 L^2 T^{-1})^z \\ M^1 L^0 T^0 &= M^z L^{y+2z} T^{x-y-z} \\ \therefore z &= 1 \\ \Rightarrow y + 2z &= 0 \\ \therefore y &= -2 \\ x - y - z &= 0 \\ x + 2 - 1 &= 0\end{aligned}$$

$$\therefore x = -1$$

$$M = [T^{-1}C^{-2}h^1]$$

Question186

The following observations were taken for determining surface tension T of water by capillary method :

Diameter of capillary, $D = 1.25 \times 10^{-2} \text{ m}$

rise of water, $h = 1.45 \times 10^{-2} \text{ m}$

Using $g = 9.80 \text{ m/s}^2$ and the simplified relation $T = \frac{r h g}{2} \times 10^3 \text{ N/m}$, the possible error in surface tension is closest to :
[2017]

Options:

A. 2.4 %

B. 10 %

C. 0.15%

D. 1.5%

Answer: D

Solution:

Solution:

Surface tension, $T = \frac{r h g}{2} \times 10^3$

Relative error in surface tension,

$$\frac{\Delta T}{T} = \frac{\Delta r}{r} + \frac{\Delta h}{h} + 0 \quad (\because g, 2 \times 10^3 \text{ are constant})$$

Percentage error

$$\begin{aligned} 100 \times \frac{\Delta T}{T} &= \left(\frac{10^{-2} \times 0.01}{1.25 \times 10^{-2}} + \frac{10^{-2} \times 0.01}{1.45 \times 10^{-2}} \right) 100 \\ &= (0.8 + 0.689) \\ &= (1.489) = 1.489\% \cong 1.5\% \end{aligned}$$

Question187

A physical quantity P is described by the relation $P = a^{\frac{1}{2}} b^2 c^3 d^{-4}$
 If the relative errors in the measurement of a, b, c and d respectively, are 2%, 1%, 3% and 5%, then the relative error in P will be :
 [Online April 9, 2017]

Options:

- A. 8%
- B. 12%
- C. 32%
- D. 25%

Answer: C

Solution:

Solution:

$$\begin{aligned} \text{Given, } P &= a^{\frac{1}{2}} b^2 c^3 d^{-4} \\ \text{Maximum relative error,} \\ \frac{\Delta P}{P} &= \frac{1}{2} \frac{\Delta a}{a} + 2 \frac{\Delta b}{b} + 3 \frac{\Delta c}{c} + 4 \frac{\Delta d}{d} \\ &= \frac{1}{2} \times 2 + 2 \times 1 + 3 \times 3 + 4 \times 5 = 32\% \end{aligned}$$

Question188

A, B, C and D are four different physical quantities having different dimensions. None of them is dimensionless. But we know that the equation $AD = C \ln(BD)$ holds true. Then which of the combination is not a meaningful quantity ?
 [Online April 10, 2016]

Options:

- A. $\frac{C}{BD} - \frac{AD^2}{C}$
- B. $A^2 - B^2 C^2$

C. $AB - C$

D. $\frac{(A - C)}{D}$

Answer: D

Solution:

Solution:

Dimension of A $\neq$ dimension of (C)

Hence $A - C$ is not possible.

Question189

**In the following 'I' refers to current and other symbols have their usual meaning, Choose the option that corresponds to the dimensions of electrical conductivity :
[Online April 9, 2016]**

Options:

A. $M^{-1}L^{-3}T^3I$

B. $M^{-1}L^{-3}T^3I^2$

C. $M^{-1}L^3T^3I$

D. $ML^{-3}T^{-3}I^2$

Answer: B

Solution:

Solution:

We know that resistivity $\rho = \frac{RA}{l}$

Conductivity $= \frac{1}{\text{resistivity}} = \frac{1}{RA}$

$= \frac{lI}{VA} (\because V = RI)$

$= \frac{[L][I]}{\left[\frac{[ML^2T^{-2}]}{[I][T]} \right] \times [L^2]} \left[\because V = \frac{W}{q} = \frac{W}{it} \right]$

$= [M^{-1}L^{-3}T^3][I^2] = [M^{-1}L^{-3}T^3I^2]$

Question190

A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line?
[2016]

Options:

- A. 0.70 mm
- B. 0.50 mm
- C. 0.75 mm
- D. 0.80 mm

Answer: D

Solution:

Solution:

$$L.C. = \frac{0.5}{50} = 0.01 \text{ mm}$$

$$\text{Zero error} = 5 \times 0.01 = 0.05 \text{ mm (Negative)}$$

$$\text{Reading} = (0.5 + 25 \times 0.01) + 0.05 = 0.80 \text{ mm}$$

Question191

A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s, and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be:
[2016]

Options:

- A. $92 \pm 1.8 \text{ s}$
- B. $92 \pm 3\text{s}$
- C. $92 \pm 1.5 \text{ s}$
- D. $92 \pm 5.0 \text{ s}$

Answer: C

Solution:

Solution:

$$\Delta T = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4|}{4}$$
$$= \frac{2 + 1 + 3 + 0}{4} = 1.5$$

As the resolution of measuring clock is 1.5 therefore the mean time should be 92 ± 1.5

Question192

If electronic charge e , electron mass m , speed of light in vacuum c and Planck's constant h are taken as fundamental quantities, the permeability of vacuum μ_0 can be expressed in units of :

[Online April 11, 2015]

Options:

- A. $\left(\frac{h}{me^2} \right)$
- B. $\left(\frac{hc}{me^2} \right)$
- C. $\left(\frac{h}{ce^2} \right)$
- D. $\left(\frac{mc^2}{he^2} \right)$

Answer: C

Solution:

Solution:

Let μ_0 related with e , m , c and h as follows.

$$\mu_0 = ke^a m^b c^c h^d$$

$$[M L T^{-2} A^{-2}] = [A T]^a [M]^b [L T^{-1}]^c [M L^2 T^{-1}]^d \\ = [M^{b+d} L^{c+2d} T^{a-c-d} A^a]$$

On comparing both sides we get

$$a = -2 \dots (i)$$

$$b + d = 1 \dots (ii)$$

$$c + 2d = 1 \dots (iii)$$

$$a - c - d = -2 \dots (iv)$$

By equation (i), (ii), (iii) & (iv) we get,

$$a = -2, b = 0, c = -1, d = 1$$

$$\therefore [\mu_0] = \left[\frac{h}{ce^2} \right]$$

Question193

If the capacitance of a nanocapacitor is measured in terms of a unit 'u' made by combining the electric charge 'e', Bohr radius 'a₀', Planck's constant 'h' and speed of light 'c' then:
[Online April 10, 2015]

Options:

A. $u = \frac{e^2 h}{a_0}$

B. $u = \frac{hc}{e^2 a_0}$

C. $u = \frac{e^2 c}{h a_0}$

D. $u = \frac{e^2 a_0}{hc}$

Answer: D

Solution:

Solution:

Let unit 'u' related with e, a₀, h and c as follows.

$$[u] = [e]^a [a_0]^b [h]^c [C]^d$$

Using dimensional method,

$$[M^{-1} L^{-2} T^{+4} A^{+2}] = [A^1 T^{-1}]^a [L]^b [M L^2 T^{-1}]^c [L T^{-1}]^d$$

$$[M^{-1} L^{-2} T^{+4} A^{+2}] = [M^c L^{b+2c+d} T^{a-c-d} A^a]$$

$$a = 2, b = 1, c = -1, d = -1$$

$$\therefore u = \frac{e^2 a_0}{hc}$$

Question194

The period of oscillation of a simple pendulum is $T = 2\pi \sqrt{\frac{L}{g}}$.

Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1s resolution. The accuracy in the determination of g is :
[2015]

Options:

A. 1%

B. 5%

C. 2%

D. 3%

Answer: D

Solution:

Solution:

$$\text{As, } g = 4\pi^2 \frac{L}{T^2}$$

$$\begin{aligned} \text{So, } \frac{\Delta g}{g} \times 100 &= \frac{\Delta L}{L} \times 100 + 2 \frac{\Delta T}{T} \times 100 \\ &= \frac{0.1}{20} \times 100 + 2 \times \frac{1}{90} \times 100 = 2.72 \simeq 3\% \end{aligned}$$

Question195

Diameter of a steel ball is measured using a Vernier callipers which has divisions of 0.1 cm on its main scale (MS) and 10 divisions of its vernier scale (VS) match 9 divisions on the main scale. Three such measurements for a ball are given as:

S.No.	MS(cm)	VS divisions
1.	0.5	8
2.	0.5	4
3.	0.5	6

**If the zero error is – 0.03 cm, then mean corrected diameter is:
[Online April 10, 2015]**

Options:

A. 0.52 cm

B. 0.59 cm

C. 0.56 cm

D. 0.53 cm

Answer: B

Solution:

Solution:

$$\text{Least count} = \frac{0.1}{10} = 0.01 \text{ cm}$$

$$d_1 = 0.5 + 8 \times 0.01 + 0.03 = 0.61 \text{ cm}$$

$$d_2 = 0.5 + 4 \times 0.01 + 0.03 = 0.57 \text{ cm}$$

$$d_3 = 0.5 + 6 \times 0.01 + 0.03 = 0.59 \text{ cm}$$

$$\text{Mean diameter} = \frac{0.61 + 0.57 + 0.59}{3} = 0.59 \text{ cm}$$

Question196

From the following combinations of physical constants (expressed through their usual symbols) the only combination, that would have the same value in different systems of units, is:

[Online April 12, 2014]

Options:

A. $\frac{ch}{2\pi\epsilon_0^2}$

B. $\frac{e^2}{2\pi\epsilon_0 G m_e^2}$ (m_e = mass of electron)

C. $\frac{\mu_0 \epsilon_0 G}{c^2 h e^2}$

D. $\frac{2\pi\sqrt{\mu_0 \epsilon_0} h}{c e^2 G}$

Answer: B

Solution:

Solution:

The dimensional formulae of

$$e = [M^0 L^0 T^1 A^1]$$

$$\epsilon_0 = [M^{-1} L^3 T^4 A^2]$$

$$G = [M^{-1} L^3 T^{-2}] \text{ and } m_e = [M^1 L^0 T^0]$$

Now, $\frac{e^2}{2\pi\epsilon_0 G m_e^2}$

$$= \frac{[M^0 L^0 T^1 A^1]^2}{2\pi[M^{-1} L^3 T^4 A^2][M^{-1} L^3 T^{-2}][M^1 L^0 T^0]^2}$$

$$= \frac{[T^2 A^2]}{2\pi[M^{-1-1+2} L^{-3+3} T^{4-2} A^2]}$$

$$= \frac{[T^2 A^2]}{2\pi[M^0 L^0 T^2 A^2]} = \frac{1}{2\pi}$$

$\therefore \frac{1}{2\pi}$ is dimensionless thus the combination $\frac{e^2}{2\pi\epsilon_0 G m_e^2}$

Question197

In terms of resistance R and time T, the dimensions of ratio $\frac{\mu}{\epsilon}$ of the permeability μ and permittivity ϵ is:
[Online April 11, 2014]

Options:

A. $[RT^{-2}]$

B. $[R^2T^{-1}]$

C. $[R^2]$

D. $[R^2T^2]$

Answer: C

Solution:

Solution:

Dimensions of $\mu = [MLT^{-2}A^{-2}]$

Dimensions of $\epsilon = [M^{-1}L^{-3}T^4A^2]$

Dimensions of R = $[ML^2T^{-3}A^{-2}]$

$$\therefore \frac{\text{Dimensions of } \mu}{\text{Dimensions of } \epsilon} = \frac{[MLT^{-2}A^{-2}]}{[M^{-1}L^{-3}T^4A^2]} = [M^2L^4T^{-6}A^{-4}] = [R^2]$$

Question198

The current voltage relation of a diode is given by

$I = (e^{1000V/T} - 1) \text{ mA}$, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring $\pm 0.01V$ while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA?

[2014]

Options:

A. 0.2 mA

B. 0.02 mA

C. 0.5 mA

D. 0.05 mA

Answer: A

Solution:

Solution:

The current voltage relation of diode is

$$I = (e^{1000V/T} - 1) \text{ mA (given)}$$

$$\text{When, } I = 5 \text{ mA, } e^{1000V/T} = 6 \text{ mA}$$

$$\text{Also, } dI = (e^{1000V/T}) \times \frac{1000}{T}$$

$$\text{Error} = \pm 0.01 \text{ (By exponential function)}$$

$$= (6 \text{ mA}) \times \frac{1000}{300} \times (0.01) = 0.2 \text{ mA}$$

Question199

A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it? [2014]

Options:

- A. A meter scale.
- B. A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.
- C. A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.
- D. A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

Answer: B

Solution:

Solution:

Measured length of rod = 3.50 cm

For Vernier Scale with 1 Main Scale Division = 1 mm

9 Main Scale Division = 10 Vernier Scale Division,

Least count = 1 MSD – 1 VSD = 0.1 mm

Question200

Match List - I (Event) with List-II (Order of the time interval for happening of the event) and select the correct option from the options given below the lists:

List - I	List - II
(1) Rotation period of earth	(i) 10^5s
(2) Revolution period of earth	(ii) 10^7s
(3) Period of light wave	(iii) 10^{-15}s
(4) Period of sound wave	(iv) 10^{-3}s

[Online April 19, 2014]

Options:

- A. (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)
- B. (1)-(ii), (2)-(i), (3)-(iv), (4)-(iii)
- C. (1)-(i), (2)-(ii), (3)-(iv), (4)-(iii)
- D. (1)-(ii), (2)-(i), (3)-(iii), (4)-(iv)

Answer: A

Solution:

Solution:

Rotation period of earth is about 24 hrs $\approx 10^5\text{s}$

Revolution period of earth is about 365 days $\approx 10^7\text{s}$

Speed of light wave $C = 3 \times 10^8\text{m/s}$

Wavelength of visible light of spectrum $\lambda = 4000 - 7800\text{\AA}$

$$C = f\lambda \left(\text{and } T = \frac{1}{f} \right)$$

Therefore period of light wave is 10^{-15}s (approx)

Question201

**In the experiment of calibration of voltmeter, a standard cell of e.m.f. 1.1 volt is balanced against 440 cm of potential wire. The potential difference across the ends of resistance is found to balance against 220 cm of the wire. The corresponding reading of voltmeter is 0.5 volt. The error in the reading of voltmeter will be:
[Online April 12, 2014]**

Options:

- A. – 0. 15 volt
- B. 0.15 volt
- C. 0.5 volt
- D. – 0.05 volt

Answer: D

Solution:

Solution:

In a voltmeter

$$V \propto l$$

$$V = kl$$

Now, it is given $E = 1.1$ volt for $l_1 = 440$ cm and $V = 0.5$ volt for $l_2 = 220$ cm

Let the error in reading of voltmeter be ΔV then,

$$1.1 = 400K \text{ and } (0.5 - \Delta V) = 220K$$

$$\Rightarrow \frac{1.1}{440} = \frac{0.5 - \Delta V}{220}$$

$$\therefore \Delta V = -0.05 \text{ volt}$$

Question202

**An experiment is performed to obtain the value of acceleration due to gravity g by using a simple pendulum of length L . In this experiment time for 100 oscillations is measured by using a watch of 1 second least count and the value is 90.0 seconds. The length L is measured by using a meter scale of least count 1 mm and the value is 20.0 cm. The error in the determination of g would be:
[Online April 9, 2014]**

Options:

A. 1.7%

B. 2.7%

C. 4.4%

D. 2.27%

Answer: B

Solution:

Solution:

According to the question.

$$t = (90 \pm 1) \text{ or, } \frac{\Delta t}{t} = \frac{1}{90}$$

$$l = (20 \pm 0.1) \text{ or, } \frac{\Delta l}{l} = \frac{0.1}{20}$$

$$\frac{\Delta g}{g} \% = ?$$

As we know,

$$t = 2\pi \sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{t^2}$$

$$\text{or, } \frac{\Delta g}{g} = \pm \left(\frac{\Delta l}{l} + 2 \frac{\Delta t}{t} \right) = \left(\frac{0.1}{20} + 2 \times \frac{1}{90} \right) = 0.027$$

$$\therefore \frac{\Delta g}{g} \% = 2.7\%$$

Question203

A metal sample carrying a current along X -axis with density J_x is subjected to a magnetic field B_z (along z-axis). The electric field E_y developed along Y-axis is directly proportional to J_x as well as B_z .

**The constant of proportionality has SI unit
[Online April 25, 2013]**

Options:

A. $\frac{m^2}{A}$

B. $\frac{m^3}{As}$

C. $\frac{\text{m}^2}{\text{As}}$

D. $\frac{\text{As}}{\text{m}^3}$

Answer: B

Solution:

Solution:

(b) According to question $E_y \propto J_x B_z \therefore$ Constant of proportionality

$$K = \frac{E_y}{B_z J_x} = \frac{C}{J_x} = \frac{\text{m}^3}{\text{As}}$$

[As $\frac{E}{B} = C$ (speed of light) and $J = I \text{ Area}$]

Question204

Let ϵ_0 denote the dimensional formula of the permittivity of vacuum. If **M** = mass, **L** = length, **T** = time and **A** = electric current, then:

[2013]

Options:

A. $\epsilon_0 = [\text{M}^{-1}\text{L}^{-3}\text{T}^2\text{A}]$

B. $\epsilon_0 = [\text{M}^{-1}\text{L}^{-3}\text{T}^4\text{A}^2]$

C. $\epsilon_0 = [\text{M}^1\text{L}^2\text{T}^1\text{A}^2]$

D. $\epsilon_0 = [\text{M}^1\text{L}^2\text{T}^1\text{A}]$

Answer: B

Solution:

Solution:

As we know, $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{R^2}$

$$\Rightarrow \epsilon_0 = \frac{q_1 q_2}{4\pi F R^2}$$

Hence, $\epsilon_0 = \frac{C^2}{N \cdot m^2} = \frac{[AT]^2}{[MLT^{-2}][L^2]}$

$$\Rightarrow \epsilon_0 = [M^{-1}L^{-3}T^4A^2]$$

Question205

If the time period t of the oscillation of a drop of liquid of density d , radius r , vibrating under surface tension s is given by the formula

$t = \sqrt{r^{2b} s^c d^{a/2}}$. It is observed that the time period is directly

proportional to $\sqrt{\frac{d}{s}}$. The value of b should therefore be:

[Online April 23, 2013]

Options:

A. $\frac{3}{4}$

B. $\sqrt{3}$

C. $\frac{3}{2}$

D. $\frac{2}{3}$

Answer: C

Solution:

Solution:

Question206

The dimensions of angular momentum, latent heat and capacitance are, respectively.

[Online April 22, 2013]

Options:

A. $M L^2 T^1 A^2, L^2 T^{-2}, M^{-1} L^{-2} T^2$

B. $M L^2 T^{-2}, L^2 T^2, M^{-1} L^{-2} T^4 A^2$

C. $M L^2 T^{-1}, L^2 T^{-2}, M L^2 T A^2$

D. $M L^2 T^{-1}, L^2 T^{-2}, M^{-1} L^{-2} T^4 A^2$

Answer: D

Solution:

Solution:

$$\text{Angular momentum} = m \times v \times r = M L^2 T^{-1}$$

$$\text{Latent heat } L = \frac{Q}{m} = \frac{M L^2 T^{-2}}{M} = L^2 T^{-2}$$

$$\text{Capacitance } C = \frac{\text{Charge}}{P \cdot d} = M^{-1} L^{-2} T^4 A^2$$

Question207

Given that K = energy, V = velocity, T = time. If they are chosen as the fundamental units, then what is dimensional formula for surface tension?

[Online May 7, 2012]

Options:

A. $[K V^{-2} T^{-2}]$

B. $[K^2 V^2 T^{-2}]$

C. $[K^2 V^{-2} T^{-2}]$

D. $[K V^2 T^2]$

Answer: A

Solution:

Solution:

$$\text{Surface tension, } T = \frac{F}{l} = \frac{F}{l} \cdot \frac{l}{l} \cdot \frac{T^2}{T^2}$$

$$(\text{As, } F \cdot l = K \text{ (energy)}) ; \frac{T^2}{l^2} = V^{-2}$$

$$\text{Therefore, surface tension} = [K V^{-2} T^{-2}]$$

Question208

Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then error in the value of resistance of the wire is
[2012]

Options:

A. 6%

B. zero

C. 1%

D. 3%

Answer: A

Solution:

Solution:

According to ohm's law, $V = I R$

$$R = \frac{V}{I}$$

$$\therefore \text{Percentage error} = \frac{\text{Absolute error}}{\text{Measurement}} \times 10^2$$

$$\text{where, } \frac{\Delta V}{V} \times 100 = \frac{\Delta I}{I} \times 100 = 3\%$$

$$\begin{aligned} \text{then, } \frac{\Delta R}{R} \times 100 &= \frac{\Delta V}{V} \times 10^2 + \frac{\Delta I}{I} \times 10^2 \\ &= 3\% + 3\% = 6\% \end{aligned}$$

Question209

A spectrometer gives the following reading when used to measure the angle of a prism.

Main scale reading : 58.5 degree

Vernier scale reading : 09 divisions

Given that 1 division on main scale corresponds to 0.5

degree. Total divisions on the Vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data is

[2012]

Options:

A. 58.59 degree

B. 58.77 degree

C. 58.65 degree

D. 59 degree

Answer: C

Solution:

Solution:

∴ Reading of Vernier = Main scale reading + Vernier scale reading × least count.

Main scale reading = 58.5

Vernier scale reading = 09 division

least count of Vernier = $\frac{0.5^\circ}{30}$

Thus, $R = 58.5^\circ + 9 \times \frac{0.5^\circ}{30} = 58.65^\circ$

Question210

N divisions on the main scale of a vernier calliper coincide with (N + 1) divisions of the vernier scale. If each division of main scale is 'a' units, then the least count of the instrument is

[Online May 19, 2012]

Options:

A. a

B. $\frac{a}{N}$

C. $\frac{N}{N+1} \times a$

D. $\frac{a}{N+1}$

Answer: D

Solution:

Solution:

No. of divisions on main scale = N

No. of divisions on vernier scale = N + 1

size of main scale division = a

Let size of vernier scale division be b
then we have

$$aN = b(N + 1) \Rightarrow b = \frac{aN}{N + 1}$$

$$\text{Least count is } a - b = a - \frac{aN}{N + 1}$$

$$= a \left[\frac{N + 1 - N}{N + 1} \right] = \frac{a}{N + 1}$$

Question211

**A student measured the diameter of a wire using a screw gauge with the least count 0.001 cm and listed the measurements. The measured value should be recorded as
[Online May 12, 2012]**

Options:

A. 5.3200 cm

B. 5.3 cm

C. 5.32 cm

D. 5.320 cm

Answer: D

Solution:

Solution:

The least count (L.C.) of a screw gauge is the smallest length which can be measured accurately with it.

As least count is $0.001 \text{ cm} = \frac{1}{1000} \text{ cm}$

Hence measured value should be recorded upto 3 decimal places i.e., 5.320 cm

Question212

A screw gauge gives the following reading when used to measure the diameter of a wire.

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1mm on main scale corresponds to 100 divisions of the circular scale. The diameter of wire from the above data is

[2011]

Options:

A. 0.052 cm

B. 0.026 cm

C. 0.005 cm

D. 0.52 cm

Answer: A

Solution:

Solution:

Least count, L.C. = $\frac{1}{100} \text{ mm}$

Diameter of wire = MSR + CSR $\times$ L.C.

$\therefore 1 \text{ mm} = 0.1 \text{ cm} = 0 + \frac{1}{100} \times 52 = 0.52 \text{ mm} = 0.052 \text{ cm}$

Question213

The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are

[2010]

Options:

A. 5, 1, 2

B. 5, 1, 5

C. 5, 5, 2

D. 4, 4, 2

Answer: A

Solution:

Solution:

Number of significant figures in $23.023 = 5$

Number of significant figures in $0.0003 = 1$

Number of significant figures in $2.1 \times 10^{-3} = 2$

So, the radiation belongs to X-rays part of the spectrum

Question214

In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half- a degree ($= 0.5^\circ$), then the least count of the instrument is:

[2009]

Options:

A. half minute

B. one degree

C. half degree

D. one minute

Answer: D

Solution:

Solution:

30 Divisions of V.S. coincide with 29 divisions of M.S.

$$\therefore 1 \text{ V.S.D} = \frac{29}{30} \text{ MSD}$$

$$\text{L.C.} = 1 \text{ MSD} - 1 \text{VSD} = 1 \text{ MSD} - \frac{29}{30} \text{ MSD}$$

$$= \frac{1}{30} \text{ MSD} = \frac{1}{30} \times 0.5^\circ = 1 \text{ minute}$$

Question215

**The dimensions of magnetic field in M, L, T and C (coulomb) is given as
[2008]**

Options:

A. $[M L T^{-1} C^{-1}]$

B. $[M T^2 C^{-2}]$

C. $[M T^{-1} C^{-1}]$

D. $[M T^{-2} C^{-1}]$

Answer: C

Solution:

Solution:

Magnitude of Lorentz formula $F = qvB \sin \theta$

$$B = \frac{F}{qv} = \frac{M L T^{-2}}{C \times L T^{-1}} = [M T^{-1} C^{-1}]$$

Question216

**A body of mass $m = 3.513 \text{ kg}$ is moving along the x-axis with a speed of 5.00 ms^{-1} . The magnitude of its momentum is recorded as
[2008]**

Options:

- A. 17.6 kg ms^{-1}
- B. $17.565 \text{ kg ms}^{-1}$
- C. 17.56 kg ms^{-1}
- D. 17.57 kg ms^{-1}

Answer: A

Solution:

Solution:

Momentum, $p = m \times v$

Given, mass of a body = 3.513 kg speed of body

$= (3.513) \times (5.00) = 17.565 \text{ kg m/s}$

$= 17.6$ (Rounding off to get three significant figures)

Question217

Two full turns of the circular scale of a screw gauge cover a distance of 1mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of -0.03 mm . While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the number of circular scale divisions in line with the main scale as 35. The diameter of the wire is
[2008]

Options:

- A. 3.32 mm
- B. 3.73 mm
- C. 3.67 mm
- D. 3.38 mm

Answer: D

Solution:

Solution:

Least count of screw gauge = 0.01 mm

$$\therefore \frac{0.5}{50} \text{ mm}$$

Reading = [M.S.R. + C.S.R. $\times$ L.C.] – (zero error)

$$= [3 + 35 \times 0.01] - (-0.03) = 3.38 \text{ mm}$$

Question218

Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$, where Q denotes the electric charge?
[2006]

Options:

A. $\frac{Wb}{m^2}$

B. Henry (H)

C. $\frac{H}{m^2}$

D. Weber (Wb)

Answer: B

Solution:

Solution:

$$\text{Mutual inductance} = \frac{\phi}{I} = \frac{BA}{I}$$

$$[\text{Henry}] = \frac{[M T^{-1} Q^{-1} L^2]}{[Q T^{-1}]} = M L^2 Q^{-2}$$

Question219

Out of the following pair, which one does NOT have identical dimensions ?
[2005]

Options:

- A. Impulse and momentum
- B. Angular momentum and planck's constant
- C. Work and torque
- D. Moment of inertia and moment of a force

Answer: D

Solution:

Solution:

Moment of Inertia, $I = M R^2$

$$[I] = [M L^2]$$

Moment of force, $\vec{\tau} = \vec{r} \times \vec{F}$

$$\vec{\tau} = [L][M L T^{-2}] = [M L^2 T^{-2}]$$

Question220

Which one of the following represents the correct dimensions of the coefficient of viscosity?

[2004]

Options:

A. $[M L^{-1} T^{-1}]$

B. $[M L T^{-1}]$

C. $[M L^{-1} T^{-2}]$

D. $[M L^{-2} T^{-2}]$

Answer: A

Solution:

Solution:

According to, Stokes law,

$$F = 6\pi\eta r v \Rightarrow \eta = \frac{F}{6\pi r v}$$

$$\eta = \frac{[M L T^{-2}]}{[L][L T^{-1}]}$$

$$\Rightarrow \eta = [M L^{-1} T^{-1}]$$

Question221

Dimensions of $\frac{1}{\mu_o \epsilon_o}$, where symbols have their usual meaning, are [2003]

Options:

A. $[L^{-1} T]$

B. $[L^{-2} T^2]$

C. $[L^2 T^{-2}]$

D. $[L T^{-1}]$

Answer: C

Solution:

Solution:

As we know, the velocity of light in free space is given by

$$c = \frac{1}{\sqrt{\mu_o \epsilon_o}}$$

$$\therefore \frac{1}{\mu_o \epsilon_o} = c^2 = [L T^{-1}]^2$$

$$\frac{1}{\sqrt{\mu_o \epsilon_o}} = C^2 [m/s]^2 = [L T^{-1}]^2 = [M^0 L^2 T^{-2}]$$

Question222

The physical quantities not having same dimensions are [2003]

Options:

A. torque and work

B. momentum and planck's constant

C. stress and young's modulus

D. speed and $(\mu_o \epsilon_o)^{-1/2}$

Answer: B

Solution:

Solution:

Momentum, $= mv = [M L T^{-1}]$

Planck's constant,

$$h = \frac{E}{\nu} = \frac{[M L^2 T^{-2}]}{[T^{-1}]} = [M L^2 T^{-1}]$$

Question 223

**Identify the pair whose dimensions are equal
[2002]**

Options:

A. torque and work

B. stress and energy

C. force and stress

D. force and work

Answer: A

Solution:

Solution:

$$\text{Work } W = \vec{F} \cdot \vec{s} = F s \cos \theta$$

$$\begin{aligned} \because \vec{A} \cdot \vec{B} &= AB \cos \theta \\ &= [M L T^{-2}][L] = [M L^2 T^{-2}] \end{aligned}$$

$$\text{Torque, } \vec{\tau} = \vec{r} \times \vec{F} \Rightarrow \tau = r F \sin \theta$$

$$\because \vec{A} \times \vec{B} = AB \sin \theta = [L][M L T^{-2}] = [M L^2 T^{-2}]$$
